Tighter Proofs for PKE-to-KEM Transformations under Average-Case Decryption Error and without γ-Spread

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Abstract

In the NIST post-quantum standardization process, Fujisaki-Okamoto-like (FO-like) transformation has become the de facto paradigm for constructing IND-CCA secure key encapsulation mechanisms (KEMs) from public-key encryption (PKE). However, most post-quantum PKE schemes exhibit decryption error, which poses significant challenges for the security proofs of FO-like PKE-to-KEM transformations, particularly in the quantum-accessible random oracle model (QROM). Hofheinz, Hövelmanns, and Kiltz (TCC 2017) gave the first QROM security proofs for PKE-to-KEM transformations under *worst-case* decryption error. To relax this to the more designer-friendly one of *average-case* decryption error, Duman et al. (PKC 2023) presented two transformations, FOAC_0 and FOAC, which are under average-case decryption error but introduce substantial loss in QROM reduction tightness ($\mathcal{O}(q^8)$ for FOAC_0 and $\mathcal{O}(q^6)$ for FOAC) and the need for the γ-spread assumption on the underlying PKEs. Very recently, Ge et al. (ePrint 2025) removed the γ-spread assumption for FOAC_0 and improved the QROM reduction tightness to $\mathcal{O}(q^4)$ for both FOAC_0 and FOAC.

In this work, we make further advances by introducing two refined variants: FOAC_0' and FOAC'. We provide new security analyses in both the ROM and the QROM, and present the following key contributions: (1) Compared with previous transformations under average-case decryption error, FOAC_0' and FOAC' exhibit tighter security proofs with QROM reduction loss of only $\mathcal{O}(q^2)$ for FOAC_0' and $\mathcal{O}(q^3)$ for FOAC' when the underlying PKE is OW-CPA secure, and just $\mathcal{O}(q)$ when it is deterministic or IND-CPA security; (2) Both FOAC_0' and FOAC' eliminate the γ-spread assumption entirely, further relaxing the requirements on the underlying PKE.

To support our QROM proofs, we provide three new QROM proof techniques that build on Zhandry's compressed oracle technique (CRYPTO 2019). These techniques may be of independent interest and could have broader applicability in post-quantum cryptography.

Keywords: Fujisaki-Okamoto transformation · Key encapsulation mechanisms · QROM · Reduction tightness.

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1 Introduction

Indistinguishability against chosen-ciphertext attacks (IND-CCA) [40] is widely recognized as the standard security notion for public-key encryption (PKE) and key encapsulation mechanisms (KEM), while the Fujisaki-Okamoto (FO) transformation [12,13] is a well-known transformation that transforms a PKE scheme with weaker security into an IND-CCA secure PKE in the random oracle model (ROM) [2] by combining it with a symmetric encryption scheme. Since Dent [9] proposed its KEM variants, the FO transformation for KEM has become a de facto standard for constructing IND-CCA secure KEMs. It has been widely adopted in submissions [3, 8] to the NIST post-quantum standardization process [34], as well as in the design of the already standardized KEM ML-KEM [35] and the soon-to-be-standardized HQC [23]. Importantly, in the post-quantum setting, many PKE schemes intended to be transformed are not perfectly correct, i.e., they occasionally fail to decrypt their own ciphertexts. Moreover, to capture adversaries with potential quantum computing capabilities, security must be established in the quantum-accessible random oracle model (QROM) [5].

The Transformation under Worst-Case Decryption Error. Addressing these challenges, Hofheinz et al. [19] provided the first QROM security proofs for FO-like KEM transformations that tolerate imperfect correctness of the PKE. They formalized four transformations, $\mathsf{FO}^{\not\perp}$, $\mathsf{FO}^{\perp}$, $\mathsf{FO}^{\not\perp}_m$, and $\mathsf{FO}^{\perp}_m$, which transform a PKE satisfying standard security notions (one-wayness against chosen-plaintext attacks (OW-CPA) or indistinguishability against chosen-plaintext attacks (IND-CPA)) into an IND-CCA secure KEM. Here, the superscripts $\not\perp$ (or $\perp$) indicate the rejection type, i.e., the scheme returns a pseudorandom value (or an explicit failure symbol $\perp$) upon decapsulation failure, while the subscript m (or its absence) specifies the KEM key is derived as $k := H(m)$ (or $k := H(m, c)$). While subsequent works have made significant progress in tightening the QROM security reductions for FO-like KEM transformations [4, 10, 14, 15, 22, 26–29, 31, 41], progress on relaxing the required properties of the underlying PKE has been comparatively modest. For a PKE scheme $\mathsf{PKE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ with message space $\mathcal{M}$ and randomness space $\mathcal{R}$, the FO-like KEM transformations require that the *worst-case* decryption error rate [19]

$$\delta_{wc} := \mathbb{E}_{(\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{Gen}(1^\lambda)}\left[\max_{m\in\mathcal{M}} \Pr_{\rho\leftarrow\$\mathcal{R}}[\mathsf{Dec}(\mathsf{sk}, \mathsf{Enc}(\mathsf{pk}, m; \rho)) \neq m]\right]$$

be negligible. This means that, to correctly apply the FO transformation, the scheme designer must control the decryption error probability for *every* possible message and bound the maximum error rate. This requirement imposes additional design effort and is not always straightforward. For instance, in the PKE schemes based on NTRU [18], reducing δ_{wc} by increasing the modulus q makes lattice-reduction algorithms more effective, thereby weakening the overall security of the scheme [11].

The Transformation under Average-Case Decryption Error. In contrast to δ_{wc}, the *average-case* decryption error rate

$$\delta_{ac} := \mathbb{E}_{(\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{Gen}(1^\lambda)}\left[\Pr_{m\leftarrow\$\mathcal{M},\rho\leftarrow\$\mathcal{R}}[\mathsf{Dec}(\mathsf{sk}, \mathsf{Enc}(\mathsf{pk}, m; \rho)) \neq m]\right]$$

is far more designer-friendly. It requires only that the decryption error is negligible for *random* messages, thereby relaxing the parameter constraints and simplifying the design of the underlying PKE. To enable the use of PKE schemes with negligible average-case decryption error δ_{ac}, Duman et al. [11] proposed and proved, in the QROM, two transformations, ACWC_0 and ACWC, that transform a PKE with negligible δ_{ac} into one with negligible δ_{wc}. By composing these with $\mathsf{FO}^{\perp}_m$, they obtained two FO-like KEM transformations under average-case decryption error: $\mathsf{FOAC}_0 := \mathsf{FO}^{\perp}_m \circ \mathsf{ACWC}_0$ and $\mathsf{FOAC} := \mathsf{FO}^{\perp}_m \circ \mathsf{ACWC}$, as illustrated in Fig. 1 and 2, respectively.

$\mathsf{Gen}(1^\lambda)$	$\mathsf{Decaps}(\mathsf{sk}, (c_1, c_2))$
$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}'(1^\lambda)$	$m := \mathsf{Dec}'(\mathsf{sk}, c_1)$
return $(\mathsf{pk}, \mathsf{sk})$	**if** $m = \perp$ **then** : **return** $\perp$
$\mathsf{Encaps}(\mathsf{pk})$	$r := c_2 \oplus F(m)$
$r \leftarrow\!\!\$\ \{0,1\}^n$	$(m', \rho, k) := H(r)$
$(m, \rho, k) \leftarrow\!\!\$\ H(r)$	**if** $r \oplus F(m') = c_2$ **and**
$c_1 := \mathsf{Enc}'(\mathsf{pk}, m; \rho)$	$\mathsf{Enc}'(\mathsf{pk}, m'; \rho) = c_1$ **then**
$c_2 := r \oplus F(m)$	**return** k
return $(k, (c_1, c_2))$	**return** $\perp$

Figure 1: $\mathsf{KEM} := \mathsf{FOAC_0}[\mathsf{PKE}', F, H]$, where $F : \mathcal{M} \to \{0,1\}^n$ and $H : \{0,1\}^n \to \mathcal{M} \times \mathcal{R} \times \mathcal{K}$ are hash functions. Here, $\mathcal{M}$ is the message space of $\mathsf{PKE}' = (\mathsf{Gen}', \mathsf{Enc}', \mathsf{Dec}')$, $\mathcal{R}$ is the randomness space of Enc', and $\mathcal{K}$ is the key space of the derived KEM KEM.

$\mathsf{Gen}(1^\lambda)$	$\mathsf{Decaps}(\mathsf{sk}, c)$
$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}'(1^\lambda)$	$m_1 \| m_2 := \mathsf{Dec}'(\mathsf{sk}, c)$
return $(\mathsf{pk}, \mathsf{sk})$	**if** $m_1 \| m_2 = \perp$ **then** : **return** $\perp$
$\mathsf{Encaps}(\mathsf{pk})$	$r := m_2 \oplus F(m_1)$
$r \leftarrow\!\!\$\ \mathcal{M}_2$	$(m_1', \rho, k) := H(r)$
$(m_1, \rho, k) \leftarrow\!\!\$\ H(r)$	**if** $\mathsf{Enc}'(\mathsf{pk}, m_1' \| m_2; \rho) = c$ **then**
$m_2 := r \oplus F(m_1)$	**return** k
$c := \mathsf{Enc}'(\mathsf{pk}, m_1 \| m_2; \rho)$	**return** $\perp$
return (k, c)	

Figure 2: $\mathsf{KEM} := \mathsf{FOAC}[\mathsf{PKE}', F, H]$, where $F : \mathcal{M}_1 \to \mathcal{M}_2$ and $H : \mathcal{M}_2 \to \mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}$ are hash functions. Here, $\mathcal{M}_1 \times \mathcal{M}_2$ is the message space of $\mathsf{PKE}' = (\mathsf{Gen}', \mathsf{Enc}', \mathsf{Dec}')$, $\mathcal{R}$ is the randomness space of Enc', and $\mathcal{K}$ is the key space of the derived KEM KEM.

Unfortunately, the reduction tightness of $\mathsf{FOAC_0}$ and FOAC given by Duman et al. [11] in both the ROM and QROM is significantly looser than that of the $\mathsf{FO}_m^{\perp}$ [21]. In particular, $\mathsf{FOAC_0}$ suffers from a QROM reduction loss factor as large as $\mathcal{O}(q^8)$, where q denotes the number of oracle queries made by the adversary, and a fourth-power degradation in the reduction loss exponent, which severely impairs its practical relevance.

State of the Art. To obtain tighter reductions, Ge et al. [16] revisited the security proofs of $\mathsf{FOAC_0}$ and FOAC and derived tighter security bounds. Moreover, they showed that the γ-spread condition previously required for $\mathsf{FOAC_0}$ can be entirely eliminated, further reducing the restrictions on the underlying PKE. Nevertheless, in the QROM, both schemes still incur a $\mathcal{O}(q^4)$ loss factor for the randomized PKE, which still remains looser than the state-of-the-art security reductions for $\mathsf{FO}_m^{\perp}$ [21]. The reduction tightness of these transformations for different types of underlying PKE, both in the ROM and the QROM, is summarized in Table 1.

Motivating Question. Although Ge et al. [16] made significant improvements over the work of Duman et al. [11], there remains considerable room for further improvement in reduction tightness. Specifically, the current security reductions for $\mathsf{FOAC_0}$ and FOAC remain looser than that of the original $\mathsf{FO}_m^{\perp}$. In practice, tight security reductions are highly desirable for

Table 1: Reduction tightness from IND-CCA secure KEM to CPA-secure PKE. ϵ_R is the advantage of the reduction w.r.t the CPA security of the underlying PKE, $\epsilon_{\mathcal{A}}$ denotes the advantage of the adversary $\mathcal{A}$ w.r.t. the IND-CCA security of the resulting KEM, q is the number of $\mathcal{A}$'s oracle queries, and rPKE (or dPKE) indicates that the underlying PKE is randomized (or deterministic).

Transformation	Model	Assump.	OW-CPA rPKE	OW-CPA dPKE	IND-CPA rPKE
FOAC_0 [11]	ROM	γ-spread	$\epsilon_R \approx \mathcal{O}(1/q^2)\epsilon_{\mathcal{A}}$	-	-
FOAC_0 [16]	ROM	/	$\epsilon_R \approx \mathcal{O}(1/q^2)\epsilon_{\mathcal{A}}$	$\epsilon_R \approx \epsilon_{\mathcal{A}}$	-
FOAC'_0(Thm.4.1)	ROM	/	$\epsilon_R \approx \mathcal{O}(1/q)\epsilon_{\mathcal{A}}$	$\epsilon_R \approx \epsilon_{\mathcal{A}}$	$\epsilon_R \approx \epsilon_{\mathcal{A}}$
FOAC_0 [11]	QROM	γ-spread	$\epsilon_R \approx \mathcal{O}(1/q^8)\epsilon_{\mathcal{A}}^4$	-	-
FOAC_0 [16]	QROM	/	$\epsilon_R \approx \mathcal{O}(1/q^4)\epsilon_{\mathcal{A}}^2$	$\epsilon_R \approx \mathcal{O}(1/q^2)\epsilon_{\mathcal{A}}^2$	-
FOAC'_0(Thm.5.4)	QROM	/	$\epsilon_R \approx \mathcal{O}(1/q^2)\epsilon_{\mathcal{A}}^2$	$\epsilon_R \approx \mathcal{O}(1/q)\epsilon_{\mathcal{A}}^2$	$\epsilon_R \approx \mathcal{O}(1/q)\epsilon_{\mathcal{A}}^2$
FOAC [11]	ROM	γ-spread	$\epsilon_R \approx \mathcal{O}(1/q^2)\epsilon_{\mathcal{A}}$	-	-
FOAC'(Thm.4.2)	ROM	/	$\epsilon_R \approx \mathcal{O}(1/q^2)\epsilon_{\mathcal{A}}$	$\epsilon_R \approx \epsilon_{\mathcal{A}}$	$\epsilon_R \approx \epsilon_{\mathcal{A}}$
FOAC [11]	QROM	γ-spread	$\epsilon_R \approx \mathcal{O}(1/q^6)\epsilon_{\mathcal{A}}^2$	-	-
FOAC [16]	QROM	γ-spread	$\epsilon_R \approx \mathcal{O}(1/q^4)\epsilon_{\mathcal{A}}^2$	$\epsilon_R \approx \mathcal{O}(1/q^2)\epsilon_{\mathcal{A}}^2$	-
FOAC'(Thm.5.5)	QROM	/	$\epsilon_R \approx \mathcal{O}(1/q^3)\epsilon_{\mathcal{A}}^2$	$\epsilon_R \approx \mathcal{O}(1/q)\epsilon_{\mathcal{A}}^2$	$\epsilon_R \approx \mathcal{O}(1/q)\epsilon_{\mathcal{A}}^2$
$\mathsf{FO}_m^{\perp}$ [21]	ROM	γ-spread	$\epsilon_R \approx \mathcal{O}(1/q)\epsilon_{\mathcal{A}}$	-	$\epsilon_R \approx \epsilon_{\mathcal{A}}$
$\mathsf{FO}_m^{\perp}$ [21]	QROM	γ-spread	$\epsilon_R \approx \mathcal{O}(1/q^2)\epsilon_{\mathcal{A}}^2$	-	$\epsilon_R \approx \mathcal{O}(1/q)\epsilon_{\mathcal{A}}^2$

efficiency reasons, as they avoid the need to increase security parameters merely to compensate for reduction loss [17,39]. This naturally raises the following question:

Is it possible to achieve a tighter PKE-to-KEM transformation under average-case decryption error with a reduction loss comparable to, or even matching, that of $\mathsf{FO}_m^{\perp}$*?*

1.1 Our Contributions

In this work, we provide an affirmative answer to the above question. We not only obtain two tighter PKE-to-KEM transformations that rely only on the average-case decryption error of the underlying PKE, but also completely eliminate the need for the γ-spread condition. To achieve this, we introduce *minor* modifications to FOAC_0 and FOAC and present new security proofs. To distinguish them from the original constructions, we denote our modified schemes by FOAC'_0 and FOAC', as illustrated in Fig. 3 and Fig. 4, respectively.

Firstly, as summarized in Table 1, we present tighter security proofs than prior works [11,16] in both the ROM and the QROM. In both models, FOAC'_0 achieves a reduction loss factor identical to that of $\mathsf{FO}_m^{\perp}$, while FOAC' (which has a more compact ciphertext) incurs an additional $\mathcal{O}(q)$ factor *only* when the underlying PKE is OW-CPA secure.

Secondly, like FOAC_0 and FOAC, both FOAC'_0 and FOAC' operate under the *average-case* decryption error δ_{ac} rather than the *worst-case* error δ_{wc} of the underlying PKE. Moreover,

$\mathsf{Gen}(1^\lambda)$

$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}'(1^\lambda)$
return $(\mathsf{pk}, \mathsf{sk})$

$\mathsf{Encaps}(\mathsf{pk})$

$r \leftarrow\!\!\$\ \{0,1\}^n$
$(m, \rho, k) \leftarrow\!\!\$\ H(r)$
$c_1 := \mathsf{Enc}'(\mathsf{pk}, m; \rho)$
$c_2 := r \oplus F(m)$
return $(k, (c_1, c_2))$

$\mathsf{Decaps}(\mathsf{sk}, (c_1, c_2))$

$m := \mathsf{Dec}'(\mathsf{sk}, c_1)$
if $m = \perp$ **then** : **return** $\perp$
$r := c_2 \oplus F(m)$
$(m', \rho, k) := H(r)$
if $m = m'$ **and** $\mathsf{Enc}'(\mathsf{pk}, m'; \rho) = c_1$ **then**
  return k
return $\perp$

Figure 3: $\mathsf{KEM} := \mathsf{FOAC}'_0[\mathsf{PKE}', F, H]$. Differences from FOAC_0 in Fig. 1 are highlighted.

$\mathsf{Gen}(1^\lambda)$

$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}'(1^\lambda)$
return $(\mathsf{pk}, \mathsf{sk})$

$\mathsf{Encaps}(\mathsf{pk})$

$r \leftarrow\!\!\$\ \mathcal{M}_2$
$(m_1, \rho, k) \leftarrow\!\!\$\ H(r)$
$m_2 := r \oplus F(m_1)$
$c := \mathsf{Enc}'(\mathsf{pk}, m_1 \| m_2; \rho)$
return (k, c)

$\mathsf{Decaps}(\mathsf{sk}, c)$

$m_1 \| m_2 := \mathsf{Dec}'(\mathsf{sk}, c)$
if $m_1 \| m_2 = \perp$ **then** : **return** $\perp$
$r := m_2 \oplus F(m_1)$
$(m'_1, \rho, k) := H(r)$
if $m_1 = m'_1$ **and** $\mathsf{Enc}'(\mathsf{pk}, m'_1 \| m_2; \rho) = c$ **then**
  return k
return $\perp$

Figure 4: $\mathsf{KEM} := \mathsf{FOAC}'[\mathsf{PKE}', F, H]$. Differences from FOAC in Fig. 2 are highlighted.

both transformations completely remove the need for the γ-spread condition, further relaxing the requirements on the PKE when constructing IND-CCA secure KEMs.

Thirdly, to support our QROM proofs, inspired by prior works [10, 15, 16, 32], we distill and formalize three novel QROM proof techniques based on the compressed oracle technique [43]. These are the *Switching Lemma* (Lemma 5.1), the *Lookup Lemma* (Lemma 5.2), and the *Puncturing Lemma* (Lemma 5.3). These tools allow us to lift ROM proof strategies to the QROM in a modular and conceptually simple manner, significantly simplifying security arguments. Due to their generality, we believe these lemmas are of independent interest and applicable to a wide range of other QROM proof scenarios.

1.2 Related Work

Research on building IND-CCA secure KEMs from weakly secure PKE has long been a major focus in academia. The FO-like KEM transformation, a KEM variant of the Fujisaki-Okamoto (FO) transform [12, 13] introduced by Dent [9] and proven secure in the random oracle model (ROM), is the most widely adopted paradigm. In the post-quantum setting, however, security must be established in the quantum-accessible random oracle model (QROM) [5] to account for quantum adversaries, and several works [42, 44] have shown that ROM security does not generally imply QROM security.

The first QROM security proof for FO-like KEM transformations, due to Hofheinz et al. [19], introduced an extra hash function and incurred a quartic loss. Subsequent works progressively improved the reduction: for OW-CPA PKE, Jiang et al. [26] removed the extra hash and achieved a quadratic loss, which was later shown to be inherent for black-box, non-rewinding reductions [29]. For IND-CPA PKE, the loss was further reduced [4, 28]. Saito et al. [41] introduced the tightly secure SXY transform via disjoint simulatability, later extended by Jiang et al. [27]. Recently, Kuchta et al. [31] and Ge et al. [14] developed the measure-rewind-measure and measure-rewind-extract techniques, respectively, to eliminate the quadratic loss. For the IND-1CCA setting[1], even more efficient transforms have been proposed and subsequently tightened [7, 24, 25].

While prior works significantly tightened the security reductions, they all require the underlying PKE to have a negligible *worst-case* decryption error δ_{wc}. To use PKEs with only negligible *average-case* decryption error δ_{ac}, Hövelmanns [20] introduced ACWC_1[2], which converts a δ_{ac}-correct PKE into a δ_{wc}-correct PKE; combined with $\mathsf{FO}_m^{\not\perp}$, this yields FOAC_1. Later, Kim and Park [30] replaced the XOR in ACWC_1 with a semi-generalized one-time pad (SOTP), obtaining ACWC_2 and FOAC_2. The latter was used to design $\mathsf{NTRU+}$ [37], a winner of the Korean Post-Quantum Cryptography Competition [38]. However, both ACWC_1 and ACWC_2 demand that the underlying PKE supports *randomness recovery*, a restriction that limits their applicability.

To lift the randomness-recovery requirement, Duman et al. [11] generalized the technique of Lyubashevsky and Seiler [33] and proposed ACWC_0 (which increases ciphertext length) and its compact variant ACWC. Composing these with $\mathsf{FO}_m^{\perp}$ gives FOAC_0 and FOAC, respectively. They proved IND-CCA security in both the ROM and the QROM, relying only on the δ_{ac}-correctness and the γ-spread property of the underlying PKE. Although these constructions remove the need for randomness recovery, their security reductions are substantially looser than that of $\mathsf{FO}_m^{\perp}$, and the additional γ-spread condition remains a burden. Subsequently, Ge et al. [16] revisited the proofs, achieving tighter bounds and eliminating the γ-spread requirement for FOAC_0. However, FOAC still relies on this condition, and the reduction loss for both schemes still lags behind that of $\mathsf{FO}_m^{\perp}$. Closing this gap is the central motivation of our work.

2 Technique Overview

2.1 Removing the γ-spread Assumption from FOAC

In this section, we first motivate the need for modifying the original FOAC_0 and FOAC and explain why, after our tweaks, the security of FOAC' no longer relies on the γ-spread condition of the underlying PKE. Before describing these modifications, we want to recall what γ-spread is and why it was required in the original security proofs of FO-like transformations.

Why $\mathsf{FO}_m^{\perp}$ requires γ-spread? For a PKE scheme $\mathsf{PKE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ with message space $\mathcal{M}$, randomness space $\mathcal{R}$ and ciphertext space $\mathcal{C}$, we say that PKE is weakly γ-spread if

$$\mathbb{E}_{(\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{Gen}(1^\lambda)}\left[\max_{m\in\mathcal{M},\, c\in\mathcal{C}} \Pr_{\rho\leftarrow\$\mathcal{R}}[\mathsf{Enc}(\mathsf{pk}, m; \rho) = c]\right] \leq 2^{-\gamma} .$$

As a concrete example, consider the security of $\mathsf{FO}_m^{\perp}$ in the ROM. Applying $\mathsf{FO}_m^{\perp}$ to PKE yields $\mathsf{KEM} := \mathsf{FO}_m^{\perp}[\mathsf{PKE}, H]$, where the hash function H is modeled as a random oracle. The reduction maintains an initially empty list List^H to record the adversary's queries. The decapsulation algorithm Decaps is shown in the left part of Fig. 5. A crucial step in the IND-CCA proof of KEM is to establish that

$$\forall m \neq \perp, c \in \mathcal{C},\ \text{if } \mathrm{List}^H(m) = \perp \text{ then } \Pr[\mathsf{Enc}(\mathsf{pk}, m; H(m)[0]) = c] \text{ is negligible},$$

[1] I.e., at most one decapsulation query is allowed in the IND-CCA security game.

[2] Originally named ACWC; we denote it as ACWC_1 to avoid confusion.

$\mathsf{Decaps}(\mathsf{sk}, c)$	$\mathsf{Enc}^0_{\mathrm{ac}}(\mathsf{pk}, m; \theta)$
$1:\ m \coloneqq \mathsf{Dec}(\mathsf{sk}, c)$	$6:\ (m', \rho) \coloneqq \theta$
$2:\ \textbf{if } m = \bot \textbf{ then}$	$7:\ c_1 \coloneqq \mathsf{Enc}'(\mathsf{pk}, m'; \rho),\ c_2 \coloneqq m \oplus F(m')$
$\quad\ \textbf{return } \bot$	$8:\ \textbf{return } c \coloneqq (c_1, c_2)$
$3:\ (\theta, k) \coloneqq H(m)$	$\mathsf{Enc}_{\mathrm{ac}}(\mathsf{pk}, m; \theta)$
$4:\ \textbf{if } \mathsf{Enc}(\mathsf{pk}, m; \theta) = c \textbf{ then}$	$9:\ (m'_1, \rho) \coloneqq \theta,\ m_2 \coloneqq m \oplus F(m'_1)$
$\quad\ \textbf{return } k$	$10: c \coloneqq \mathsf{Enc}'(\mathsf{pk}, m'_1 \| m_2; \rho)$
$5:\ \textbf{return } \bot$	$11: \textbf{return } c$

Figure 5: Decapsulation algorithm Decaps of $\mathsf{FO}^{\bot}_{m}$, and encryption algorithms $\mathsf{Enc}^0_{\mathrm{ac}}$ and $\mathsf{Enc}_{\mathrm{ac}}$ used by FOAC_0 and FOAC, respectively.

where $H(m)[0]$ denotes the first component of $H(m)$. This ensures that, for any ciphertext passing the re-encryption check in line 4 of Fig. 5, its plaintext has been stored in List^H with overwhelming probability. Consequently, the reduction can answer decapsulation queries without sk by retrieving the matching m from List^H.

When $\mathrm{List}^H(m) = \bot$ (i.e., the adversary has not queried $H(m)$), the value $H(m)$ is uniformly random by the random oracle model. The weak γ-spread condition then can directly imply $\Pr\big[\mathsf{Enc}(\mathsf{pk}, m; H(m)[0]) = c\big] \le 2^{-\gamma}$.

Why the proof of FOAC_0 in [16] does not require γ-spread? For $\mathsf{KEM} \coloneqq \mathsf{FOAC}_0[\mathsf{PKE}', F, H] = \mathsf{FO}^{\bot}_{m}\big[\mathsf{ACWC}_0[\mathsf{PKE}', F], H\big]$, both the hash functions F and H are modeled as random oracles, with initially empty lists List^F and List^H maintained by the reduction. The encryption algorithm $\mathsf{Enc}^0_{\mathrm{ac}}$ used for the re-encryption step in Decaps is depicted in the upper right part of Fig. 5. The requirement on $\mathsf{Enc}^0_{\mathrm{ac}}$ now becomes:

$$\begin{aligned}&\forall m \neq \bot, c = (c_1, c_2),\ \text{if } \mathrm{List}^H(m) = \bot\\ &\qquad\qquad \text{then } \Pr[\mathsf{Enc}'(\mathsf{pk}, m'; \rho) = c_1 \wedge m \oplus F(m') = c_2] \text{ is negligible ,}\end{aligned}$$

where $((m', \rho), k) \coloneqq H(m)$. If $\mathrm{List}^H(m) = \bot$, the adversary has not queried $H(m)$, so m' is uniformly random. Since a PPT adversary makes at most polynomially many queries to F, and the space of m' is sufficiently large, the probability that m' has been queried to F is also negligible. Hence, with overwhelming probability, $F(m')$ is also uniformly random. That is, for fixed m and c_2,

$$\Pr[m \oplus F(m') = c_2] \le 1/|\mathcal{Y}| ,$$

where $\mathcal{Y}$ denotes the codomain of F. Consequently,

$$\Pr[\mathsf{Enc}'(\mathsf{pk}, m'; \rho) = c_1 \wedge m \oplus F(m') = c_2] \le \Pr[m \oplus F(m') = c_2] \le 1/|\mathcal{Y}| ,$$

and no γ-spread assumption is required.

Why the same technique does not apply to FOAC? For $\mathsf{KEM} \coloneqq \mathsf{FOAC}[\mathsf{PKE}', F, H] = \mathsf{FO}^{\bot}_{m}\big[\mathsf{ACWC}[\mathsf{PKE}', F], H\big]$, F and H are again modeled as random oracles, maintaining lists List^F and List^H. The encryption algorithm $\mathsf{Enc}_{\mathrm{ac}}$ used in the re-encryption step is shown in the lower right part of Fig. 5. The condition now becomes:

$$\begin{aligned}&\forall m \neq \bot, c \in \mathcal{C},\ \text{if } \mathrm{List}^H(m) = \bot\\ &\qquad\qquad \text{then } \Pr[\mathsf{Enc}'(\mathsf{pk}, m'_1 \| (m \oplus F(m'_1)); \rho) = c] \text{ is negligible ,}\end{aligned}$$

where $((m_1', \rho), k) := H(m)$. Here, if $\text{List}^H(m) = \perp$, m_1' is uniformly random, and with overwhelming probability, m_1' has not been queried to F, so $F(m_1')$ is also uniform, as in the case of FOAC_0. Nevertheless, even under these conditions we cannot bound the above probability without invoking γ-spread.

In FOAC_0, the event "$m \oplus F(m') = c_2$" depends on a single random value; for fixed m and c_2, exactly one value of $F(m')$ satisfies the equality. In contrast, for FOAC, the event involves three random variables $(m_1', F(m_1'), \rho)$, and the set of tuples $(m_1', F(m_1'), \rho)$ satisfying $\mathsf{Enc}'(\mathsf{pk}, m_1' \| (m \oplus F(m_1')); \rho) = c$ is not a singleton. Its cardinality, and hence the probability, cannot be bounded a priori without a γ-spread assumption for fixed m and c.

How to modify FOAC to eliminate γ-spread? The fundamental reason for requiring $\Pr[\mathsf{Enc}(\mathsf{pk}, m; H(m)[0]) = c]$ to be negligible when $\text{List}^H(m) = \perp$ is to guarantee that, whenever the decapsulation algorithm outputs a non-$\perp$ key k, the adversary must have queried H at the corresponding m with non-negligible probability. This allows the reduction, without sk, to retrieve the appropriate entry from List^H and answer the decapsulation query.

Our modification is simple: we insert an extra check $m_1 = m_1'$ at line 5 of Fig. 2, requiring that the value m_1' obtained from $H(r)$ equal the value m_1 computed by Dec' in line 1. This modification is shown in line 5 of Fig. 4. With this additional check, for any $m_1 \| m_2 \neq \perp$ and any ciphertext c, if $\text{List}^H(r) = \perp$, the probability that line 5 passes is at most $1/|\mathcal{M}_1|$, where $\mathcal{M}_1$ is the space of m_1. This yields a clean, parameter-independent bound. In fact, relative to FOAC, the only case excluded by FOAC' is the pathological situation where $m_1 \| m_2 = \mathsf{Dec}'(\mathsf{sk}, \mathsf{Enc}'(\mathsf{pk}, m_1' \| m_2, \rho))$ for some $m_1' \neq m_1$. In this scenario, the two parties attempting to establish a shared key via the KEM would very likely obtain inconsistent keys. Thus, our modification also enhances the correctness of the scheme.

We can also apply an analogous modification to FOAC_0; the revised condition is shown in line 5 of Fig. 3. This change both eliminates the pathological case where $m \neq m'$ yet $m = \mathsf{Dec}'(\mathsf{sk}, \mathsf{Enc}'(\mathsf{pk}, m'; \rho))$, thereby enhancing correctness, and removes one hash evaluation, which makes the proof cleaner.

2.2 Tighter Proofs for FOAC_0' and FOAC'

In the security analyses of $\mathsf{FOAC}_0 := \mathsf{FO}_m^{\perp} \circ \mathsf{ACWC}_0$ and $\mathsf{FOAC} := \mathsf{FO}_m^{\perp} \circ \mathsf{ACWC}$ due to Duman et al. [11], a *modular* proof strategy is employed: the security of the intermediate transformations ACWC_0 and ACWC is established first, and then combined with the existing bounds for $\mathsf{FO}_m^{\perp}$ [10] to obtain the overall security guarantees. Ge et al. [16] adopt a similar modular approach for FOAC, but treat FOAC_0 differently: they first reduce its IND-CCA security to IND-CPA security, and then further reduce IND-CPA security to the OW-CPA security of the underlying PKE.

The appeal of modular analysis lies in its reusability: it allows one to directly leverage established results, requiring proofs only for the missing or altered components. However, this modularity also obscures the algorithmic details of the underlying constructions, making it difficult to exploit their specific structural properties within the reduction. Such information loss often leads to unnecessary tightness degradation. As illustrated in the preceding discussion on eliminating the γ-spread assumption, a γ-spread-free security proof hinges on the fine-grained characteristics of the underlying PKE, which is precisely the kind of detail that modular reasoning tends to discard. Thus, in this work we adopt a *holistic* perspective: we treat FOAC_0' and FOAC' as fully integrated constructions and analyze their IND-CCA security directly, in both the ROM and the QROM.

ROM proofs for FOAC_0' and FOAC'. At a high level, our ROM security proofs for FOAC_0' and FOAC' consist of three main stages.

Stage 1: Simulating decapsulation without the secret key. This stage is essential for reducing

IND-CCA security to the CPA security of the underlying PKE. We first argue that, whenever a decapsulation query returns a non-$\perp$ key k, all random oracle calls involved in its computation (e.g., $(m', \rho, k) \coloneqq H(r)$) must have already been made. The reduction can then retrieve the corresponding entry from the random oracle's query history and compute the response *without* sk.
Stage 2: Removing challenger's footprint from the random oracle. In the IND-CCA game, the challenger itself queries the random oracle when generating the challenge ciphertext and the challenge key. These queries leave entries in the oracle history that are not attributable to the adversary. Since later reduction steps rely on the fact that all relevant oracle entries originate from the adversary, we must *erase* or *puncture* these challenger-induced records.
Stage 3: Reprogramming random oracle to decouple the challenge. In this stage, we reprogram the random oracle at a carefully selected point so that the generation of the challenge ciphertext and the challenge key becomes *independent* of the oracle's response. We show that, unless the adversary queries the reprogrammed point, this modification is undetectable. By linking the reprogramming point to the solution of the CPA game of the underlying PKE, we can leverage the adversary's ability to hit that point to construct a successful adversary against the CPA security of the PKE.

QROM proofs for FOAC_0' and FOAC'. In the ROM, the reduction can freely read and exploit the adversary's classical query history. Lifting such an argument to the QROM is non-trivial: quantum adversaries can query the random oracle in superposition, and the no-cloning theorem prevents us from simply recording the queries without disturbing the adversary's state. Fortunately, Zhandry's *compressed oracle* technique [43] allows us to *implicitly* and *imperfectly* record the adversary's queries in the database register. To turn this record-keeping into a usable reduction, we develop three new technical lemmas, each tailored to one of the three stages of our ROM proof.
Switching Lemma (Lemma 5.1). This lemma allows us to *switch* each call to the quantum-accessible random oracle (QRO) in the decapsulation procedure into a check against the compressed oracle's database register. It corresponds to the first part of Stage 1, ensuring that a non-$\perp$ decapsulation response implies the relevant oracle query is already recorded in the database.
Lookup Lemma (Lemma 5.2). Once all QRO calls have been switched to database checks, this lemma bounds the adversarial advantage when we replace the decapsulation logic with a direct *lookup* of a matching entry in the database, without using sk. This captures the second part of Stage 1 in the QROM setting.
Puncturing Lemma (Lemma 5.3). This lemma implements Stage 2 in the QROM. It enables us to *remove* from the compressed oracle's database those entries inserted by the challenger during the generation of the challenge ciphertext and challenge key, thereby ensuring that the database reflects only the adversary's queries.

For the final stage (oracle reprogramming) and any other situation requiring reprogramming of the random oracle, we directly employ existing One-Way to Hiding (O2H) lemmas [1, 15] adapted to the QROM. With these three new tools in hand, we are able to lift the entire ROM proof strategy to the quantum setting, yielding tighter QROM security proofs for both FOAC_0' and FOAC'.

3 Preliminaries

This section reviews the basic notation, the (quantum-accessible) random oracle model, relevant cryptographic primitives, the compressed oracle technique, and some key lemmas.

3.1 Notation

We write a function H with domain $\mathcal{X}$ and codomain $\mathcal{Y}$ as $H : \mathcal{X} \to \mathcal{Y}$, and denote the set of all such functions by Ω_H. The cardinality of a set $\mathcal{S}$ is denoted by $|\mathcal{S}|$. The notation $s \leftarrow\!\!\$\ \mathcal{S}$ means that s is sampled uniformly at random from $\mathcal{S}$, while $z \leftarrow \mathfrak{D}_z$ indicates that z is sampled according to the distribution $\mathfrak{D}_z$. For an n-tuple $T := (t_0, \ldots, t_{n-1})$, we denote by $T[i]$ the $(i+1)$-th element, i.e., t_i, where $0 \leq i \leq n-1$. For a probabilistic (or deterministic) algorithm A on input x, we write $y \leftarrow A(x)$ (or $y := A(x)$) to denote that $A(x)$ outputs y. An oracle algorithm with classical (or quantum) access to an oracle H is denoted by A^H (or $A^{|H\rangle}$). The notation $[a = b]$ denotes the indicator function that outputs 1 if $a = b$ and 0 otherwise. The security parameter is denoted by λ, and PPT stands for *probabilistic polynomial time*. The base of logarithm log is 2, unless stated otherwise.

3.2 The (Quantum-Accessible) Random Oracle Model

We refer the reader to [36] for a standard introduction to quantum computation. In brief, the *state space* of a quantum system is a complex Hilbert space equipped with an inner product. Dirac notation is used to denote state vectors: $|\psi\rangle$ represents a unit vector in the state space, and its dual is denoted $\langle\psi|$. The space admits an orthonormal basis, called the *computational basis*. The joint state of two systems in states $|\psi\rangle$ and $|\phi\rangle$ is given by the tensor product $|\psi\rangle \otimes |\phi\rangle$, often abbreviated as $|\psi\rangle\,|\phi\rangle$ or $|\psi, \phi\rangle$. The norm of $|\psi\rangle$ is defined as $\||\psi\rangle\| = \sqrt{\langle\psi|\psi\rangle}$, where $\langle\psi|\phi\rangle$ denotes the inner product of $|\psi\rangle$ and $|\phi\rangle$.

The random oracle model (ROM), introduced by Bellare and Rogaway [2], idealizes a hash function as a publicly accessible random oracle. In this model, an adversary can only obtain the value of a hash function $\mathsf{H}(x)$ by querying the oracle at x; as if H were a truly random function.

The quantum-accessible ROM (QROM) extends the ROM by allowing adversaries to query the random oracle in superposition. A quantum-accessible random oracle (QRO) H is implemented as a unitary operator U_{H} acting as $|x, y\rangle \mapsto |x, y \oplus \mathsf{H}(x)\rangle$. Classical queries remain admissible; we write $y := \mathsf{H}(x)$ to denote a classical query to H at x that returns a classical output y. This can be viewed as querying the QRO with the state $|x, 0\rangle$ and then measuring the second register in the computational basis to obtain the classical outcome [10].

3.3 Cryptographic Primitives

Definition 3.1 (Public-Key Encryption, PKE)**.** The public-key encryption (PKE) scheme consists of three PPT algorithms with a security parameter λ, a message space $\mathcal{M}$, and a ciphertext space $\mathcal{C}$: (1) Key generation algorithm $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$ is a probabilistic algorithm that takes as input 1^λ and outputs a public/private key pair $(\mathsf{pk}, \mathsf{sk})$, where pk implicitly defines a randomness space $\mathcal{R} := \mathcal{R}(\mathsf{pk})$. (2) Encryption algorithm $c \leftarrow \mathsf{Enc}(\mathsf{pk}, m)$ is a probabilistic algorithm that takes as input pk and a message $m \in \mathcal{M}$ and outputs a ciphertext $c \in \mathcal{C}$. When necessary, we explicitly denote the randomness used for encryption as $c := \mathsf{Enc}(\mathsf{pk}, m; \rho)$, where $\rho \leftarrow\!\!\$\ \mathcal{R}$ and $\mathcal{R}$ is the randomness space. (3) Decryption algorithm $m := \mathsf{Dec}(\mathsf{sk}, c)$ is a deterministic algorithm that takes as input sk and a ciphertext $c \in \mathcal{C}$ and outputs either a message $m \in \mathcal{M}$ or a special symbol $\perp \notin \mathcal{M}$. We say that $\mathsf{PKE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ is defined over $(\mathcal{M}, \mathcal{C})$.

The correctness requirement of a PKE scheme is that for all $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$ and for all $c \leftarrow \mathsf{Enc}(\mathsf{pk}, m)$, we have $\mathsf{Dec}(\mathsf{sk}, c) = m$. A PKE scheme is said to be *deterministic* if Enc is a deterministic algorithm.

Definition 3.2 (Worst/Average-Case Correctness of PKE [11])**.** For a PKE scheme $\mathsf{PKE} = (\mathsf{Gen}, \mathsf{Enc}, \mathsf{Dec})$ defined over $(\mathcal{M}, \mathcal{C})$, we say it is δ_{wc}-worst-case-correct or has worst-case decryption error rate δ_{wc} if

$$\mathbb{E}_{(\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{Gen}(1^\lambda)}\left[\max_{m\in\mathcal{M}}\Pr_{\rho\leftarrow\$\mathcal{R}}[\mathsf{Dec}(\mathsf{sk},\mathsf{Enc}(\mathsf{pk},m;\rho))\neq m]\right]\leq\delta_{wc}\ . \tag{1}$$

We say it is δ_{ac}-average-case-correct or has average-case decryption error rate δ_{ac} if

$$\mathbb{E}_{(\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{Gen}(1^\lambda)}\left[\Pr_{m\leftarrow\$\mathcal{M},\rho\leftarrow\$\mathcal{R}}[\mathsf{Dec}(\mathsf{sk},\mathsf{Enc}(\mathsf{pk},m;\rho))\neq m]\right]\leq\delta_{ac}\ . \tag{2}$$

Definition 3.3 ((Weak) γ-spreadness of PKE [10])**.** For the PKE scheme $\mathsf{PKE}=(\mathsf{Gen},\mathsf{Enc},\mathsf{Dec})$ defined over $(\mathcal{M},\mathcal{C})$, we say it is weakly γ-spread if

$$\mathbb{E}_{(\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{Gen}(1^\lambda)}\left[\max_{m\in\mathcal{M},\,c\in\mathcal{C}}\Pr_{\rho\leftarrow\$\mathcal{R}}[\mathsf{Enc}(\mathsf{pk},m;\rho)=c]\right]\leq 2^{-\gamma}\ .$$

When the context is clear, we omit the qualifier "weak" or "weakly".

Definition 3.4 (Security of PKE)**.** A PKE scheme $\mathsf{PKE}=(\mathsf{Gen},\mathsf{Enc},\mathsf{Dec})$ defined over $(\mathcal{M},\mathcal{C})$ is said to be OW-CPA (or IND-CPA) secure if for any PPT adversary $\mathcal{A}$, the advantage $\mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{PKE}}(\mathcal{A})$ (or $\mathsf{Adv}^{\text{IND-CPA}}_{\mathsf{PKE}}(\mathcal{A})$) is negligible, where the security notions are defined via a game played between a challenger and the adversary $\mathcal{A}$ as shown in Fig. 6. The adversary's ad-

OW-CPA Game:	IND-CPA Game:
$1:\ (\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{Gen}(1^\lambda)$	$1:\ (\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{Gen}(1^\lambda)$
$2:\ m^*\leftarrow\$\ \mathcal{M}$	$2:\ (m_0,m_1,\mathsf{st})\leftarrow\mathcal{A}(\mathsf{pk})$
$3:\ c^*\leftarrow\mathsf{Enc}(\mathsf{pk},m^*)$	$3:\ b\leftarrow\$\ \{0,1\},\ c^*\leftarrow\mathsf{Enc}(\mathsf{pk},m_b)$
$4:\ \hat{m}\leftarrow\mathcal{A}(\mathsf{pk},c^*)$	$4:\ \hat{b}\leftarrow\mathcal{A}(\mathsf{pk},c^*,\mathsf{st})$
$5:\ \textbf{return}\ [m^*=\hat{m}]$	$5:\ \textbf{return}\ [b=\hat{b}]$

Figure 6: Security games for OW-CPA and IND-CPA security of $\mathsf{PKE}=(\mathsf{Gen},\mathsf{Enc},\mathsf{Dec})$.

vantages are defined as $\mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{PKE}}(\mathcal{A}):=\Pr[m^*=\hat{m}]$ (or $\mathsf{Adv}^{\text{IND-CPA}}_{\mathsf{PKE}}(\mathcal{A}):=|\Pr[b=\hat{b}]-1/2|$). The message m^* and ciphertext c^* generated by the challenger are referred to as the challenge message and challenge ciphertext, respectively.

Definition 3.5 (Key Encapsulation Mechanism, KEM)**.** The key encapsulation mechanism (KEM) consists of three PPT algorithms with a security parameter λ, a key space $\mathcal{K}$, and a ciphertext space $\mathcal{C}$: (1) Key generation algorithm $(\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{Gen}(1^\lambda)$ is a probabilistic algorithm that takes as input 1^λ and outputs a public/private key pair $(\mathsf{pk},\mathsf{sk})$. (2) Encapsulation algorithm $(k,c)\leftarrow\mathsf{Encaps}(\mathsf{pk})$ is a probabilistic algorithm that takes as input pk and outputs a key-ciphertext pair $(k,c)\in\mathcal{K}\times\mathcal{C}$. (3) Decapsulation algorithm $k:=\mathsf{Decaps}(\mathsf{sk},c)$ is a deterministic algorithm that takes as input sk and $c\in\mathcal{C}$ and outputs $k\in\mathcal{K}\cup\{\perp\}$. We say that $\mathsf{KEM}=(\mathsf{Gen},\mathsf{Encaps},\mathsf{Decaps})$ is defined over $(\mathcal{K},\mathcal{C})$.

The correctness requirement of a KEM is that for all $(\mathsf{pk},\mathsf{sk})\leftarrow\mathsf{Gen}(1^\lambda)$ and for all $(k,c)\leftarrow\mathsf{Encaps}(\mathsf{pk})$, we have $\mathsf{Decaps}(\mathsf{sk},c)=k$. A KEM is said to be with explicit rejection if $\perp\notin\mathcal{K}$, or with implicit rejection if $\perp\in\mathcal{K}$ denotes a random value.

Definition 3.6 (IND-CCA Security of KEM)**.** The IND-CCA security of $\mathsf{KEM}=(\mathsf{Gen},\mathsf{Encaps},\mathsf{Decaps})$ defined over $(\mathcal{K},\mathcal{C})$ is defined via a game played between a challenger and an adversary $\mathcal{A}$ as shown in Fig. 7. The adversary's advantage is defined as $\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}):=|\Pr[b=\hat{b}]-1/2|$. If this advantage is negligible for any PPT adversary, then KEM is said to be IND-CCA secure. The keys k_0,k_1 and ciphertext c generated by the challenger are referred to as the challenge keys and challenge ciphertext, respectively.

IND-CCA Game:	Decapsulation oracle $O_{\mathsf{Dec}}(c \in \mathcal{C})$:
$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$	**if** $c = c^*$ **then**
$(k_0, c^*) \leftarrow \mathsf{Encaps}(\mathsf{pk})$	**return** $\perp$
$k_1 \leftarrow\!\!\$\ \mathcal{K},\ b \leftarrow\!\!\$\ \{0,1\}$	**else**
$\hat{b} \leftarrow \mathcal{A}^{O_{\mathsf{Dec}}}(\mathsf{pk}, c^*, k_b)$	$k := \mathsf{Decaps}(\mathsf{sk}, c)$
return $[b = \hat{b}]$	**return** k

Figure 7: Security games for IND-CCA security of $\mathsf{KEM} = (\mathsf{Gen}, \mathsf{Encaps}, \mathsf{Decaps})$.

3.4 The Compressed Oracle Technique

In the classical ROM, reductions can record adversaries' queries. In the QROM, this ability was long considered unattainable due to the quantum no-cloning principle: any direct measurement of a quantum register inevitably disturbs the adversary's state. However, Zhandry [43] circumvented this "recording barrier" by proposing the compressed oracle technique. The core idea is to purify the quantum random oracle and then record query information on the purified quantum random oracle.

Definition 3.7 (Compressed Standard Oracle)**.** Let D be a database comprising q entries of the form $(x, y) \in (\mathcal{X} \times \mathcal{Y}) \cup \{(\perp, 0^n)\}$, where $n := \log |\mathcal{Y}|$ and q is an upper bound on the number of quantum random oracle queries. The structure of D is

$$D = ((x_1, y_1), (x_2, y_2), \ldots, (x_l, y_l), (\perp, 0^n), \ldots, (\perp, 0^n)) \ ,$$

with $0 \le l \le q$, $(x_i, y_i) \in \mathcal{X} \times \mathcal{Y}$, $x_1 < \cdots < x_l$, and exactly $q - l$ trailing copies of $(\perp, 0^n)$. We write $\mathcal{D}$ for the set of all such databases. We sometimes explicitly denote a database with q entries by D_q, and let $\mathcal{D}_q$ be the set of such databases. For any $x \in \mathcal{X}$, if there exists a pair (x, y) in D we set $D(x) = y$; otherwise $D(x) = \perp$. By construction, no two distinct entries share the same x. The quantity $|D|$ counts the entries with $x \neq \perp$. When $|D| < q$ and $D(x) = \perp$, the operation $D \cup (x, y)$ removes one $(\perp, 0^n)$ and inserts (x, y) while maintaining the ascending order of the x's.

Let D be a register with state space $\mathcal{H} = \mathbb{C}[\mathcal{D}]$. For a basis state $|D\rangle$ ($D \in \mathcal{D}$), we define a unitary decompression operator F_x as follows:

- If $D(x) = \perp$ and $|D| < q$,

$$F_x |D\rangle = 2^{-n/2} \sum\nolimits_y |D \cup (x, y)\rangle \ , \quad F_x\Big(2^{-n/2} \sum\nolimits_y |D \cup (x, y)\rangle\Big) = |D\rangle \ ,$$
$$F_x\Big(2^{-n/2} \sum\nolimits_y (-1)^{z \cdot y} |D \cup (x, y)\rangle\Big) = 2^{-n/2} \sum\nolimits_y (-1)^{z \cdot y} |D \cup (x, y)\rangle \ \ (z \neq 0).$$

- If $D(x) = \perp$ but $|D| = q$, then $F_x |D\rangle = |D\rangle$.

Let X and Y be the input and output registers of the quantum random oracle. We define a unitary O_x acting on YD by

$$O_x : |y, D\rangle_{\mathsf{YD}} \mapsto |y \oplus D(x), D\rangle_{\mathsf{YD}} \ ,$$

where the XOR operation $\oplus$ is extended to $\perp$ following the convention of [7], i.e., $0^n \oplus \perp = \perp$, $\perp \oplus 0^n = \perp$, $\perp \oplus \perp = 0^n$, and for any $y \neq 0^n$, $y \oplus \perp = y$, $\perp \oplus y = \perp$. Finally, the compressed standard oracle acting on XYD is defined as

$$\mathsf{CStO} := \sum\nolimits_x |x\rangle\langle x| \otimes F_x O_x F_x \ .$$

When the context permits, we refer to the compressed standard oracle simply as the *compressed oracle*. Zhandry [43] proved that the compressed standard oracle is perfectly indistinguishable from the quantum-accessible random oracle.

Lemma 3.1 (Lemma 4 in [43]). *The compressed oracle as defined in Definition 3.7 with* D *set as* $\bigotimes_{i=1}^{q}(\perp, 0^n)$ *initially is perfectly indistinguishable from a quantum random oracle* $H : \mathcal{X} \to \mathcal{Y}$ *for any quantum adversary making at most* q *random oracle queries.*

3.5 The One-Way to Hiding Lemma

Definition 3.8 (Semi-Classical Oracle [1], adapted). For a subset $S \subseteq \mathcal{X}$, define an indicator function $f : \mathcal{X} \to \{0,1\}$ with $f(x) = 1$ iff $x \in S$. The *semi-classical oracle* $\mathcal{O}_f^{SC}$ is a quantum-accessible oracle that, on a query state $\sum_{x\in\mathcal{X}} \alpha_x |x\rangle$, performs the following:

1. Initialize a single-qubit register B in $|0\rangle_{\mathsf{B}}$.
2. Transform $\sum_x \alpha_x |x\rangle |0\rangle_{\mathsf{B}} \mapsto \sum_x \alpha_x |x\rangle |f_S(x)\rangle_{\mathsf{B}}$ and immediately measure B in the computational basis.

The measurement outcome is output by the oracle. Sometimes we also denote this semi-classical oracle by $\mathcal{O}_S^{SC}$. During the execution of a quantum oracle algorithm $A^{|\mathcal{O}_S^{SC}\rangle}$, we denote by Find the classical event that the measurement of $\mathcal{O}_S^{SC}$ ever yields outcome 1. It is not hard to see that once the event Find occurs, immediately measuring the input register yields a value $x \in S$.

Lemma 3.2 (Semi-Classical One-Way to Hiding [1, Theorem 1], adapted). *Let* $S \subseteq \mathcal{X}$ *be a random set and let* $G, H : \mathcal{X} \to \mathcal{Y}$ *be random functions such that* $G(x) = H(x)$ *for all* $x \notin S$. *Let* z *be a random bitstring. The tuple* (G, H, S, z) *may follow an arbitrary joint distribution* $\mathfrak{D}$. *Denote by* $H \backslash S$ *a punctured oracle that first queries* $\mathcal{O}_S^{SC}$ *on the input register of* H *and then queries* H *itself. Let* A *be a quantum oracle algorithm that makes at most* q *queries. Define*

$$
\begin{aligned}
P_{\text{left}} &:= \Pr[1 \leftarrow A^{|H\rangle}(z) : (G, H, S, z) \leftarrow \mathfrak{D}] \ , \\
P_{\text{right}} &:= \Pr[1 \leftarrow A^{|G\rangle} : (G, H, S, z) \leftarrow \mathfrak{D}] \ , \\
P_{\text{find}} &:= \Pr[\mathsf{Find} : A^{|G\backslash S\rangle}(z), (G, H, S, z) \leftarrow \mathfrak{D}] \ .
\end{aligned}
$$

Then we have

$$
|P_{\text{left}} - P_{\text{right}}| \leq 2\sqrt{(q+1)\cdot P_{\text{find}}} \ , \qquad \left|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}\right| \leq 2\sqrt{(q+1)\cdot P_{\text{find}}} \ .
$$

Lemma 3.3 (Search in a Semi-Classical Oracle [1, Theorem 2], adapted). *Let* $S \subseteq \mathcal{X}$ *be a random set and let* z *be a random bitstring. The pair* (S, z) *may follow an arbitrary joint distribution* $\mathfrak{D}$. *Let* A *be any quantum oracle algorithm that makes at most* q *queries to a semi-classical oracle with domain* $\mathcal{X}$. *Define an algorithm* B *which, on input* z, *proceeds as follows: it picks* $i^* \leftarrow\!\$\, \{1, \dots, q\}$ *uniformly at random, runs* $A^{|\mathcal{O}_\emptyset^{SC}\rangle}(z)$ *until (just before) its* i^**-th query, measures the input register of that query in the computational basis, and outputs the measured value* x. *Then*

$$
\Pr[\mathsf{Find} : A^{|\mathcal{O}_S^{SC}\rangle}(z), (S, z) \leftarrow \mathfrak{D}] \leq 4q \cdot \Pr[x \in S : x \leftarrow B(z), (S, z) \leftarrow \mathfrak{D}] \ .
$$

If S *and* z *are independent, define* $P_{\max} := \max_{x\in\mathcal{X}} \Pr[x \in S]$. *Then we have*

$$
\Pr[x \in S : x \leftarrow B(z), (S, z) \leftarrow \mathfrak{D}] \leq P_{\max} \ .
$$

Definition 3.9 (Compressed Semi-Classical Oracle [15], adapted)**.** Let $\mathcal{D}_q$ be a set of databases and $S \subseteq \mathcal{D}_q$. Define a function $f_S : \mathcal{D}_q \to \{0,1\}$ such that $f_S(D) = 1$ if $D \in S$ and $f_S(D) = 0$ otherwise. The compressed semi-classical oracle $\mathcal{O}_S^{CSC}$ operates on an input state $\sum_{z\in\{0,1\}^*,\, D\in\mathcal{D}_q} \alpha_{z,D} |z, D\rangle$ as follows:

1. Initialize a single qubit register L as $|0\rangle_{\mathsf{L}}$.
2. Transform $\sum_{z,D} \alpha_{z,D} |z, D\rangle |0\rangle_{\mathsf{L}} \mapsto \sum_{z,D} \alpha_{z,D} |z, D\rangle |f_S(D)\rangle_{\mathsf{L}}$ and immediately measure L in the computational basis.

The measurement outcome is output by the oracle. During the execution of a quantum oracle algorithm $A^{|\mathcal{O}_S^{CSC}\rangle}$, we denote by Find the classical event that the measurement of $\mathcal{O}_S^{CSC}$ (on L) ever has outcome 1.

Lemma 3.4 (Compressed Semi-Classical One-Way to Hiding [15, Theorem 1])**.** *Let D_q be the database register defined over $\mathcal{D}_q$ and let $\mathsf{H_c} : \{0,1\}^m \to \{0,1\}^n$ be the compressed standard oracle implemented on D_q. Let $S \subseteq \mathcal{D}_q$ be a subset such that $D_\perp \notin S$, where $D_\perp$ denotes the database containing only q pairs $(\perp, 0^n)$, and let z be a random string. The pair (S, z) may follow an arbitrary joint distribution $\mathfrak{D}$. Define $\mathsf{H_c}\backslash S$ as an oracle that first queries $\mathsf{H_c}$ and then queries $\mathcal{O}_S^{CSC}$ on the database register D_q. Let A be a quantum oracle algorithm (not necessarily unitary) that has oracle access to $\mathsf{H_c}$ and oRead_f, and suppose A makes at most $q_H \le q$ queries to $\mathsf{H_c}$ and q_R queries to oRead_f, where oRead_f (with f possibly depending on z) is simulated by the database read operation Read_f defined as $\mathsf{Read}_f : |x, D, y\rangle_{\mathsf{XD}_q\mathsf{Y}} \mapsto |x, D, y \oplus f(x, D)\rangle_{\mathsf{XD}_q\mathsf{Y}}$. Define*

$$P_{\text{left}} := \Pr[1 \leftarrow A^{|\mathsf{H_c}\rangle, |\mathsf{oRead}_f\rangle}(z) : (S, z) \leftarrow \mathfrak{D}] \ ,$$
$$P_{\text{right}} := \Pr[1 \leftarrow A^{|\mathsf{H_c}\backslash S\rangle, |\mathsf{oRead}_f\rangle}(z) : (S, z) \leftarrow \mathfrak{D}] \ ,$$
$$P_{\text{find}} := \Pr[\mathsf{Find} : A^{|\mathsf{H_c}\backslash S\rangle, |\mathsf{oRead}_f\rangle}(z),\ (S, z) \leftarrow \mathfrak{D}] \ .$$

Then we have

$$|P_{\text{left}} - P_{\text{right}}| \le \sqrt{(q_H + 1) \cdot P_{\text{find}}}, \qquad \left|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}\right| \le \sqrt{(q_H + 1) \cdot P_{\text{find}}}.$$

Define a projector on the database register D_q as $J_S := \sum_{D\in S} |D\rangle\langle D|$. Then the probability P_{find} defined above and the compressed standard oracle CStO given in Definition 3.7 satisfy

$$P_{\text{find}} \le q_H \cdot \mathbb{E}_{(S,z)\leftarrow\mathfrak{D}}[\|[J_S, \mathsf{CStO}]\|^2] \ ,$$

where $[A, B] := AB - BA$ denotes the commutator between A and B [36, (2.66)].

3.6 Difference Lemma

Lemma 3.5 (Difference Lemma [6, Theorem 4.7])**.** *Let W_1, W_2, and Z be events defined over some probability space, and let $\overline{Z}$ denote the complement of Z. If $\Pr[W_1 \wedge \overline{Z}] = \Pr[W_2 \wedge \overline{Z}]$, then*

$$\left|\Pr[W_2] - \Pr[W_1]\right| \le \Pr[Z] \ .$$

4 The ROM Proofs

4.1 $\mathsf{KEM} := \mathsf{FOAC}_0'[\mathsf{PKE}', F, H]$

Theorem 4.1 (ROM Security of FOAC_0')**.** *Let $F : \mathcal{M} \to \{0,1\}^n$ and $H : \{0,1\}^n \to \mathcal{M} \times \mathcal{R} \times \mathcal{K}$ be modeled as random oracles. If the underlying PKE PKE' is δ_{ac}-average-case-correct, for any*

Games G_0 to G_8:

$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda),\ r^* \leftarrow\!\!\$\ \{0,1\}^n$

initialize empty associative arrays

$\quad Map_1 : \mathcal{M} \rightarrow \{0,1\}^n$

$\quad Map_2 : \{0,1\}^n \rightarrow \mathcal{M} \times \mathcal{R} \times \mathcal{K}$

$(m^*, \rho^*, k_0) \leftarrow\!\!\$\ \mathcal{M} \times \mathcal{R} \times \mathcal{K}$

$\quad u^* \leftarrow\!\!\$\ \{0,1\}^n,\ \hat{m}^* \leftarrow\!\!\$\ \mathcal{M}$

$Map_2[r^*] := (m^*, \rho^*, k_0)$ // G_0–G_3

$Map_1[m^*] := u^*$ // G_0–G_5

$c_1^* := \mathsf{Enc}'(\mathsf{pk}, m^*; \rho^*)$ // G_0–G_7

$c_1^* := \mathsf{Enc}'(\mathsf{pk}, \hat{m}^*; \rho^*)$ // G_8

$c_2^* := r^* \oplus u^*$

$b \leftarrow\!\!\$\ \{0,1\},\ k_1 \leftarrow\!\!\$\ \mathcal{K}$

$\hat{b} \leftarrow \mathcal{A}^{F,H,O_{\mathsf{Dec}}}(\mathsf{pk}, (c_1^*, c_2^*), k_b)$

return $[b = \hat{b}]$

Random Oracle $F(m)$:

if $m = m^*$ **then**

$\quad$ **return** u^* // G_6

if $m \notin \mathrm{Domain}(Map_1)$ **then**

$\quad Map_1[m] \leftarrow\!\!\$\ \{0,1\}^n$

return $Map_1[m]$

Random Oracle $H(r)$:

if $r = r^*$ **then**

$\quad$ **return** (m^*, ρ^*, k_0) // G_4

if $r \notin \mathrm{Domain}(Map_2)$ **then**

$\quad Map_2[r] \leftarrow\!\!\$\ \mathcal{M} \times \mathcal{R} \times \mathcal{K}$

return $Map_2[r]$

Decapsulation Oracle $O_{\mathsf{Dec}}(c_1, c_2)$:

if $(c_1, c_2) = (c_1^*, c_2^*)$ **then**

$\quad$ **return** $\perp$

$m := \mathsf{Dec}'(\mathsf{sk}, c_1)$ // G_0–G_2

if $m = \perp$ **then**

$\quad$ **return** $\perp$ // G_0–G_2

if $m \notin \mathrm{Domain}(Map_1)$

$\quad$ **return** $\perp$ // G_2

$r := c_2 \oplus F(m)$ // G_0–G_2

if $r \notin \mathrm{Domain}(Map_2)$

$\quad$ **return** $\perp$ // G_1–G_2

$(m', \rho, k) := H(r)$ // G_0–G_2

if $m = m'$ **and**

$\quad \mathsf{Enc}'(\mathsf{pk}, m'; \rho) = c_1$ **then**

$\quad$ **return** k // G_0–G_2

if $\exists (r, (m, \rho, k)) \in Map_2$ **and**

$\quad (m, u) \in Map_1$ s.t. $r \oplus u = c_2$ **and**

$\quad \mathsf{Enc}'(\mathsf{pk}, m; \rho) = c_1$ **then**

$\quad$ **return** k // G_3–G_8

return $\perp$

Figure 8: Games G_0 to G_8 for the proof of Theorem 4.1.

PPT *adversary* $\mathcal{A}$ *against the* IND-CCA *security of* $\mathsf{KEM} := \mathsf{FOAC}_0'[\mathsf{PKE}', F, H]$, *there exists a* PPT OW-CPA *(resp.* IND-CPA*) adversary* $\mathcal{B}$ *(resp.* $\mathcal{B}'$*) against* PKE' *such that*

$$\mathsf{Adv}_{\mathsf{KEM}}^{\text{IND-CCA}}(\mathcal{A}) \leq q_F \mathsf{Adv}_{\mathsf{PKE}'}^{\text{OW-CPA}}(\mathcal{B}) + (q_H + 1)\delta_{ac} + (q_D + q_H)/|\mathcal{M}| + \Delta\ ,$$

$$\mathsf{Adv}_{\mathsf{KEM}}^{\text{IND-CCA}}(\mathcal{A}) \leq 2\mathsf{Adv}_{\mathsf{PKE}'}^{\text{IND-CPA}}(\mathcal{B}') + (q_H + 1)\delta_{ac} + (q_D + q_H + q_F)/|\mathcal{M}| + \Delta\ ,$$

where q_F, q_H, q_D *bound the number of queries made by* $\mathcal{A}$ *to the random oracles* F, H *and the decapsulation oracle, respectively, and* $\Delta \leq (q_D + 1)(q_H + 1)/2^n$.

If PKE' *is deterministic, the bound can be improved to*

$$\mathsf{Adv}_{\mathsf{KEM}}^{\text{IND-CCA}}(\mathcal{A}) \leq \mathsf{Adv}_{\mathsf{dPKE}'}^{\text{OW-CPA}}(\mathcal{B}) + (q_H + 3)\delta_{ac} + (q_D + q_H)/|\mathcal{M}| + \Delta\ .$$

Theorem 4.1. We define a sequence of games G_j (for $j = 0, \ldots, 8$) played between a challenger and an adversary $\mathcal{A}$, as shown in Fig. 8. In each game, b is a random bit sampled by the challenger, and $\hat{b}$ is the bit output by $\mathcal{A}$. Let W_j be the event that $b = \hat{b}$ in game G_j.

Game G_0. In this game, the challenger explicitly initializes two associative arrays, $Map_1 : \mathcal{M} \rightarrow \{0,1\}^n$ and $Map_2 : \{0,1\}^n \rightarrow \mathcal{M} \times \mathcal{R} \times \mathcal{K}$, to simulate the random oracles F and H, respectively. During the initialization phase, the challenger randomly samples $(m^*, \rho^*, k_0) \leftarrow\!\!\$$

$\mathcal{M} \times \mathcal{R} \times \mathcal{K}$ and $u^* \leftarrow\$ \{0,1\}^n$, and stores them in $Map_2[r^*]$ and $Map_1[m^*]$, respectively. This effectively sets the responses of the random oracles H (resp. F) at the points r^* (resp. m^*) to (m^*, ρ^*, k_0) (resp. u^*). Note that although the challenger samples an extra $\hat{m}^* \leftarrow\$ \mathcal{M}$, it is never used. We can see that, apart from the additional bookkeeping for the random oracle responses, the challenger's behavior is identical to that in the IND-CCA game of the KEM $\mathsf{KEM} := \mathsf{FOAC}_0'[\mathsf{PKE}', F, H]$. Therefore,

$$|\Pr[W_0] - 1/2| = \mathsf{Adv}_{\mathsf{KEM}}^{\text{IND-CCA}}(\mathcal{A}) \ .$$

Game G_1. In this game, we modify the decapsulation oracle as follows. For a query $(c_1, c_2) \neq (c_1^*, c_2^*)$, after computing $m := \mathsf{Dec}'(\mathsf{sk}, c_1)$ $(\neq \perp)$ and $r := c_2 \oplus F(m)$, if $r \notin \text{Domain}(Map_2)$, the oracle directly returns $\perp$ (as shown in line 23 of Fig. 8). Let Z_1 be the event that $m = m'$ in this case, where $(m', \rho, k) := H(r)$. Clearly, the behavior of the decapsulation oracle in Game G_1 is identical to that in Game G_0, if the event Z_1 does not occur. However, by the simulation of the random oracle H, when $r \notin \text{Domain}(Map_2)$, the value m' is sampled uniformly at random from $\mathcal{M}$. Hence, for each decapsulation oracle query, $\Pr[Z_1]$ is at most $1/|\mathcal{M}|$. Let q_D be the bounded number of decapsulation oracle queries made by $\mathcal{A}$. By the Difference Lemma and a union bound, we have

$$|\Pr[W_1] - \Pr[W_0]| \leq q_D \cdot \Pr[Z_1] \leq q_D/|\mathcal{M}| \ .$$

Game G_2. This game further modifies the decapsulation oracle. For a query $(c_1, c_2) \neq (c_1^*, c_2^*)$, after computing $m := \mathsf{Dec}'(\mathsf{sk}, c_1)$, if $m \notin \text{Domain}(Map_1)$, the oracle returns $\perp$ directly (as shown in line 21 of Fig. 8). Let Z_2 be the event that, in this case, it holds that $r := c_2 \oplus F(m) \in \text{Domain}(Map_2)$. It is clear to see that if the event Z_2 does not occur, the behavior of the decapsulation oracle in Game G_2 is identical to that in Game G_1. According to the simulation of the random oracle F, when $m \notin \text{Domain}(Map_1)$, the value $F(m)$ is sampled uniformly at random from $\{0,1\}^n$. Let q_H be the bounded number of queries $\mathcal{A}$ makes to the random oracle H. Then we have $|\text{Domain}(Map_2)| \leq q_H + 1$ (where the extra 1 comes from the r^* stored during the challenger's initialization). Therefore, for each decapsulation oracle query, $\Pr[Z_2] \leq (q_H + 1)/2^n$. By the Difference Lemma and a union bound, we obtain

$$|\Pr[W_2] - \Pr[W_1]| \leq q_D \cdot \Pr[Z_2] \leq q_D(q_H + 1)/2^n \ .$$

Game G_3. In this game, for a decapsulation oracle query $(c_1, c_2) \neq (c_1^*, c_2^*)$, the challenger no longer computes the return value via $m := \mathsf{Dec}'(\mathsf{sk}, c_1)$ (i.e., lines 19–25 in Fig. 8 are removed). Instead, it directly searches whether there exist entries $(r, (m, \rho, k)) \in Map_2$ and $(m, u) \in Map_1$ such that $r \oplus u = c_2$ and $\mathsf{Enc}'(\mathsf{pk}, m; \rho) = c_1$: if such entries exist, it returns k; otherwise, it returns $\perp$.

First, if the decapsulation oracle in Game G_3 returns $\perp$, then the oracle in Game G_2 must also return $\perp$. This holds because, if the oracle in Game G_2 did not return $\perp$, then Map_1 and Map_2 would necessarily contain entries satisfying the above conditions, which would force the oracle in Game G_3 not to return $\perp$.

Now the difference between Game G_3 and Game G_2 lies in the situation where the oracle in Game G_3 returns some $k \neq \perp$, but the oracle in Game G_2 does not return that k. This means that the m found in Game G_3 is not equal to $\mathsf{Dec}'(\mathsf{sk}, c_1)$, i.e., there exists an entry $(r, (m, \rho, k)) \in Map_2$ such that $\mathsf{Dec}'(\mathsf{sk}, \mathsf{Enc}'(\mathsf{pk}, m; \rho)) \neq m$. Denote this event by Z_3.

Given fixed $(\mathsf{pk}, \mathsf{sk})$, let $\delta(\mathsf{pk}, \mathsf{sk}) = \Pr[\mathsf{Dec}'(\mathsf{sk}, \mathsf{Enc}'(\mathsf{pk}, m; \rho)) \neq m]$, where the probability is taken over $(m, \rho) \leftarrow\$ \mathcal{M} \times \mathcal{R}$. Thus, $\Pr[Z_3 : (\mathsf{pk}, \mathsf{sk})] \leq |\text{Domain}(Map_2)| \cdot \delta(\mathsf{pk}, \mathsf{sk}) = (q_H + 1)\delta(\mathsf{pk}, \mathsf{sk})$. Then, by the Difference Lemma, we have $|\Pr[W_3 : (\mathsf{pk}, \mathsf{sk})] - \Pr[W_2 : (\mathsf{pk}, \mathsf{sk})]| \leq \Pr[Z_3 : (\mathsf{pk}, \mathsf{sk})]$. By averaging over $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$, we can finally obtain

$$|\Pr[W_3] - \Pr[W_2]| \leq \mathbb{E}[\Pr[Z_3 : (\mathsf{pk}, \mathsf{sk})]] \leq \mathbb{E}[(q_H + 1)\delta(\mathsf{pk}, \mathsf{sk})] \leq (q_H + 1)\delta_{ac} \ ,$$

where the last inequality follows from the definition of the δ_{ac}-average-case correctness of PKE' (see (2)), i.e., $\mathbb{E}[\delta(\mathsf{pk}, \mathsf{sk})] \leq \delta_{ac}$, with the expectation taken over $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$.

Game G_4. Starting from Game G_3, the simulation of the decapsulation oracle no longer uses the secret key sk. We now focus on reducing $\mathcal{A}$'s advantage in distinguishing k_0 and k_1 to breaking the CPA security of the underlying PKE'. In this game, the challenger no longer stores the tuple $(r^*, (m^*, \rho^*, k_0^*))$ in Map_2 (i.e., line 4 in Fig. 8 is removed), but it still uses (m^*, ρ^*, k_0^*) as the value of $H(r^*)$ (i.e., line 15 in Fig. 8). This means that in Game G_4, line 26 will not use $(r^*, (m^*, \rho^*, k))$ to compute responses to decapsulation oracle queries. Note that in Game G_3, line 26 also would not use $(r^*, (m^*, \rho^*, k))$ to compute responses to decapsulation oracle queries; otherwise, it would imply that for some decapsulation query $(c_1, c_2) \neq (c_1^*, c_2^*)$, there exist $(r^*, (m^*, \rho^*, k)) \in Map_2$ and $(m^*, u^*) \in Map_1$ such that $r^* \oplus u^* = c_2$ and $\mathsf{Enc}'(\mathsf{pk}, m^*; \rho^*) = c_1$, which would contradict $(c_1, c_2) \neq (c_1^*, c_2^*)$. Therefore, from $\mathcal{A}$'s view, this modification does not actually change the behavior of the challenger in Game G_3. Hence,

$$\Pr[W_4] = \Pr[W_3] \ .$$

Game G_5. In this game, the challenger no longer uses (m^*, ρ^*, k_0) as the response to the random oracle query $H(r^*)$. Let X_j be the event that $\mathcal{A}$ makes a random oracle H query on r^* in Game G_j. We can see that this game and Game G_4 proceed identically until X_5 occurs. By the Difference Lemma, we have

$$|\Pr[W_5] - \Pr[W_4]| \leq \Pr[X_5] \ .$$

Note that now both k_0 and k_1 are sampled independently and uniformly from $\mathcal{K}$, and are not used elsewhere in the game. Consequently, the bit b is independent and uniformly random from $\mathcal{A}$'s view. Therefore,

$$\Pr[W_5] = 1/2 \ .$$

Game G_6. Next, we perform a similar treatment for $Map_1[m^*] \coloneqq u^*$. We delete line 5 in Fig. 8, i.e., we no longer store (m^*, u^*) in Map_1, while still setting u^* as the value of $F(m^*)$ as indicated in line 12. Similarly, this means that in Game G_6, line 26 will no longer use (m^*, u^*) to compute responses for the decapsulation oracle queries. It is worth noting that Game G_5 might still use (m^*, u^*) for computing the decapsulation oracle's response when there exists an entry $(r, (m = m^*, \rho, k)) \in Map_2$. However, since for every $(r, (m, \rho, k))$ in Map_2, the value m is sampled uniformly at random from $\mathcal{M}$, and the number of queries $\mathcal{A}$ makes to the random oracle H is at most q_H, the probability that there exists an entry $(r, (m = m^*, \rho, k)) \in Map_2$ is at most $q_H/|\mathcal{M}|$. Therefore,

$$|\Pr[X_6] - \Pr[X_5]| \leq q_H/|\mathcal{M}| \ .$$

Game G_7. In this game, the challenger no longer uses u^* to answer the random oracle query $F(m^*)$, i.e., line 12 in Fig. 8 is removed. Let Y_j denote the event that the adversary $\mathcal{A}$ makes a query to the random oracle F on m^* in Game G_j. Clearly, Game G_7 and Game G_6 proceed identically until the event Y_7 occurs. By the Difference Lemma, we obtain

$$|\Pr[X_7] - \Pr[X_6]| \leq \Pr[Y_7] \ .$$

Note that in this game, both r^* and u^* are uniformly random values sampled from their respective spaces. They are used only to compute $c_2^* \coloneqq r^* \oplus u^*$ throughout the entire game, which can be viewed as a one-time pad encryption of r^* using u^*. Hence, the value of r^* is perfectly hidden from $\mathcal{A}$, so the probability that $\mathcal{A}$ queries $H(r^*)$ in Game G_7 is

$$\Pr[X_7] \leq q_H/2^n \ .$$

Since the challenger in Game G_7 no longer requires the specific values of m^*, ρ^*, and sk to interact with $\mathcal{A}$, if Y_7 occurs, we can construct an adversary $\mathcal{B}$ to attack the OW-CPA security of the underlying PKE'. Specifically, given (pk, c^*) from the OW-CPA challenger, $\mathcal{B}$ sets $c_1^* := c^*$, randomly chooses $k \leftarrow\$ \mathcal{K}$ and $c_2^* \leftarrow\$ \{0,1\}^n$, sends $(\mathsf{pk}, (c_1^*, c_2^*), k)$ to $\mathcal{A}$, and then simulates the random oracles F, H and the decapsulation oracle to interact with $\mathcal{A}$ according to the strategy of Game G_7. At the end of the game, if Y_7 occurs, $\mathcal{B}$ randomly selects an m from Domain(Map_1) and outputs it. Since $\mathcal{A}$ makes at most q_F queries to the random oracle F, we have

$$\Pr[Y_7]/q_F \leq \mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{PKE}'}(\mathcal{B}) \ .$$

When PKE' is deterministic, let $\mathcal{R} = \emptyset$. The reduction above still holds. The only difference is that when event Y_7 occurs, $\mathcal{B}$ no longer randomly selects an m from Domain(Map_1) to output, but instead selects an m from it that satisfies $\mathsf{Enc}'(\mathsf{pk}, m) = c^*$ and outputs that m. Clearly, such an m equals m^*, unless for $m^* \leftarrow\$ \mathcal{M}$, there exists an $m' \neq m^*$ such that $\mathsf{Enc}'(\mathsf{pk}, m') = \mathsf{Enc}'(\mathsf{pk}, m^*)$. According to [16, Lemma G1], for a deterministic PKE', the probability of this event is at most $2\delta_{ac}$. Therefore, we have

$$\Pr[Y_7] \leq \mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{dPKE}'}(\mathcal{B}) + 2\delta_{ac} \ .$$

Game G_8. We now further reduce the advantage to the IND-CPA security of the underlying PKE'. In this game, the challenger uses a fresh random message $\hat{m}^*$ to compute $c_1^* := \mathsf{Enc}'(\mathsf{pk}, \hat{m}^*; \rho^*)$. Consequently, m^* is never used throughout the game, which implies that the probability that $\mathcal{A}$ queries $F(m^*)$ satisfies

$$\Pr[Y_8] \leq q_F/|\mathcal{M}| \ .$$

Now, we proceed to construct an adversary $\mathcal{B}'$ against the IND-CPA security of PKE'. Specifically, upon receiving pk from the IND-CPA challenger, $\mathcal{B}'$ randomly samples $m^*, \hat{m}^* \leftarrow\$ \mathcal{M}$, sets $m_0 := m^*$ and $m_1 := \hat{m}^*$, and sends (m_0, m_1) to the challenger. When receiving the challenge ciphertext $c^* = \mathsf{Enc}'(\mathsf{pk}, m_{b'})$ (where b' is the challenger's random bit), $\mathcal{B}'$ sets $c_1^* := c^*$, samples $k \leftarrow\$ \mathcal{K}$ and $c_2^* \leftarrow\$ \{0,1\}^n$, and sends $(\mathsf{pk}, (c_1^*, c_2^*), k)$ to $\mathcal{A}$. It then simulates the random oracles F, H, and the decapsulation oracle for $\mathcal{A}$ exactly as in Game G_7. At the end of the game, $\mathcal{B}'$ checks whether $m^* \in$ Domain(Map_1): if yes, it outputs $\hat{b}' := 1$; otherwise, it outputs $\hat{b}' := 0$.

Observe that when $b' = 0$, we have $c_1^* = \mathsf{Enc}'(\mathsf{pk}, m^*)$; hence, $\mathcal{B}'$ simulates the challenger of Game G_7. When $b' = 1$, we have $c_1^* = \mathsf{Enc}'(\mathsf{pk}, \hat{m}^*)$; thus, $\mathcal{B}'$ simulates the challenger of Game G_8. Therefore, $\Pr[\hat{b}' = 1 | b' = 0] = \Pr[Y_7]$ and $\Pr[\hat{b}' = 1 | b' = 1] = \Pr[Y_8]$. Consequently,

$$\begin{aligned}
\mathsf{Adv}^{\text{IND-CPA}}_{\mathsf{PKE}'}(\mathcal{B}') &\geq \left|\Pr[b' = \hat{b}'] - 1/2\right| \\
&= \left|\Pr[b' = \hat{b}' \wedge b' = 0] + \Pr[b' = \hat{b}' \wedge b' = 1] - 1/2\right| \\
&= \frac{1}{2}\left|\Pr[\hat{b}' = 0 | b' = 0] + \Pr[\hat{b}' = 1 | b' = 1] - 1\right| \\
&= \frac{1}{2}\left|\Pr[\hat{b}' = 1 | b' = 1] - \Pr[\hat{b}' = 1 | b' = 0]\right| \\
&= \frac{1}{2}|\Pr[Y_8] - \Pr[Y_7]| \ .
\end{aligned}$$

Putting the bounds together, we have

$$\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) \leq q_F \mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{PKE}'}(\mathcal{B}) + (q_H + 1)\delta_{ac} + (q_D + q_H)/|\mathcal{M}| + \Delta \ ,$$

$$\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) \leq \mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{dPKE}'}(\mathcal{B}) + (q_H + 3)\delta_{ac} + (q_D + q_H)/|\mathcal{M}| + \Delta \ ,$$

and

$$\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) \leq 2\mathsf{Adv}^{\text{IND-CPA}}_{\mathsf{PKE}'}(\mathcal{B}') + (q_H + 1)\delta_{ac} + (q_D + q_H + q_F)/|\mathcal{M}| + \Delta \ ,$$

where $\Delta \leq (q_D + 1)(q_H + 1)/2^n$. □

4.2 $\mathsf{KEM} := \mathsf{FOAC}'[\mathsf{PKE}', F, H]$

Theorem 4.2 (ROM Security of FOAC')**.** *Let $F : \mathcal{M}_1 \to \mathcal{M}_2$ and $H : \mathcal{M}_2 \to \mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}$ be modeled as random oracles. If the underlying PKE scheme PKE' is δ_{ac}-average-case correct, then for any* PPT *adversary $\mathcal{A}$ against the* IND-CCA *security of the KEM $\mathsf{KEM} := \mathsf{FOAC}'[\mathsf{PKE}', F, H]$, there exists a* PPT *adversary $\mathcal{B}$ (resp. $\mathcal{B}'$) against the* OW-CPA *(resp.* IND-CPA*) security of PKE' such that*

$$\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) \leq q_F q_H \mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{PKE}'}(\mathcal{B}) + (q_H + 1)\delta_{ac} + (q_D + q_H)/|\mathcal{M}_1| + \Delta \ ,$$

$$\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) \leq 2\mathsf{Adv}^{\text{IND-CPA}}_{\mathsf{PKE}'}(\mathcal{B}') + (q_H + 1)\delta_{ac} + (q_D + q_H + q_F)/|\mathcal{M}_1| + \Delta \ ,$$

where q_F, q_H, q_D bound the number of queries made by $\mathcal{A}$ to the random oracles F, H and the decapsulation oracle, respectively, and $\Delta \leq (q_D + 1)(q_H + 1)/|\mathcal{M}_2|$.

If PKE' is deterministic, the bound can be improved to

$$\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) \leq \mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{dPKE}'}(\mathcal{B}) + (q_H + 3)\delta_{ac} + (q_D + q_H)/|\mathcal{M}_1| + \Delta \ .$$

Proof Sketch. The proof of Theorem 4.2 follows a similar structure to that of Theorem 4.1. We first simulate the decapsulation oracle without sk by retrieving responses from the random oracle query history. Next, we remove from the oracle history those entries that were inserted by the challenger during the generation of the challenge ciphertext and the challenge key. Then, we reprogram the oracle to decouple the challenge, and finally reduce to the CPA security of the underlying PKE via the adversary's ability to detect the reprogrammed point.

The main difference lies in the OW-CPA reduction. In FOAC', both r^* and m_1^* are tied to the challenge ciphertext c^* produced by the underlying PKE, whereas in FOAC_0', only m^* is linked to the PKE ciphertext c_1^*. Hence, the reduction must extract both r^* and m_1^* from the query history, incurring an $\mathcal{O}(q^2)$ loss. For deterministic or IND-CPA secure PKE, the reduction can retrieve the exact values as in Theorem 4.1, preserving an $\mathcal{O}(1)$ loss.

Theorem 4.2. We define a sequence of games G_j (for $j = 0, \ldots, 7$) played between a challenger and an adversary $\mathcal{A}$, as shown in Fig. 9. In each game, b denotes the random bit sampled by the challenger, and $\hat{b}$ is the bit output by $\mathcal{A}$. We use W_j to denote the event that $b = \hat{b}$ in game G_j.

Game G_0. The challenger's behavior in this game is depicted in Fig. 9. It initializes two associative arrays, $Map_1 : \mathcal{M}_1 \to \mathcal{M}_2$ and $Map_2 : \mathcal{M}_2 \to \mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}$, to simulate the random oracles F and H, respectively. The challenger first randomly samples $(m_1^*, \rho^*, k_0) \leftarrow\!\$\ \mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}$ and $u^* \leftarrow\!\$\ \mathcal{M}_2$, and immediately stores them in $Map_2[r^*]$ and $Map_1[m_1^*]$, respectively. This is equivalent to setting $H(r^*) := (m_1^*, \rho^*, k_0)$ and $F(m_1^*) := u^*$. It also samples $\hat{m}^* \leftarrow\!\$\ \mathcal{M}_1$, but this value is never used. From $\mathcal{A}$'s view, the challenger's behavior is identical to that in the IND-CCA game for $\mathsf{KEM} := \mathsf{FOAC}'[\mathsf{PKE}', F, H]$. Therefore, we have

$$|\Pr[W_0] - 1/2| = \mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) \ .$$

Game G_1. This game modifies the decapsulation oracle as follows. For a decapsulation oracle query $c \neq c^*$, after computing $m_1 \| m_2 := \mathsf{Dec}'(\mathsf{sk}, c)$ $(\neq \perp)$ and $r := m_2 \oplus F(m_1)$, if

Games G_0 to G_7:

1 : $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$, $r^* \leftarrow\!\!\$\ \mathcal{M}_2$

2 : initialize empty associative arrays

$\quad Map_1 : \mathcal{M}_1 \rightarrow \mathcal{M}_2$

$\quad Map_2 : \mathcal{M}_2 \rightarrow \mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}$

3 : $(m_1^*, \rho^*, k_0) \leftarrow\!\!\$\ \mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}$

$\quad u^* \leftarrow\!\!\$\ \mathcal{M}_2$, $\hat{m}_1^* \leftarrow\!\!\$\ \mathcal{M}_1$

4 : $Map_2[r^*] := (m_1^*, \rho^*, k_0)$ // G_0–G_3

5 : $Map_1[m_1^*] := u^*$ // G_0–G_5

6 : $m_2^* := r^* \oplus u^*$

7 : $c^* := \mathsf{Enc}'(\mathsf{pk}, m_1^* \| m_2^*; \rho^*)$ // G_0–G_6

8 : $c^* := \mathsf{Enc}'(\mathsf{pk}, \hat{m}_1^* \| m_2^*; \rho^*)$ // G_7

9 : $b \leftarrow\!\!\$\ \{0,1\}$, $k_1 \leftarrow\!\!\$\ \mathcal{K}$

10 : $\hat{b} \leftarrow \mathcal{A}^{F,H,O_{\mathsf{Dec}}}(\mathsf{pk}, c^*, k_b)$

11 : **return** $[b = \hat{b}]$

Random Oracle $F(m_1)$:

12 : **if** $m_1 = m_1^*$ **then**

$\quad Map_1[m_1^*] := u^*$ // G_6–G_7

13 : **if** $m_1 \notin \mathrm{Domain}(Map_1)$ **then**

$\quad Map_1[m_1] \leftarrow\!\!\$\ \mathcal{M}_2$

14 : **return** $Map_1[m_1]$

Random Oracle $H(r)$:

15 : **if** $r = r^*$ **then**

$\quad$ **return** (m_1^*, ρ^*, k_0) // G_4

16 : **if** $r \notin \mathrm{Domain}(Map_2)$ **then**

$\quad Map_2[r] \leftarrow\!\!\$\ \mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}$

17 : **return** $Map_2[r]$

Decapsulation Oracle $O_{\mathsf{Dec}}(c)$:

18 : **if** $c = c^*$ **then**

$\quad$ **return** $\perp$

19 : $m_1 \| m_2 := \mathsf{Dec}'(\mathsf{sk}, c)$ // G_0–G_2

20 : **if** $m_1 \| m_2 = \perp$ **then**

$\quad$ **return** $\perp$ // G_0–G_2

21 : **if** $m_1 \notin \mathrm{Domain}(Map_1)$

$\quad$ **return** $\perp$ // G_2

22 : $r := m_2 \oplus F(m_1)$ // G_0–G_2

23 : **if** $r \notin \mathrm{Domain}(Map_2)$

$\quad$ **return** $\perp$ // G_1–G_2

24 : $(m_1', \rho, k) := H(r)$ // G_0–G_2

25 : **if** $m_1 = m_1'$ **and**

$\quad \mathsf{Enc}'(\mathsf{pk}, m_1' \| m_2; \rho) = c$ **then**

$\quad$ **return** k // G_0–G_2

26 : **if** $\exists (r, (m_1, \rho, k)) \in Map_2$ **and**

$\quad (m_1, u) \in Map_1$ s.t.

$\quad \mathsf{Enc}'(\mathsf{pk}, m_1 \| (r \oplus u); \rho) = c$ **then**

$\quad$ **return** k // G_3–G_7

27 : **return** $\perp$

Figure 9: Games G_0 to G_7 for the proof of Theorem 4.2.

$r \notin \mathrm{Domain}(Map_2)$, the oracle directly returns $\perp$ (as shown in line 23 of Fig. 9). Let Z_1 be the event that $r \notin \mathrm{Domain}(Map_2)$ yet $m_1' = m_1$, where $(m_1', \rho, k) := H(r)$. According to the simulation of the random oracle H, since m_1' is sampled uniformly at random from $\mathcal{M}_1$, we have $\Pr[Z_1] \leq 1/|\mathcal{M}_1|$. If Z_1 does not occur, then Games G_1 and G_0 proceed identically. By the Difference Lemma and a union bound, we obtain

$$|\Pr[W_1] - \Pr[W_0]| \leq q_D \cdot \Pr[Z_1] \leq q_D / |\mathcal{M}_1| ,$$

where q_D is the bounded number of decapsulation oracle queries made by $\mathcal{A}$.

Game G_2. This game continues the modification of the decapsulation oracle. For a decapsulation oracle query $c \neq c^*$, after computing $m_1 \| m_2 := \mathsf{Dec}'(\mathsf{sk}, c)$ $(\neq \perp)$, if $m_1 \notin \mathrm{Domain}(Map_1)$, the oracle directly outputs $\perp$ (as shown in line 21 of Fig. 9). Let Z_2 be the event that $m_1 \notin \mathrm{Domain}(Map_1)$ but $r := m_2 \oplus F(m_1) \in \mathrm{Domain}(Map_2)$. Clearly, if event Z_2 does not occur, the challenger's behavior in Game G_2 is identical to that in Game G_1. According to the simulation of the random oracle F, when $m_1 \notin \mathrm{Domain}(Map_1)$, the value $F(m_1)$ is uniformly random over $\mathcal{M}_2$. Therefore, for a fixed m_2, the probability that $m_2 \oplus F(m_1)$ lies in $\mathrm{Domain}(Map_2)$ is at most $(q_H+1)/|\mathcal{M}_2|$. Here, q_H is the bounded number of queries $\mathcal{A}$ makes to the random oracle H, and the extra $+1$ in the numerator comes from the entry $(r^*, (m^*, \rho^*, k_0))$ stored in Map_2 during the challenger's initialization, which ensures $|\mathrm{Domain}(Map_2)| \leq q_H + 1$.

Consequently, by the Difference Lemma and a union bound, we have

$$|\Pr[W_2] - \Pr[W_1]| \leq q_D \cdot \Pr[Z_2] \leq q_D(q_H + 1)/|\mathcal{M}_2| \ .$$

Game $\boldsymbol{G_3}$. From this game onward, the decapsulation oracle no longer uses the computation $m_1\|m_2 := \mathsf{Dec}'(\mathsf{sk}, c)$ to obtain its response. Specifically, lines 19–25 of Fig. 9 are removed and replaced by a search through Map_2 and Map_1 for entries $(r, (m_1, \rho, k))$ and (m_1, u) satisfying $\mathsf{Enc}'(\mathsf{pk}, m_1\|(r \oplus u); \rho) = c$ (as indicated in line 26 of Fig. 9): if such a pair of entries is found, the oracle returns k; otherwise, it returns $\perp$. It is clear that for the same decapsulation oracle query $c \neq c^*$, if the oracle in Game G_3 returns $\perp$, then the oracle in Game G_2 must also return $\perp$. This is because if the oracle in Game G_2 did not return $\perp$, it implies that Map_2 and Map_1 must contain entries $(r, (m_1, \rho, k))$ and (m_1, u) satisfying $\mathsf{Enc}'(\mathsf{pk}, m_1\|(r \oplus u); \rho) = c$, which would in turn prevent the oracle in Game G_3 from returning $\perp$.

Therefore, a difference in the behavior of the decapsulation oracles for a query $c \neq c^*$ between Games G_3 and G_2 can only occur when the oracle in Game G_3 returns some $k \neq \perp$, while the oracle in Game G_2 does not return this k. This implies that the value $m_1\|(r \oplus u)$ found in Game G_3 is not equal to $\mathsf{Dec}'(\mathsf{sk}, c)$. In other words, there exists an entry $(r, (m_1, \rho, k)) \in Map_2$ (with a corresponding $(m_1, u) \in Map_1$) such that $\mathsf{Dec}'(\mathsf{sk}, \mathsf{Enc}'(\mathsf{pk}, m_1\|(r \oplus u); \rho)) \neq m_1\|(r \oplus u)$. We denote this event by Z_3.

Given $(\mathsf{pk}, \mathsf{sk})$, define $\delta(\mathsf{pk}, \mathsf{sk}) = \Pr\big[\mathsf{Dec}'(\mathsf{sk}, \mathsf{Enc}'(\mathsf{pk}, m_1\|m_2; \rho)) \neq m_1\|m_2\big]$, where the probability is taken over $m_1\|m_2 \leftarrow\!\!\$\ \mathcal{M}_1 \times \mathcal{M}_2$ and $\rho \leftarrow\!\!\$\ \mathcal{R}$. For any fixed $r \in \mathcal{M}_2$, the random variable $r \oplus u$, where $u \leftarrow\!\!\$\ \mathcal{M}_2$, is uniformly distributed over $\mathcal{M}_2$. Thus, we have $\Pr[Z_3 : (\mathsf{pk}, \mathsf{sk})] \leq |\mathrm{Domain}(Map_2)| \cdot \delta(\mathsf{pk}, \mathsf{sk}) = (q_H + 1)\,\delta(\mathsf{pk}, \mathsf{sk})$. By the Difference Lemma and a union bound, it follows that $|\Pr[W_3 : (\mathsf{pk}, \mathsf{sk})] - \Pr[W_2 : (\mathsf{pk}, \mathsf{sk})]| \leq \Pr[Z_3 : (\mathsf{pk}, \mathsf{sk})]$. By averaging over $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$, we can obtain

$$|\Pr[W_3] - \Pr[W_2]| \leq \mathbb{E}\big[\Pr[Z_3 : (\mathsf{pk}, \mathsf{sk})]\big] \leq \mathbb{E}\big[(q_H + 1)\delta(\mathsf{pk}, \mathsf{sk})\big] \leq (q_H + 1)\delta_{ac} \ ,$$

where the last inequality follows from the definition of the δ_{ac}-average-case correctness of PKE' (see (2)), i.e., $\mathbb{E}[\delta(\mathsf{pk}, \mathsf{sk})] \leq \delta_{ac}$, under the distribution $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$.

Game $\boldsymbol{G_4}$. We now begin modifying the random oracles. In this game, the challenger no longer stores $(r^*, (m_1^*, \rho^*, k_0))$ in Map_2 during initialization (i.e., line 4 of Fig. 9 is removed), but still uses (m_1^*, ρ^*, k_0) as the value of $H(r^*)$ (as shown in line 15 of Fig. 9). Consequently, in the simulation of the decapsulation oracle in Game G_4, line 26 will not use $(r^*, (m_1^*, \rho^*, k_0))$ to compute responses to decapsulation oracle queries. It is important to note, however, that line 26 in Game G_3 would not have used $(r^*, (m_1^*, \rho^*, k_0))$ for such computations either. If it did, this would imply that for some decapsulation oracle query $c \neq c^*$, there exist entries $(r^*, (m_1^*, \rho^*, k_0)) \in Map_2$ and $(m_1^*, u^*) \in Map_1$ such that $\mathsf{Enc}'(\mathsf{pk}, m_1^*\|(r^* \oplus u^*); \rho^*) = c$, which contradicts $c \neq c^*$. Therefore, from $\mathcal{A}$'s perspective, the challenger's behavior in Game G_4 is identical to that in Game G_3. Hence, we have

$$\Pr[W_4] = \Pr[W_3] \ .$$

Game $\boldsymbol{G_5}$. In this game, the challenger no longer uses (m_1^*, ρ^*, k_0) as the value of $H(r^*)$, i.e., line 15 of Fig. 9 is removed. Let X_j be the event that $\mathcal{A}$ queries r^* to the random oracle H in Game G_j. Clearly, if X_j does not occur, then Games G_5 and G_4 proceed identically. Hence, by the Difference Lemma, we have

$$|\Pr[W_5] - \Pr[W_4]| \leq \Pr[X_5] \ .$$

Furthermore, note that at this point, both k_0 and k_1 are sampled independently and uniformly from $\mathcal{K}$ and are not used elsewhere in the game. Therefore, the bit b is independent and uniformly random from $\mathcal{A}$'s view. Consequently, we have

$$\Pr[W_5] = 1/2 \ .$$

Game G_6. We now analyze X_j. In the initialization phase, the challenger no longer stores (m_1^*, u^*) in Map_1 (i.e., line 5 of Fig. 9 is removed). However, when $\mathcal{A}$ queries $F(m_1^*)$, the random oracle F still returns u^* as the value of $F(m_1^*)$ (as in line 12 of Fig. 9). Note that, in Game G_6, although $(m_1^*, u^*) \notin Map_1$ at the beginning of the game, once $\mathcal{A}$ queries $F(m_1^*)$, the pair (m_1^*, u^*) enters Map_1. This modification causes line 26 in the decapsulation oracle of Game G_6 not to use (m_1^*, u^*) for response computation before $\mathcal{A}$ queries $F(m_1^*)$. In Game G_5, however, line 26 might still use (m_1^*, u^*) when there exists an entry $(r, (m_1 = m_1^*, \rho, k)) \in Map_2$. Note that for all $(r, (m_1, \rho, k)) \in Map_2$, the value m_1 is uniformly random over $\mathcal{M}_1$. Let q_H bound the number of queries $\mathcal{A}$ makes to the random oracle H. The probability that there exists an entry $(r, (m_1 = m_1^*, \rho, k)) \in Map_2$ is at most $q_H/|\mathcal{M}_1|$. Therefore,

$$|\Pr[X_6] - \Pr[X_5]| \leq q_H/|\mathcal{M}_1| \ .$$

Recall that X_6 is the event in Game G_6 where $\mathcal{A}$ queries r^* to the random oracle H, with $r^* = m_2^* \oplus u^* = m_2^* \oplus F(m_1^*)$. Importantly, u^* is used only as the value of $F(m_1^*)$, and r^* is not used elsewhere in the game. Thus, if $\mathcal{A}$ has not queried m_1^* to F, then u^* is independent of $\mathcal{A}$'s view, and consequently r^* is also independent of $\mathcal{A}$'s view. Let Y_j be the event that $\mathcal{A}$ queries m_1^* to F in Game G_j. This implies $\Pr[X_6 \wedge \overline{Y_6}] \leq q_H/|\mathcal{M}_2|$, and so

$$\Pr[X_6] = \Pr[X_6 \wedge Y_6] + \Pr[X_6 \wedge \overline{Y_6}] \leq \Pr[X_6 \wedge Y_6] + q_H/|\mathcal{M}_2| \ .$$

When the event $X_6 \wedge Y_6$ occurs, it means $\mathcal{A}$ has queried m_1^* to F and r^* to H, where $r^* = m_2^* \oplus F(m_1^*)$. In this case, we can construct an adversary $\mathcal{B}$ against the OW-CPA security of the underlying PKE'. Specifically, given (pk, c^*) from the OW-CPA challenger, $\mathcal{B}$ samples $k \leftarrow\$ \mathcal{K}$, sends (pk, c^*, k) to $\mathcal{A}$, and then simulates the random oracle H and the decapsulation oracle for $\mathcal{A}$ exactly as in Game G_6. For the random oracle F, $\mathcal{B}$ simulates it as in Game G_6 but with line 12 removed. Note that here $\mathcal{B}$ implicitly chooses a random u^*; when $\mathcal{A}$ queries $F(m_1^*)$, the oracle responds with this u^*. From $\mathcal{A}$'s view, $\mathcal{B}$'s behavior is identical to the challenger in Game G_6. At the end of the game, if $X_6 \wedge Y_6$ occurs, $\mathcal{B}$ first randomly selects an m_1 from Domain(Map_1) and an r from Domain(Map_2), and then outputs $m_1 \| (r \oplus F(m_1))$ as its solution to the OW-CPA challenge (pk, c^*). Since $|\text{Domain}(Map_1)| \leq q_F$ and $|\text{Domain}(Map_2)| \leq q_H$, we have

$$\Pr[X_6 \wedge Y_6]/q_F q_H \leq \mathsf{Adv}_{\mathsf{PKE}'}^{\text{OW-CPA}}(\mathcal{B}) \ .$$

When PKE' is deterministic, let $\mathcal{R} = \emptyset$. The above reduction still holds. The only difference is that when $X_6 \wedge Y_6$ occurs, $\mathcal{B}$ does not randomly select m_1 and r; instead, it searches for a pair $(m_1 \in \text{Domain}(Map_1), r \in \text{Domain}(Map_2))$ satisfying $\mathsf{Enc}'(\mathsf{pk}, m_1 \| (r \oplus F(m_1))) = c^*$. Clearly, such $m_1 \| (r \oplus F(m_1)) = m_1^* \| m_2^*$, unless for $m_1^* \| m_2^* \leftarrow\$ \mathcal{M}_1 \times \mathcal{M}_2$, there exists $m_1 \| m_2 \neq m_1^* \| m_2^*$ such that $\mathsf{Enc}'(\mathsf{pk}, m_1 \| m_2) = \mathsf{Enc}'(\mathsf{pk}, m_1^* \| m_2^*)$. According to [16, Lemma G1], for a deterministic PKE', this probability is at most $2\delta_{ac}$. Therefore,

$$\Pr[X_6 \wedge Y_6] \leq \mathsf{Adv}_{\mathsf{dPKE}'}^{\text{OW-CPA}}(\mathcal{B}) + 2\delta_{ac} \ .$$

Game G_7. We now reduce the advantage further to the IND-CPA security of the underlying PKE'. In this game, the challenger uses another random message $\hat{m}_1^* \leftarrow\$ \mathcal{M}_1$ to compute $c^* := \mathsf{Enc}'(\mathsf{pk}, \hat{m}_1^* \| m_2^*; \rho^*)$ (as shown in line 8 of Fig. 9), instead of using m_1^* (i.e., line 7 is removed). Note that in this game, before $\mathcal{A}$ queries $F(m_1^*)$, the value m_1^* is not involved in any computation and is independent of $\mathcal{A}$'s view. Therefore, we have

$$\Pr[X_7 \wedge Y_7] \leq \Pr[Y_7] \leq q_F/|\mathcal{M}_1| \ .$$

We now construct a new adversary $\mathcal{B}'$ against the IND-CPA security of PKE'. Upon receiving pk from the IND-CPA challenger, $\mathcal{B}'$ randomly samples $m_1^*, \hat{m}_1^* \leftarrow\$ \mathcal{M}_1$ and $r^*, u^* \leftarrow\$ \mathcal{M}_2$. It sets $m_2^* := r^* \oplus u^*$, $m_0 := m_1^* \| m_2^*$, $m_1 := \hat{m}_1^* \| m_2^*$, and sends (m_0, m_1) to the challenger. Upon receiving the challenge ciphertext $c^* = \mathsf{Enc}'(\mathsf{pk}, m_{b'})$ (where b' is the challenger's random bit), $\mathcal{B}'$ samples $k \leftarrow\$ \mathcal{K}$, sends (pk, c^*, k) to $\mathcal{A}$, and simulates the random oracles F, H, and the decapsulation oracle for $\mathcal{A}$ exactly as in Game G_6. At the end of the game, $\mathcal{B}'$ checks if both $m_1^* \in \text{Domain}(Map_1)$ and $r^* \in \text{Domain}(Map_2)$ hold. If yes, it outputs $\hat{b}' := 1$; otherwise, it outputs $\hat{b}' := 0$.

Observe that when $b' = 0$, we have $c^* = \mathsf{Enc}'(\mathsf{pk}, m_1^* \| m_2^*)$, and $\mathcal{B}'$ perfectly simulates the challenger of Game G_6. When $b' = 1$, $c^* = \mathsf{Enc}'(\mathsf{pk}, \hat{m}_1^* \| m_2^*)$, and $\mathcal{B}'$ perfectly simulates the challenger of Game G_7. Therefore, $\Pr[\hat{b}' = 1 | b' = 0] = \Pr[X_6 \wedge Y_6]$ and $\Pr[\hat{b}' = 1 | b' = 1] = \Pr[X_7 \wedge Y_7]$. Consequently,

$$\begin{aligned}
\mathsf{Adv}^{\text{IND-CPA}}_{\mathsf{PKE}'}(\mathcal{B}') &\geq \left| \Pr[b' = \hat{b}'] - 1/2 \right| \\
&= \left| \Pr[b' = \hat{b}' \wedge b' = 0] + \Pr[b' = \hat{b}' \wedge b' = 1] - 1/2 \right| \\
&= \frac{1}{2} \left| \Pr[\hat{b}' = 0 | b' = 0] + \Pr[\hat{b}' = 1 | b' = 1] - 1 \right| \\
&= \frac{1}{2} \left| \Pr[\hat{b}' = 1 | b' = 1] - \Pr[\hat{b}' = 1 | b' = 0] \right| \\
&= \frac{1}{2} \left| \Pr[X_7 \wedge Y_7] - \Pr[X_6 \wedge Y_6] \right| .
\end{aligned}$$

Combining all bounds, we have

$$\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) \leq q_F q_H \mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{PKE}'}(\mathcal{B}) + (q_H + 1)\delta_{ac} + (q_D + q_H)/|\mathcal{M}_1| + \Delta ,$$

$$\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) \leq \mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{dPKE}'}(\mathcal{B}) + (q_H + 3)\delta_{ac} + (q_D + q_H)/|\mathcal{M}_1| + \Delta ,$$

and

$$\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) \leq 2\mathsf{Adv}^{\text{IND-CPA}}_{\mathsf{PKE}'}(\mathcal{B}') + (q_H + 1)\delta_{ac} + (q_D + q_H + q_F)/|\mathcal{M}_1| + \Delta ,$$

where $\Delta \leq (q_D + 1)(q_H + 1)/|\mathcal{M}_2|$. □

5 The QROM Proofs

5.1 Some Useful Lemmas

We now introduce three lemmas that serve as the core technical tools for our QROM security proofs. As motivated in Section 2, lifting the ROM proof strategy for FOAC_0' and FOAC' to the quantum setting requires mechanisms to record and manage superposition queries to the random oracle. Such mechanisms are naturally supplied by the compressed oracle technique [43], and the lemmas below distill its essential features into three reusable statements. Each lemma directly mirrors a key step of the corresponding ROM argument and is tailored to the quantum setting. Their role will become evident in the detailed security analysis of FOAC_0' and FOAC' presented in the next sections.

The proof of each lemma proceeds by tracking the evolution of quantum states across all relevant registers, quantifying the effect of modifying the oracle behavior, and applying a general bound by Ambainis et al. [1, Lemma 4] that translates state distance into a bound on the distinguishing advantage. This yields clean, modular estimates that are readily applicable to our later proofs. The full proofs are provided in Appendix A.

Lemma 5.1 (Switching Lemma)**.** *Let $\mathsf{H} : \mathcal{X} \to \mathcal{Y}$ be a quantum-accessible random oracle, and let O_0 be a quantum oracle acting on the input registers WX (defined over $\mathcal{W} \times \mathcal{X}$) and an output register $\bar{\mathsf{Y}}$ (defined over $\bar{\mathcal{Y}}$) as*

$$\mathsf{O}_0 : |w, x, \bar{y}\rangle_{\mathsf{WX\bar{Y}}} \mapsto |w, x, \bar{y} \oplus f(\mathsf{H}(x))\rangle_{\mathsf{WX\bar{Y}}} \text{ with } f(y) = \begin{cases} g_{w,x}(y), & \text{if } y \in S_{w,x} \\ g_{w,x}(\perp), & \text{otherwise} \end{cases},$$

where both the function $g_{w,x} : \mathcal{Y} \cup \{\perp\} \to \bar{\mathcal{Y}}$ and the set $S_{w,x} \subseteq \mathcal{Y}$ are determined by (w, x). Let $\mathsf{H_c} : \mathcal{X} \to \mathcal{Y}$ be a compressed standard oracle with a database register D initialized to $\bigotimes_{i=1}^{q_{\mathsf{H}}} |(\perp, 0^{\log|\mathcal{Y}|}\rangle$, and let O_1 be the quantum oracle

$$\mathsf{O}_1 : |w, x, D, \bar{y}\rangle_{\mathsf{WXD\bar{Y}}} \mapsto |w, x, D, \bar{y} \oplus f(D(x))\rangle_{\mathsf{WXD\bar{Y}}}$$

which uses the same input/output registers $\mathsf{WX}/\bar{\mathsf{Y}}$ as O_0 and additionally employs D as a work register.

Let A be a quantum oracle algorithm (not necessarily unitary) making at most q_{H} queries to $\mathsf{H}/\mathsf{H_c}$ and at most q_{O} queries to $\mathsf{O}_0/\mathsf{O}_1$. For an arbitrary classical event Ev and a random z distributed according to $\mathfrak{D}_z$, define

$$P_{\text{left}} \coloneqq \Pr[\mathsf{Ev} : A^{|\mathsf{H}\rangle, |\mathsf{O}_0\rangle}(z)] \text{ and } P_{\text{right}} \coloneqq \Pr[\mathsf{Ev} : A^{|\mathsf{H_c}\rangle, |\mathsf{O}_1\rangle}(z)],$$

where the probability is taken over $z \leftarrow \mathfrak{D}_z$ and the internal randomness of A (including all quantum measurements). Setting $\Gamma_S \coloneqq \max_{w,x} |S_{w,x}|$, we have

$$|P_{\text{left}} - P_{\text{right}}| \leq 6q_{\mathsf{O}}\sqrt{\Gamma_S/|\mathcal{Y}|} \quad \text{and} \quad \left|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}\right| \leq 6q_{\mathsf{O}}\sqrt{\Gamma_S/|\mathcal{Y}|}.$$

Lemma 5.2 (Lookup Lemma)**.** *Let $\mathsf{H_c} : \mathcal{X} \to \mathcal{Y}$ be a compressed standard oracle with a database register D initialized to $\bigotimes_{i=1}^{q_{\mathsf{H}}} |(\perp, 0^{\log|\mathcal{Y}|})\rangle$, and let O_0 be a quantum oracle acting on the input register W (defined over $\mathcal{W}$), the database register D, and an output register $\bar{\mathsf{Y}}$ (defined over $\bar{\mathcal{Y}}$) as*

$$\mathsf{O}_0 : |w, D, \bar{y}\rangle_{\mathsf{WD\bar{Y}}} \mapsto |w, D, \bar{y} \oplus f_0(w, D)\rangle_{\mathsf{WD\bar{Y}}}$$

with

$$f_0(w, D) = \begin{cases} g_{w,D}(x), & \text{if } (x, D(x)) \in S_w, \text{ where } x \coloneqq d(w) \\ g_{w,D}(\perp), & \text{otherwise} \end{cases},$$

where each function $g_{w,D} : \mathcal{X} \cup \{\perp\} \to \bar{\mathcal{Y}}$ and each set $S_w \subseteq \mathcal{X} \times \mathcal{Y}$ are determined by (w, D) and by w, respectively, and $d : \mathcal{W} \to \mathcal{X}$ is a fixed function. Let O_1 be the quantum oracle

$$\mathsf{O}_1 : |w, D, \bar{y}\rangle_{\mathsf{WD\bar{Y}}} \mapsto |w, D, \bar{y} \oplus f_1(w, D)\rangle_{\mathsf{WD\bar{Y}}}$$

with

$$f_1(w, D) = \begin{cases} g_{w,D}(x), & \text{if } \exists (x, y) \in D \text{ s.t. } (x, y) \in S'_w \\ & \text{and } \forall x' < x,\ (x', y') \in D \implies (x', y') \notin S'_w \\ g_{w,D}(\perp), & \text{otherwise} \end{cases},$$

which uses the same registers $\mathsf{WD\bar{Y}}$ and the same function $g_{w,D}$ as O_0. It should be note that the set $S'_w \subseteq \mathcal{X} \times \mathcal{Y}$ determined by w differs from S_w used in O_0.

Let A be a quantum oracle algorithm (not necessarily unitary) making at most q_{H} queries to $\mathsf{H_c}$ and at most q_{O} queries to $\mathsf{O}_0/\mathsf{O}_1$. For an arbitrary classical event Ev and a random z distributed according to $\mathfrak{D}_z$, define

$$P_{\text{left}} \coloneqq \Pr[\mathsf{Ev} : A^{|\mathsf{H_c}\rangle, |\mathsf{O}_0\rangle}(z)] \text{ and } P_{\text{right}} \coloneqq \Pr[\mathsf{Ev} : A^{|\mathsf{H_c}\rangle, |\mathsf{O}_1\rangle}(z)],$$

where the probability is taken over $z \leftarrow \mathfrak{D}_z$ and the internal randomness of A (including all quantum measurements). For each $w \in \mathcal{W}$, define the set

$$T_w \coloneqq \{(x,y) \mid (x,y) \in S_w \text{ and } x = d(w)\} \ .$$

Then, let $R \subseteq \mathcal{X} \times \mathcal{Y}$ be the binary relation given by

$$R \coloneqq \{(x,y) \mid \exists w \in \mathcal{W} \text{ s.t. } (x,y) \in T_w \triangle S'_w\} \ ,$$

where $\triangle$ denotes the symmetric difference, i.e., $T_w \triangle S'_w \coloneqq (T_w \setminus S'_w) \cup (S'_w \setminus T_w)$. Let $\Gamma_R \coloneqq \max_x |\{y \mid (x,y) \in R\}|$. Then, we have

$$|P_{\text{left}} - P_{\text{right}}| \leq 16(q_{\mathsf{H}}+1)\sqrt{\Gamma_R/|\mathcal{Y}|}, \ \left|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}\right| \leq 16(q_{\mathsf{H}}+1)\sqrt{\Gamma_R/|\mathcal{Y}|} \ .$$

Lemma 5.3 (Puncturing Lemma). *Let $\mathsf{H_c} : \mathcal{X} \to \mathcal{Y}$ be a compressed standard oracle with a database register D initialized to $\bigotimes_{i=1}^{q_{\mathsf{H}}} |(\perp, 0^{\log|\mathcal{Y}|})\rangle$. For a fixed pair $(x^*, y^*) \in \mathcal{X} \times \mathcal{Y}$, define a punctured oracle*

$$\mathsf{H'_c} : |x,y,D\rangle \mapsto \begin{cases} |x, y \oplus y^*, D\rangle \ , & \text{if } x = x^* \ , \\ \mathsf{H_c}|x,y,D\rangle \ , & \text{otherwise} \ , \end{cases}$$

which answers queries to x^* without updating the database D. Let O_{x^*,y^*} be a quantum oracle acting on the input register W (defined over $\mathcal{W}$), the database D, and an output register $\bar{\mathsf{Y}}$ (defined over $\bar{\mathcal{Y}}$) as

$$\mathsf{O}_{x^*,y^*} : |w, D, \bar{y}\rangle_{\mathsf{WD\bar{Y}}} \mapsto |w, D, \bar{y} \oplus f_{x^*,y^*}(w,D)\rangle_{\mathsf{WD\bar{Y}}} \ ,$$

where $f_{x^*,y^*} : \mathcal{W} \times \mathcal{D} \to \bar{\mathcal{Y}}$ satisfies

$$f_{x^*,y^*}(w,D) = f_{x^*,y^*}\big(w, D \cup (x^*,y^*)\big) \ \text{ for every } D \text{ with } D(x^*) = \perp \ .$$

Let A be a quantum oracle algorithm (not necessarily unitary) making at most q_{H} queries to $\mathsf{H_c}/\mathsf{H'_c}$ and at most q_{O} queries to O_{x^*,y^*}. For an arbitrary classical event Ev, define

$$P_{\text{left}} \coloneqq \Pr[\mathsf{Ev} : x^* \leftarrow\!\$\ \mathcal{X},\ y^* \coloneqq \mathsf{H_c}(x^*),\ z \leftarrow \mathfrak{D}_z,\ A^{|\mathsf{H_c}\rangle, |\mathsf{O}_{x^*,y^*}\rangle}(y^*, z)] \ ,$$
$$P_{\text{right}} \coloneqq \Pr[\mathsf{Ev} : x^* \leftarrow\!\$\ \mathcal{X},\ y^* \leftarrow\!\$\ \mathcal{Y},\ z \leftarrow \mathfrak{D}_z,\ A^{|\mathsf{H'_c}\rangle, |\mathsf{O}_{x^*,y^*}\rangle}(y^*, z)] \ ,$$

where the probability is taken over all random coins. Then, we have

$$|P_{\text{left}} - P_{\text{right}}| \leq 2q_{\mathsf{O}}/\sqrt{|\mathcal{Y}|} \quad \text{and} \quad \left|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}\right| \leq 2q_{\mathsf{O}}/\sqrt{|\mathcal{Y}|} \ .$$

5.2 $\mathsf{KEM} \coloneqq \mathsf{FOAC}'_0[\mathsf{PKE}', F, H]$

Theorem 5.4 (QROM Security of FOAC'_0). *Assume that $F : \mathcal{M} \to \{0,1\}^n$ and $H : \{0,1\}^n \to \mathcal{M} \times \mathcal{R} \times \mathcal{K}$ are modeled as quantum-accessible random oracles. For any efficient adversary $\mathcal{A}$ attacking the* IND-CCA *security of $\mathsf{KEM} \coloneqq \mathsf{FOAC}'_0[\mathsf{PKE}', F, H]$ with quantum access to the random oracles F and H, there exists an efficient adversary $\mathcal{B}$ (resp. $\mathcal{B}'$) against the* OW-CPA *(resp.* IND-CPA*) security of PKE', such that*

$$\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) \leq 4(q_F + q_H + 1)\sqrt{\mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{PKE}'}(\mathcal{B})} + 16(q_H+1)\sqrt{\delta_{ac}} + \frac{16(q_D + q_H + 1)}{\sqrt{|\mathcal{M}|}} + \Delta,$$

Games G_0 to G_8:

1 : $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda),\ r^* \leftarrow\!\!\$\ \{0,1\}^n$

2 : $\mathsf{F} \leftarrow\!\!\$\ \Omega_F,\ \mathsf{H} \leftarrow\!\!\$\ \Omega_H$

3 : $(m^*, \rho^*, k_0) := H(r^*)$ // G_0–G_1

4 : $(m^*, \rho^*, k_0) \leftarrow\!\!\$\ \mathcal{M} \times \mathcal{R} \times \mathcal{K}$ // G_2–G_8

5 : $u^* := F(m^*)$ // G_0–G_5

6 : $u^* \leftarrow\!\!\$\ \{0,1\}$ // G_6–G_8

7 : $c_1^* := \mathsf{Enc}'(\mathsf{pk}, m^*; \rho^*),\ c_2^* := r^* \oplus u^*$

8 : $b \leftarrow\!\!\$\ \{0,1\},\ k_1 \leftarrow\!\!\$\ \mathcal{K}$

9 : $\hat{b} \leftarrow \mathcal{A}^{|F\rangle, |H\rangle, O_{\mathsf{Dec}}}(\mathsf{pk}, (c_1^*, c_2^*), k_b)$

10 : **return** $[b = \hat{b}]$

Random Oracle $F(|x, y\rangle)$:

11 : **return** $|x, y \oplus \mathsf{F}(x)\rangle$ // G_0–G_2

12 : **if** $x = m^*$ **then**
    return $|x, y \oplus u^*\rangle$ // G_6–G_7

13 : **perform** $\mathsf{F_c}$ // G_3–G_8

Random Oracle $H(|x, y\rangle)$:

14 : **return** $|x, y \oplus \mathsf{H}(x)\rangle$ // G_0

15 : **if** $x = r^*$ **then**
    return $|x, y \oplus (m^*, \rho^*, k_0)\rangle$ // G_2–G_7

16 : **perform** $\mathsf{H_c}$ // G_1–G_8

17 : **perform** $\mathcal{O}_S^{CSC}$ // G_5–G_6

Decapsulation Oracle $O_{\mathsf{Dec}}(c_1, c_2)$:

18 : **if** $(c_1, c_2) = (c_1^*, c_2^*)$ **then**
    return $\perp$

19 : $m := \mathsf{Dec}'(\mathsf{sk}, c_1)$ // G_0–G_3

20 : **if** $m = \perp$ **then**
    return $\perp$ // G_0–G_3

21 : $r := c_2 \oplus F(m)$ // G_0–G_2

22 : $r := c_2 \oplus \mathsf{D_F}(m)$ // G_3

23 : $(m', \rho, k) := H(r)$ // G_0

24 : $(m', \rho, k) := \mathsf{D_H}(r)$ // G_1–G_3

25 : **if** $m = m'$ **and**
  $\mathsf{Enc}'(\mathsf{pk}, m'; \rho) = c_1$ **then**
    return k // G_0–G_3

26 : **return** $\perp$ // G_0–G_3

27 : **return** $f_1((c_1, c_2, \mathsf{D_F}), \mathsf{D_H})$ // G_4–G_8

Figure 10: Games G_0 to G_8 for the proof of Theorem 5.4.

$$\mathsf{Adv}_{\mathsf{KEM}}^{\text{IND-CCA}}(\mathcal{A}) \leq 2\sqrt{2(q_F + q_H + 1)\mathsf{Adv}_{\mathsf{PKE}'}^{\text{IND-CPA}}(\mathcal{B}')} + 16(q_H + 1)\sqrt{\delta_{ac}} \\ + 20(q_D + q_H + q_F + 1)/\sqrt{|\mathcal{M}|} + \Delta\ ,$$

where q_F, q_H, and q_D are the bounded numbers of $\mathcal{A}$'s queries to F, H, and the decapsulation oracle, respectively, and $\Delta \leq 8(q_D + q_H)\sqrt{q_F + q_H + 1}/\sqrt{2^n}$,

If PKE' is deterministic, the bound can be improved to

$$\mathsf{Adv}_{\mathsf{KEM}}^{\text{IND-CCA}}(\mathcal{A}) \leq 2\sqrt{(q_F + q_H + 1)\mathsf{Adv}_{\mathsf{dPKE}'}^{\text{OW-CPA}}(\mathcal{B})} + 20(q_F + q_H + 1)\sqrt{\delta_{ac}} \\ + 16(q_D + q_H + 1)/\sqrt{|\mathcal{M}|} + \Delta\ .$$

Theorem 5.4. We define a series of games and describe in Fig. 10 how the challenger is simulated for the adversary $\mathcal{A}$ in games G_j, where $j = 0, \ldots, 8$. In each game, let b denote the random bit sampled by the challenger and $\hat{b}$ the bit output by $\mathcal{A}$. We define W_j as the event that $b = \hat{b}$ in game G_j.

Game G_0. At the beginning of the game, the challenger samples two functions $\mathsf{F} \leftarrow\!\!\$\ \Omega_F$ and $\mathsf{H} \leftarrow\!\!\$\ \Omega_H$ to model the random oracles F and H, respectively. It is clear that the challenger's behavior in Game G_0 is identical to that in the IND-CCA game for $\mathsf{KEM} := \mathsf{FOAC}_0'[\mathsf{PKE}', F, H]$. Hence, we have

$$|\Pr[W_0] - 1/2| = \mathsf{Adv}_{\mathsf{KEM}}^{\text{IND-CCA}}(\mathcal{A})\ .$$

Game G_1. In this game, the random oracle H is simulated using a compressed oracle $\mathsf{H_c}$ (as shown in line 16 of Fig. 10), with a database register $\mathsf{D_H}$. Note that in the decapsulation oracle of Game G_0, after computing $m := \mathsf{Dec}'(\mathsf{sk}, c_1)$ $(\neq \perp)$ and $r := c_2 \oplus F(m)$, the oracle's behavior is equivalent to first applying a unitary operator

$$|(c_1, m), r, 0^{\log|\mathcal{K}|}\rangle \mapsto \begin{cases} |(c_1, m), r, 0^{\log|\mathcal{K}|} \oplus H(r)[2]\rangle\,, & \text{if } H(r) \in S_{(c_1,m),r}\ , \\ |(c_1, m), r, 0^{\log|\mathcal{K}|} \oplus \perp\rangle\,, & \text{otherwise}\ , \end{cases}$$

where $S_{(c_1,m),r} := \{(m', \rho, k) \mid m' = m \wedge \mathsf{Enc}'(\mathsf{pk}, m'; \rho) = c_1\}$, and then measuring the last register in the computational basis to obtain a classical outcome (i.e., $k := H(r)[2]$ or $\perp$). In this game, we replace $H(r)$ in this unitary with $D_{\mathsf{H}}(r)$, where $|D_{\mathsf{H}}\rangle$ is the computational basis state of $\mathsf{D_H}$. Line 24 of Fig. 10 indicates that in the unitary operator, the tuple (m', ρ, k) is now obtained from the value stored at position r in $\mathsf{D_H}$, rather than from computing $H(r)$. Since

$$\begin{aligned} \Gamma_{S_{(c_1,m),r}} &= \max_{(c_1,m),r}\big|\{(m', \rho, k) \mid m' = m \wedge \mathsf{Enc}'(\mathsf{pk}, m'; \rho) = c_1\}\big| \\ &\le \max_{(c_1,m),r}\big|\{(m', \rho, k) \mid m' = m\}\big| \\ &= |\{m\} \times \mathcal{R} \times \mathcal{K}| = |\mathcal{R} \times \mathcal{K}|\ , \end{aligned}$$

by the Switching Lemma (Lemma 5.1), we have

$$|\Pr[W_1] - \Pr[W_0]| \le 6q_D\sqrt{\Gamma_{S_{(c_1,m),r}}/|\mathcal{M} \times \mathcal{R} \times \mathcal{K}|} = 6q_D\sqrt{1/|\mathcal{M}|}\ .$$

Game G_2. Observe that in Game G_1, the decapsulation oracle can be viewed as follows: given a query $(c_1, c_2) \neq (c_1^*, c_2^*)$ and the database register $\mathsf{D_H}$, after computing $m := \mathsf{Dec}'(\mathsf{sk}, c_1)$ $(\neq \perp)$ and $r := c_2 \oplus F(m)$, it applies a unitary operator

$$|(c_1, m, r), D_{\mathsf{H}}, 0^{\log|\mathcal{K}|}\rangle \mapsto |(c_1, m, r), D_{\mathsf{H}}, 0^{\log|\mathcal{K}|} \oplus f\big((c_1, m, r), D_{\mathsf{H}}\big)\rangle$$

where

$$f\big((c_1, m, r), D_{\mathsf{H}}\big) = \begin{cases} D_{\mathsf{H}}(r)[2], & \text{if } m' = m \text{ and } \mathsf{Enc}'(\mathsf{pk}, m'; \rho) = c_1 \\ & \text{where } (m', \rho, k) := D_{\mathsf{H}}(r)\ , \\ \perp, & \text{otherwise}\ , \end{cases}$$

and then measures the last register in the computational basis to obtain the response of O_{Dec}. Note that by the definition of f, we have

$$f\big((c_1, m, r), D_{\mathsf{H}}\big) = f\big((c_1, m, r), D_{\mathsf{H}} \cup (r^*, (m^*, \rho^*, k_0))\big)$$

for every D_{H} with $D_{\mathsf{H}}(r^*) = \perp$. Otherwise, this would imply the existence of some $(c_1, c_2) \neq (c_1^*, c_2^*)$ such that $r^* = c_2 \oplus F(m^*)$ and $\mathsf{Enc}'(\mathsf{pk}, m^*; \rho^*) = c_1$, which contradicts $(c_1, c_2) \neq (c_1^*, c_2^*)$. Consequently, the function f here satisfies the property required for f_{x^*,y^*} in the Puncturing Lemma (Lemma 5.3), with $x^* := r^*$ and $y^* := (m^*, \rho^*, k_0)$.

In this game, we sample (m^*, ρ^*, k_0) uniformly at random (as shown in line 4 of Fig. 10) and modify the behavior of the random oracle H to

$$|x, y, D_{\mathsf{H}}\rangle \mapsto \begin{cases} |x, y \oplus (m^*, \rho^*, k_0), D_{\mathsf{H}}\rangle\ , & \text{if } x = r^*\ , \\ H\,|x, y, D_{\mathsf{H}}\rangle\,, & \text{otherwise}\ . \end{cases}$$

That is, we add line 15 in Fig. 10.

Note that this modification is a direct application of the Puncturing Lemma. Therefore, by Lemma 5.3, we have

$$|\Pr[W_2] - \Pr[W_1]| \le 2q_D/\sqrt{|\mathcal{M} \times \mathcal{R} \times \mathcal{K}|}\ .$$

Game G_3. In this game, we simulate the random oracle F using a compressed oracle $\mathsf{F_c}$ (as shown in line 13 of Fig. 10), with a database register $\mathsf{D_F}$. In the decapsulation oracle, after computing $m := \mathsf{Dec}'(\mathsf{sk}, c_1)$ $(\neq \bot)$, the behavior of the decapsulation oracle in Game G_2 can be viewed as first applying a unitary operator

$$|(c_1, c_2, D_{\mathsf{H}}), m, 0^{\log|\mathcal{K}|}\rangle \mapsto \begin{cases} |(c_1, c_2, D_{\mathsf{H}}), m, 0^{\log|\mathcal{K}|} \oplus D_{\mathsf{H}}(c_2 \oplus F(m))[2]\rangle\,, & \\ \qquad\qquad\qquad\qquad\qquad \text{if } F(m) \in S_{(c_1,c_2,D_{\mathsf{H}}),m}, & \\ |(c_1, c_2, D_{\mathsf{H}}), m, 0^{\log|\mathcal{K}|} \oplus \bot\rangle\,, \qquad\qquad \text{otherwise}\,, & \end{cases}$$

where $S_{(c_1,c_2,D_{\mathsf{H}}),m} := \{u \mid (m', \rho, k) := D_{\mathsf{H}}(c_2 \oplus u) \text{ s.t. } m' = m \wedge \mathsf{Enc}'(\mathsf{pk}, m'; \rho) = c_1\}$, and then measuring the last register in the computational basis to obtain the classical response of O_{Dec}. In this game, we replace $F(m)$ in this unitary with $D_{\mathsf{F}}(m)$, where $|D_{\mathsf{F}}\rangle$ is the computational basis state of $\mathsf{D_F}$. We indicate this modification in Fig. 10 by using $r := c_2 \oplus \mathsf{D_F}(m)$ (line 22) instead of $r := c_2 \oplus F(m)$ (line 21). Because

$$\begin{aligned} \Gamma_{S_{(c_1,c_2,D_{\mathsf{H}}),m}} &= \max\nolimits_{(c_1,c_2,D_{\mathsf{H}}),m} \left|S_{(c_1,c_2,D_{\mathsf{H}}),m}\right| \\ &\leq \max\nolimits_{(c_1,c_2,D_{\mathsf{H}}),m} |\{u \mid D_{\mathsf{H}}(c_2 \oplus u) \neq \bot\}| \leq q_H + 1 \end{aligned}$$

(where q_H is the number of queries made by $\mathcal{A}$ to the random oracle H, and the extra 1 comes from the challenger's generation of the challenge key), applying the Switching Lemma (Lemma 5.1), we have

$$|\Pr[W_3] - \Pr[W_2]| \leq 6q_D\sqrt{\Gamma_{S_{(c_1,c_2,D_{\mathsf{H}}),m}}/2^n} \leq 6q_D\sqrt{(q_H + 1)/2^n}\ .$$

Game G_4. In Game G_3, the decapsulation oracle can be viewed as follows: given a query $(c_1, c_2) \neq (c_1^*, c_2^*)$ and given the database registers $\mathsf{D_F}$ and $\mathsf{D_H}$, it applies the unitary operator

$$|(c_1, c_2, D_{\mathsf{F}}), D_{\mathsf{H}}, 0^{\log|\mathcal{K}|}\rangle \mapsto |(c_1, c_2, D_{\mathsf{F}}), D_{\mathsf{H}}, 0^{\log|\mathcal{K}|} \oplus f_0\big((c_1, c_2, D_{\mathsf{F}}), D_{\mathsf{H}}\big)\rangle\ ,$$

with

$$f_0\big((c_1, c_2, D_{\mathsf{F}}), D_{\mathsf{H}}\big) = \begin{cases} D_{\mathsf{H}}(r)[2], & \text{if } (r, D_{\mathsf{H}}(r)) \in S_{(c_1,c_2,D_{\mathsf{F}})}\,, \\ & \text{where } r = c_2 \oplus D_{\mathsf{F}}(\mathsf{Dec}'(\mathsf{sk}, c_1))\ , \\ \bot, & \text{otherwise,} \end{cases}$$

for $S_{(c_1,c_2,D_{\mathsf{F}})} := \{(r, (m', \rho, k)) \mid m' = \mathsf{Dec}'(\mathsf{sk}, c_1) \wedge \mathsf{Enc}'(\mathsf{pk}, m'; \rho) = c_1\}$, and then measures the last register in the computational basis to obtain the classical response of O_{Dec}. In this game, we replace f_0 with

$$f_1\big((c_1, c_2, D_{\mathsf{F}}), D_{\mathsf{H}}\big) = \begin{cases} D_{\mathsf{H}}(r)[2], & \text{if } \exists (r, D_{\mathsf{H}}(r)) \in S'_{(c_1,c_2,D_{\mathsf{F}})}, \\ & \text{and } \forall r' < r, (r', D_{\mathsf{H}}(r')) \notin S'_{(c_1,c_2,D_{\mathsf{F}})}\ , \\ \bot, & \text{otherwise}\ , \end{cases}$$

where $S'_{(c_1,c_2,D_{\mathsf{F}})} := \{(r, (m', \rho, k)) \mid r \oplus D_{\mathsf{F}}(m') = c_2 \wedge \mathsf{Enc}'(\mathsf{pk}, m'; \rho) = c_1\}$.

According to the Lookup Lemma (Lemma 5.2), we can first define a set

$$T_{(c_1,c_2,D_{\mathsf{F}})} := \{(r, (m', \rho, k)) \in S_{c_1,c_2,D_{\mathsf{F}}} \mid r = c_2 \oplus D_{\mathsf{F}}(\mathsf{Dec}'(\mathsf{sk}, c_1))\}$$

for each $(c_1, c_2, D_{\mathsf{F}})$, and then define the binary relation

$$R := \big\{\big(r, (m', \rho, k)\big) \mid \exists (c_1, c_2, D_{\mathsf{F}}) \text{ s.t. } (r, (m', \rho, k)) \in T_{(c_1,c_2,D_{\mathsf{F}})} \triangle S'_{(c_1,c_2,D_{\mathsf{F}})}\big\}\ .$$

For any $(r, (m', \rho, k)) \in T_{(c_1,c_2,D_{\mathsf{F}})}$, it necessarily holds that $r = c_2 \oplus D_{\mathsf{F}}(m')$ and $\mathsf{Enc}'(\mathsf{pk}, m'; \rho) = c_1$. This implies $(r, (m', \rho, k)) \in S'_{(c_1,c_2,D_{\mathsf{F}})}$; hence $T_{(c_1,c_2,D_{\mathsf{F}})} \setminus S'_{(c_1,c_2,D_{\mathsf{F}})} = \emptyset$. On the other hand, for any $(r,(m',\rho,k)) \in S'_{(c_1,c_2,D_\mathsf{F})}$ with $(r,(m',\rho,k)) \notin T_{(c_1,c_2,D_\mathsf{F})}$, we must have $\mathsf{Dec}'(\mathsf{sk}, \mathsf{Enc}'(\mathsf{pk}, m';\rho)) \neq m'$. Therefore, we have

$$\begin{aligned}\Gamma_R &= \max_r \left|\{(m',\rho,k) \mid (r,(m',\rho,k)) \in R\}\right| \\ &\le \max_r \left|\{(m',\rho,k) \mid \exists (c_1,c_2,D_\mathsf{F}) \text{ s.t. } (r,(m',\rho,k)) \in T_{(c_1,c_2,D_\mathsf{F})} \setminus S'_{(c_1,c_2,D_\mathsf{F})}\}\right| \\ &\quad + \max_r \left|\{(m',\rho,k) \mid \exists (c_1,c_2,D_\mathsf{F}) \text{ s.t. } (r,(m',\rho,k)) \in S'_{(c_1,c_2,D_\mathsf{F})} \setminus T_{(c_1,c_2,D_\mathsf{F})}\}\right| \\ &\le \left|\{(m',\rho) \mid \mathsf{Dec}'(\mathsf{sk}, \mathsf{Enc}'(\mathsf{pk}, m';\rho)) \neq m'\}\right| \cdot |\mathcal{K}|.\end{aligned}$$

For a given key pair $(\mathsf{pk}, \mathsf{sk})$, define $\delta(\mathsf{pk}, \mathsf{sk}) := \Pr[\mathsf{Dec}'(\mathsf{sk}, \mathsf{Enc}'(\mathsf{pk}, m;\rho)) \neq m]$, where the probability is taken over $(m,\rho) \leftarrow\$\, \mathcal{M} \times \mathcal{R}$. Then, we have

$$\frac{\Gamma_R}{|\mathcal{M} \times \mathcal{R} \times \mathcal{K}|} \le \frac{\left|\{(m',\rho) \mid \mathsf{Dec}'(\mathsf{sk}, \mathsf{Enc}'(\mathsf{pk}, m';\rho)) \neq m'\}\right|}{|\mathcal{M} \times \mathcal{R}|} \le \delta(\mathsf{pk}, \mathsf{sk}) \ .$$

By the Lookup Lemma (Lemma 5.2), we can obtain

$$\begin{aligned}|\Pr[W_4 : (\mathsf{pk}, \mathsf{sk})] - \Pr[W_3 : (\mathsf{pk}, \mathsf{sk})]| &\le 16(q_H+1)\sqrt{\Gamma_R / |\mathcal{M} \times \mathcal{R} \times \mathcal{K}|} \\ &\le 16(q_H+1)\sqrt{\delta(\mathsf{pk}, \mathsf{sk})} \ .\end{aligned}$$

Averaging over $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$, we obtain

$$\begin{aligned}|\Pr[W_4] - \Pr[W_3]| &\le 16(q_H+1)\mathbb{E}\left[\sqrt{\delta(\mathsf{pk}, \mathsf{sk})}\right] \\ &\overset{(a)}{\le} 16(q_H+1)\sqrt{\mathbb{E}\left[\delta(\mathsf{pk}, \mathsf{sk})\right]} \overset{(b)}{=} 16(q_H+1)\sqrt{\delta_{ac}} \ ,\end{aligned}$$

where (a) follows from Jensen's inequality, and (b) holds by the definition of the δ_{ac}-average-case correctness of PKE'.

Note that from this game onward, O_Dec can be simulated without sk.

Game G_5. We first define a set

$$S := \{D_\mathsf{H} \mid \exists (r,(m',\rho,k)) \in D_\mathsf{H} \text{ such that } m' = m^*\} \ .$$

In this game, after each query to the random oracle H, we immediately query the compressed semi-classical oracle $\mathcal{O}_S^{CSC}$ (as defined in Definition 3.9) on the database register $\mathsf{D_H}$ of $\mathsf{H_c}$ (as in Line 17 of Fig. 10). This effectively checks, for each computational basis state $|D_\mathsf{H}\rangle$, whether D_H contains an entry $(r,(m',\rho,k))$ with $m' = m^*$. Let Find be the event that $\mathcal{O}_S^{\mathrm{CSC}}$ ever outputs 1. By the Compressed Semi-Classical One-Way to Hiding Lemma (Lemma 3.4), we obtain[3]

$$\left|\Pr[W_5 \wedge \overline{\mathsf{Find}}] - \Pr[W_4]\right| \le \sqrt{q_H(q_H+1)\mathbb{E}[\|[J_S, \mathsf{CStO}]\|^2]}$$

where $J_S := \sum_{D \in S} |D\rangle\langle D|$, CStO is the unitary implementing $\mathsf{H_c}$, and the expectation is taken over the randomness of S and the input to $\mathcal{A}$. Next, we define a binary relation

$$R := \{(r,(m',\rho,k)) \mid m' = m^*\} \ .$$

Clearly, $\Gamma_R := \max_r |\{(m',\rho,k) | (r,(m',\rho,k)) \in R\}| = |\{m^*\} \times \mathcal{R} \times \mathcal{K}| = |\mathcal{R} \times \mathcal{K}|$.

Define the projector $\Sigma^\perp := \sum_{D_\mathsf{H} : \nexists (r, D_\mathsf{H}(r)) \in R} |D_\mathsf{H}\rangle\langle D_\mathsf{H}|$. It is evident that $I - J_S = \Sigma^\perp$, where $I := \sum_{D_\mathsf{H}} |D_\mathsf{H}\rangle\langle D_\mathsf{H}|$. Consequently,

$$\|[J_S, \mathsf{CStO}]\| \overset{(c)}{=} \|[I - J_S, \mathsf{CStO}]\| = \left\|[\Sigma^\perp, \mathsf{CStO}]\right\| \overset{(d)}{\le} 8\sqrt{\frac{\Gamma_R}{|\mathcal{M} \times \mathcal{R} \times \mathcal{K}|}} = 8\sqrt{\frac{1}{|\mathcal{M}|}} \ ,$$

[3] When applying Lemma 3.4, we can view $\mathcal{A}$ as outputting 1 if $b = \hat{b} \wedge \overline{\mathsf{Find}}$, and 0 otherwise. By the definition of Find, if the compressed semi-classical oracle $\mathcal{O}_S^{CSC}$ is never queried, Find never occurs.

where (c) follows from basic properties of commutators, and (d) follows from [15, Lemma 4]. Therefore, we have

$$\left|\Pr[W_5 \wedge \overline{\mathsf{Find}}] - \Pr[W_4]\right| \leq 8(q_H + 1)\sqrt{1/|\mathcal{M}|} \ .$$

Game G_6. When $\overline{\mathsf{Find}}$ occurs, in the computational basis every component $|D_{\mathsf{H}}\rangle$ of the state in register $\mathsf{D_H}$ is a database D_{H} that contains *no* entry $(r, (m', \rho, k))$ with $m' = m^*$. Hence, by the definition of the function f_1 used in O_{Dec}, we have

$$f_1(c_1, c_2, D_{\mathsf{F}}, D_{\mathsf{H}}) = f_1(c_1, c_2, D_{\mathsf{F}} \cup \{(m^*, u^*)\}, D_{\mathsf{H}}) \text{ for every } D_{\mathsf{F}} \text{ with } D_{\mathsf{F}}(m^*) = \perp.$$

Thus, the conditions of the Puncturing Lemma is satisfied. We therefore sample u^* uniformly (line 6 of Fig. 10) and modify the behavior of the random oracle F to

$$|x, y, D_{\mathsf{F}}\rangle \mapsto \begin{cases} |x, y \oplus u^*, D_{\mathsf{F}}\rangle \,, & \text{if } x = m^* \ , \\ F\,|x, y, D_{\mathsf{F}}\rangle \,, & \text{otherwise} \ . \end{cases}$$

By Lemma 5.3 we obtain

$$\left|\Pr[W_6 \wedge \overline{\mathsf{Find}}] - \Pr[W_5 \wedge \overline{\mathsf{Find}}]\right| \leq 2q_D/\sqrt{2^n} \ .$$

Game G_7. In this game, we remove the queries to the semi-classical oracle $\mathcal{O}_S^{CSC}$ that followed each call to the random oracle H (i.e., we delete line 17 of Fig. 10). This effectively reverts the modification made in Game G_5 relative to Game G_4. Repeating the same analysis as in Game G_5 and applying the Compressed Semi-Classical One-Way to Hiding Lemma (Lemma 3.4), we obtain

$$\left|\Pr[W_7] - \Pr[W_6 \wedge \overline{\mathsf{Find}}]\right| \leq 8(q_H + 1)\sqrt{1/|\mathcal{M}|} \ .$$

Game G_8. This game is identical to Game G_7, except that we restore the random oracles F and H to be simulated *solely* by the compressed oracles $\mathsf{F_c}$ and $\mathsf{H_c}$, respectively. That is, we remove lines 12 and 15 from Fig. 10, which is equivalent to reprogramming the random oracles F at m^* and H at r^* simultaneously. To capture both reprogrammings in a unified manner, we define a combined random oracle $G : \mathcal{M} \times \{0,1\}^n \to \{0,1\}^n \times (\mathcal{M} \times \mathcal{R} \times \mathcal{K})$ as

$$G(x) := \begin{cases} F(x), & \text{if } x \in \mathcal{M} \ , \\ H(x), & \text{if } x \in \{0,1\}^n \ , \end{cases}$$

where F and H are simulated by the compressed oracles $\mathsf{F_c}$ and $\mathsf{H_c}$, respectively, as in Game G_8. Let G' be another oracle such that $G'(x) = G(x)$ for all $x \notin \{m^*, r^*\}$, while $G'(m^*) = u^*$ and $G'(r^*) = (m^*, \rho^*, k_0)$, matching the behavior in Game G_7. Now, we can construct two oracle algorithms $A^{|G\rangle, O_{\mathsf{Dec}}}(z)$ and $A^{|G'\rangle, O_{\mathsf{Dec}}}(z)$ to simulate Games G_8 and G_7, respectively, and then use the Semi-Classical One-Way to Hiding Lemma (Lemma 3.2) to bound $|\Pr[W_8] - \Pr[W_7]|$.

Let $z := (\mathsf{pk}, c_1^*, c_2^*, k_0, k_1, b)$, where $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$, $r^*, u^* \leftarrow\!\!\$ \ \{0,1\}^n$, $(m^*, \rho^*, k_0) \leftarrow\!\!\$ \ \mathcal{M} \times \mathcal{R} \times \mathcal{K}$, $c_1^* := \mathsf{Enc}'(\mathsf{pk}, m^*; \rho^*)$, $c_2^* := r^* \oplus u^*$, $k_1 \leftarrow\!\!\$ \ \mathcal{K}$, and $b \leftarrow\!\!\$ \ \{0,1\}$. The algorithm $A^{|G\rangle, O_{\mathsf{Dec}}}(z)$ (resp. $A^{|G'\rangle, O_{\mathsf{Dec}}}(z)$) runs $\hat{b} \leftarrow\!\!\$ \ \mathcal{A}^{|F\rangle, |H\rangle, O_{\mathsf{Dec}}}(\mathsf{pk}, c_1^*, c_2^*, k_b)$ and outputs $[b = \hat{b}]$, where A uses the random oracle G (resp. G') to answer $\mathcal{A}$'s queries to F and H, and O_{Dec} behaves as in Game G_7. Clearly, we have

$$\Pr[1 \leftarrow A^{|G\rangle, O_{\mathsf{Dec}}}(z)] = \Pr[W_8] \text{ and } \Pr[1 \leftarrow A^{|G'\rangle, O_{\mathsf{Dec}}}(z)] = \Pr[W_7] \ .$$

By the Semi-Classical One-Way to Hiding lemma (Lemma 3.2), we have[4]

$$\begin{aligned} &\left|\Pr[1 \leftarrow A^{|G\rangle, O_{\mathsf{Dec}}}(z)] - \Pr[1 \leftarrow A^{|G'\rangle, O_{\mathsf{Dec}}}(z)]\right| \\ &\qquad \leq 2\sqrt{(q_F + q_H + 1)\Pr[\mathsf{Find}_{m^*, r^*} : A^{|G \setminus \{m^*, r^*\}\rangle, O_{\mathsf{Dec}}}(z)]} \ , \end{aligned}$$

[4] When applying Lemma 3.2, O_{Dec} is treated as internal oracle of A, and the total number of queries to the combined oracle G made by $A^{|G\rangle, O_{\mathsf{Dec}}}(z)$ is $q_F + q_H$, comprising $\mathcal{A}$'s q_F queries to F and q_H queries to H.

Games G_9 to G_{10}:	Punctured Oracle $F\backslash\{m^*\}(\|x,y\rangle)$:
$(\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda),\ r^* \leftarrow\!\!\$\ \{0,1\}^n$	**perform** $\mathcal{O}^{SC}_{\{m^*\}}$ // G_9, G_{10}
$(m^*,\rho^*,k_0) \leftarrow\!\!\$\ \mathcal{M}\times\mathcal{R}\times\mathcal{K}$	**perform** $\mathcal{O}^{SC}_{[\mathsf{Enc}'(\mathsf{pk},\cdot)=c_1^*]}$ // G_9'
$u^* \leftarrow\!\!\$\ \{0,1\}^n,\ \hat{m}^* \leftarrow\!\!\$\ \mathcal{M}$	**perform** $\mathsf{F_c}$
$c_1^* := \mathsf{Enc}'(\mathsf{pk}, m^*;\rho^*)$	Punctured Oracle $H\backslash\{r^*\}(\|x,y\rangle)$:
$c_1^* := \mathsf{Enc}'(\mathsf{pk}, \hat{m}^*;\rho^*)$ // G_{10}	**perform** $\mathcal{O}^{SC}_{\{r^*\}}$
$c_2^* := r^* \oplus u^*$	**perform** $\mathsf{H_c}$
$b \leftarrow\!\!\$\ \{0,1\},\ k_1 \leftarrow\!\!\$\ \mathcal{K}$	Decapsulation Oracle $O_{\mathsf{Dec}}(c_1,c_2)$:
$\hat{b} \leftarrow \mathcal{A}^{\|F\backslash\{m^*\}\rangle,\|H\backslash\{r^*\}\rangle,O_{\mathsf{Dec}}}(\mathsf{pk},(c_1^*,c_2^*), k_b)$	**if** $(c_1,c_2)=(c_1^*,c_2^*)$ **then** **return** $\perp$
if Find_{m^*} **or** Find_{r^*} **then** : **return** 1 **else** : **return** 0	**return** $f_1(c_1,c_2,\mathsf{D_F},\mathsf{D_H})$

Figure 11: Games G_9 to G_{10} for the proof of Theorem 5.4.

where the notation $A^{|G\backslash\{m^*,r^*\}\rangle,O_{\mathsf{Dec}}}(z)$ means that before each query to the oracle G, A first applies a semi-classical oracle $\mathcal{O}^{SC}_{\{m^*,r^*\}}$ (as defined in Definition 3.8) to its input register, and then proceeds with the usual query to G; Find_{m^*,r^*} is the classical event that the measurement of $\mathcal{O}^{SC}_{\{m^*,r^*\}}$ ever outputs 1 (as defined in Definition 3.8).

Note that in Game G_8, the values k_0 and k_1 are sampled independently and uniformly at random, and are independent of all oracle responses and other public information. Hence, the bit b is uniformly random and independent of $\mathcal{A}$'s view, yielding

$$\Pr[W_8] = 1/2\ .$$

Next, we define games G_j (for $j = 9, 10$) played by the challenger against the adversary $\mathcal{A}$, as shown in Fig. 11.

Game G_9. This game is defined in Fig. 11. In this game, before each query to the random oracle F (resp. H), the adversary $\mathcal{A}$ first applies the semi-classical oracle $\mathcal{O}^{SC}_{\{m^*\}}$ (resp. $\mathcal{O}^{SC}_{\{r^*\}}$) to its input register, and then proceeds with the usual query to F (resp. H). Let $\mathsf{Find}_{\{m^*\}}$ (resp. $\mathsf{Find}_{\{r^*\}}$) denote the event that the measurement of $\mathcal{O}^{SC}_{\{m^*\}}$ (resp. $\mathcal{O}^{SC}_{\{r^*\}}$) ever outputs 1 (as defined in Definition 3.8). Note that the challenger samples a random $\hat{m}^* \leftarrow\!\!\$\ \mathcal{M}$, but it is not used within this game. The rest of the challenger's behavior is identical to that in Game G_8. From the definitions of the oracles G and $G\backslash\{m^*,r^*\}$ and the events Find_{m^*,r^*} in Game G_8, we can see that

$$\Pr[\mathsf{Find}_{m^*} \vee \mathsf{Find}_{r^*} : \mathcal{A}^{|F\backslash\{m^*\}\rangle,|H\backslash\{r^*\}\rangle,O_{\mathsf{Dec}}}(\mathsf{pk},(c_1^*,c_2^*),k_b)] \\ = \Pr[\mathsf{Find}_{m^*,r^*} : A^{|G\backslash\{m^*,r^*\}\rangle,O_{\mathsf{Dec}}}(z)]\ .$$

Note that in Game G_9, the value r^* is used only to compute $c_2^* := r^* \oplus u^*$, which can be viewed as a one-time pad encryption of r^* using u^*. Since u^* is chosen uniformly at random by the challenger and never used elsewhere, c_2^* perfectly hides r^*. Since r^* is independent of all oracles except $H\backslash\{r^*\}$ and other public values, by Lemma 3.3, we have

$$\Pr[\mathsf{Find}_{r^*} : \mathcal{A}^{|F\backslash\{m^*\}\rangle,|H\backslash\{r^*\}\rangle,O_{\mathsf{Dec}}}(\mathsf{pk},(c_1^*,c_2^*),k_b)] \le 4q_H/2^n\ .$$

Let Z_j be the event that Find_{m^*} occurs in Game G_j. Then, we have

$$\begin{aligned}
&\Pr[\mathsf{Find}_{m^*} \vee \mathsf{Find}_{r^*} : \mathcal{A}^{|F\backslash\{m^*\}\rangle,|H\backslash\{r^*\}\rangle,O_{\mathsf{Dec}}}(\mathsf{pk},(c_1^*,c_2^*),k_b)] \\
\leq\ &\Pr[\mathsf{Find}_{m^*} : \mathcal{A}^{|F\backslash\{m^*\}\rangle,|H\backslash\{r^*\}\rangle,O_{\mathsf{Dec}}}(\mathsf{pk},(c_1^*,c_2^*),k_b)] \\
&\qquad\qquad + \Pr[\mathsf{Find}_{r^*} : \mathcal{A}^{|F\backslash\{m^*\}\rangle,|H\backslash\{r^*\}\rangle,O_{\mathsf{Dec}}}(\mathsf{pk},(c_1^*,c_2^*),k_b)] \\
\leq\ &\Pr[Z_9] + 4q_H/2^n \ .
\end{aligned}$$

Next, we bound $\Pr[Z_9]$. Let $B^{|F\rangle,|H\backslash\{r^*\}\rangle,O_{\mathsf{Dec}}}(z')$ be the oracle algorithm defined as follows: on input $z' := (\mathsf{pk},(c_1^*,c_2^*),k_b)$, it picks a random $i^* \leftarrow\$ \{1,\ldots,q_F\}$, runs the adversary $\mathcal{A}^{|F\rangle,|H\backslash\{r^*\}\rangle,O_{\mathsf{Dec}}}(z')$ until (just before) its i^*-th query to the random oracle F, measures the input register in the computational basis, and outputs the measurement outcome. Here, F behaves the same as $F\backslash\{m^*\}$ except that the semi-classical oracle $\mathcal{O}_{m^*}^{SC}$ is not applied. By Lemma 3.3, we have

$$\Pr[Z_9] \leq 4q_F \cdot \Pr[\hat{m} = m^* : \hat{m} \leftarrow B^{|F\rangle,|H\backslash\{r^*\}\rangle,O_{\mathsf{Dec}}}(z')] \ .$$

Since the simulation of all oracles for B does not require (m^*,ρ^*) nor sk, we can argue that whenever the event $\hat{m} = m^*$ occurs, we can construct an adversary $\mathcal{B}$ that breaks the OW-CPA security of the underlying PKE'. Concretely, upon receiving (pk,c^*) from its OW-CPA challenger, $\mathcal{B}$ sets $c_1^* := c^*$, generates c_2^*, k_0, k_1, b as in Game G_9, runs $\hat{m} \leftarrow B^{|F\rangle,|H\backslash\{r^*\}\rangle,O_{\mathsf{Dec}}}(z')$ with input $z' := (\mathsf{pk},(c_1^*,c_2^*),k_b)$, and outputs $\hat{m}$ as the solution to the OW-CPA challenge (pk,c^*). Consequently, we have

$$\Pr[\hat{m} = m^* : \hat{m} \leftarrow B^{|F\rangle,|H\backslash\{r^*\}\rangle,O_{\mathsf{Dec}}}(z')] \leq \mathsf{Adv}_{\mathsf{PKE}'}^{\text{OW-CPA}}(\mathcal{B}) \ .$$

Therefore,

$$\Pr[Z_9] \leq 4q_F \cdot \mathsf{Adv}_{\mathsf{PKE}'}^{\text{OW-CPA}}(\mathcal{B}) \ .$$

Putting everything together, we obtain

$$\begin{aligned}
\mathsf{Adv}_{\mathsf{KEM}}^{\text{IND-CCA}}(\mathcal{A}) &\leq 2\sqrt{(q_F+q_H+1)(4q_F\mathsf{Adv}_{\mathsf{PKE}'}^{\text{OW-CPA}}(\mathcal{B})+\frac{4q_H}{2^n})} + 16(q_H+1)\sqrt{\delta_{ac}} \\
&\quad + \frac{6q_D + 16(q_H+1)}{\sqrt{|\mathcal{M}|}} + \frac{2q_D}{\sqrt{|\mathcal{M}\times\mathcal{R}\times\mathcal{K}|}} + \frac{6q_D\sqrt{q_H+1}+2q_D}{\sqrt{2^n}} \\
&\leq 4(q_F+q_H+1)\sqrt{\mathsf{Adv}_{\mathsf{PKE}'}^{\text{OW-CPA}}(\mathcal{B})} + 16(q_H+1)\sqrt{\delta_{ac}} \\
&\quad + 16(q_D+q_H+1)/\sqrt{|\mathcal{M}|} + 8(q_D+q_H)\sqrt{q_F+q_H+1}/\sqrt{2^n} \ .
\end{aligned}$$

When PKE' is deterministic (so that $\mathcal{R} = \emptyset$), the same reduction essentially remains valid. We can further replace the semi-classical oracle $\mathcal{O}_{\{m^*\}}^{SC}$ in the punctured oracle $F\backslash\{m^*\}$ with $\mathcal{O}_{[\mathsf{Enc}'(\mathsf{pk},\cdot)=c_1^*]}^{SC}$, i.e., a semi-classical oracle defined via the indicator function $f(\cdot) := [\mathsf{Enc}'(\mathsf{pk},\cdot) = c_1^*]$ (see Definition 3.8). We denote this modified game by G_9', as illustrated in Figure 11. Clearly, $\mathcal{O}_{\{m^*\}}^{SC}$ and $\mathcal{O}_{[\mathsf{Enc}'(\mathsf{pk},\cdot)=c_1^*]}^{SC}$ are equivalent unless for the randomly sampled m^* there exists a different $m \in \mathcal{M}$ such that $\mathsf{Enc}'(\mathsf{pk},m) = \mathsf{Enc}'(\mathsf{pk},m^*)$. For a deterministic PKE', the probability of such a collision is at most $2\delta_{ac}$ according to [16, Lemma G1]. Hence,

$$\left|\Pr[Z_9'] - \Pr[Z_9]\right| \leq 2\delta_{ac} \ .$$

Here, Z_9' is the event that the measurement of $\mathcal{O}_{[\mathsf{Enc}'(\mathsf{pk},\cdot)=c_1^*]}^{SC}$ ever outputs 1 in Game G_9'.

Since in Game G_9' the simulation of all oracles for $\mathcal{A}$ does not require m^* nor sk, upon receiving (pk, c^*) from its OW-CPA challenger, the adversary $\mathcal{B}$ can set $c_1^* := c^*$, obtain c_2^*, k_b exactly as the challenger does in Game G_9', and then simulate the oracles as in Game G_9' while interacting with $\mathcal{A}^{|F\backslash\{m^*\}\rangle, |H\backslash\{r^*\}\rangle, O_{\mathsf{Dec}}}(\mathsf{pk},\ (c_1^*, c_2^*), k_b)$. Whenever Z_9' occurs, i.e., the measurement of $\mathcal{O}^{SC}_{[\mathsf{Enc}'(\mathsf{pk},\cdot)=c_1^*]}$ outputs 1, $\mathcal{B}$ can immediately measure $\mathcal{A}$'s input register to obtain $\hat{m} \in \mathcal{M}$ as the solution to the OW-CPA challenge. In this case, $\hat{m} = m^*$ unless for the randomly sampled m^* there exists a different $m \in \mathcal{M}$ with $\mathsf{Enc}'(\mathsf{pk}, m) = \mathsf{Enc}'(\mathsf{pk}, m^*)$. This probability is at most $2\delta_{ac}$ according to [16, Lemma G1]. Therefore,

$$\Pr[Z_9'] \leq \mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{dPKE}'}(\mathcal{B}) + 2\delta_{ac} \ .$$

Putting all the bounds together, we have

$$\begin{aligned}
\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) &\leq 2\sqrt{(q_F+q_H+1)(\mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{dPKE}'}(\mathcal{B})+4\delta_{ac}+\frac{4q_H}{2^n})}+16(q_H+1)\sqrt{\delta_{ac}} \\
&\quad + \frac{6q_D+16(q_H+1)}{\sqrt{|\mathcal{M}|}} + \frac{2q_D}{\sqrt{|\mathcal{M}\times\mathcal{R}\times\mathcal{K}|}} + \frac{6q_D\sqrt{q_H+1}+2q_D}{\sqrt{2^n}} \\
&\leq 2\sqrt{(q_F+q_H+1)\mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{dPKE}'}(\mathcal{B})} + 20(q_F+q_H+1)\sqrt{\delta_{ac}} \\
&\quad + 16(q_D+q_H+1)/\sqrt{|\mathcal{M}|} + 8(q_D+q_H)\sqrt{q_F+q_H+1}/\sqrt{2^n} \ .
\end{aligned}$$

Game G_{10}. We now further reduce the advantage to the IND-CPA security of PKE'. This game is identical to Game G_9, except that the challenger uses a fresh random message $\hat{m}^*$ to compute $c_1^* := \mathsf{Enc}'(\mathsf{pk}, \hat{m}^*; \rho^*)$, as shown in line 5 of Fig. 11. Since m^* is never used in the game except within the punctured oracle $F\backslash\{m^*\}$, by Lemma 3.3, we have

$$\Pr[Z_{10}] \leq 4q_F/|\mathcal{M}| \ .$$

Note that in both Games G_9 and G_{10}, all oracles can be simulated without sk. We now argue that if $\mathcal{A}$ can distinguish between G_9 and G_{10}, we can construct an adversary $\mathcal{B}'$ that breaks the IND-CPA security of PKE'. Concretely, upon receiving pk from its IND-CPA challenger, $\mathcal{B}'$ first picks two random messages $m_0 := m^*$ and $m_1 := \hat{m}^*$ from $\mathcal{M}$, and sends (m_0, m_1) to the challenger. Upon receiving the challenge ciphertext $c^* = \mathsf{Enc}'(\mathsf{pk}, m_{b'})$ (where b' is the challenger's random bit), $\mathcal{B}'$ sets $c_1^* := c^*$, generates c_2^*, k_0, k_1, b as in Game G_9, and runs $\mathcal{A}^{|F\backslash\{m^*\}\rangle, |H\backslash\{r^*\}\rangle, O_{\mathsf{Dec}}}(\mathsf{pk}, (c_1^*, c_2^*), k_b)$. The oracles $F\backslash\{m^*\}$, $H\backslash\{r^*\}$, and O_{Dec} are simulated by $\mathcal{B}'$ exactly as in Game G_9 without using sk. Whenever the event Find_{m^*} occurs, $\mathcal{B}'$ outputs $\hat{b}' := 1$.

Note that if $b' = 0$, then $c_1^* = \mathsf{Enc}'(\mathsf{pk}, m^*)$, and $\mathcal{B}'$ behaves exactly like the challenger in Game G_9; if $b' = 1$, then $c_1^* = \mathsf{Enc}'(\mathsf{pk}, \hat{m}^*)$, and $\mathcal{B}'$ behaves exactly like the challenger in Game G_{10}. Thus, we have $\Pr[\hat{b}' = 1 | b' = 0] = \Pr[Z_9]$ and $\Pr[\hat{b}' = 1 | b' = 1] = \Pr[Z_{10}]$. Now, we have

$$\begin{aligned}
\mathsf{Adv}^{\text{IND-CPA}}_{\mathsf{PKE}'}(\mathcal{B}') &\geq \Big|\Pr[b' = \hat{b}'] - 1/2\Big| \\
&= \Big|\Pr[b' = \hat{b}' \wedge b' = 0] + \Pr[b' = \hat{b}' \wedge b' = 1] - 1/2\Big| \\
&= \frac{1}{2}\Big|\Pr[\hat{b}' = 0 | b' = 0] + \Pr[\hat{b}' = 1 | b' = 1] - 1\Big| \\
&= \frac{1}{2}\Big|\Pr[\hat{b}' = 1 | b' = 1] - \Pr[\hat{b}' = 1 | b' = 0]\Big| \\
&= \frac{1}{2}|\Pr[Z_{10}] - \Pr[Z_9]| \ .
\end{aligned}$$

Combining all bounds, we obtain

$$\begin{aligned}
\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) \leq & 2\sqrt{(q_F+q_H+1)(2\mathsf{Adv}^{\text{IND-CPA}}_{\mathsf{PKE}'}(\mathcal{B}')+\frac{4q_F}{|\mathcal{M}|}+\frac{4q_H}{2^n})+16(q_H+1)\sqrt{\delta_{ac}}} \\
& + \frac{6q_D + 16(q_H+1)}{\sqrt{|\mathcal{M}|}} + \frac{2q_D}{\sqrt{|\mathcal{M}\times\mathcal{R}\times\mathcal{K}|}} + \frac{6q_D\sqrt{q_H+1}+2q_D}{\sqrt{2^n}} \\
\leq & 2\sqrt{2(q_F+q_H+1)\mathsf{Adv}^{\text{IND-CPA}}_{\mathsf{PKE}'}(\mathcal{B}') + 16(q_H+1)\sqrt{\delta_{ac}}} \\
& + 20(q_D+q_H+q_F+1)/\sqrt{|\mathcal{M}|} + 8(q_D+q_H)\sqrt{q_F+q_H+1}/\sqrt{2^n}.
\end{aligned}$$

□

5.3 $\mathsf{KEM} := \mathsf{FOAC}'[\mathsf{PKE}', F, H]$

Theorem 5.5 (QROM Security of FOAC'). *Assume that $F : \mathcal{M}_1 \to \mathcal{M}_2$ and $H : \mathcal{M}_2 \to \mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}$ are modeled as quantum-accessible random oracles. For any efficient adversary $\mathcal{A}$ with quantum access to the random oracles F and H that attacks the* IND-CCA *security of* $\mathsf{KEM} := \mathsf{FOAC}'[\mathsf{PKE}', F, H]$*, there exist efficient adversaries $\mathcal{B}$ (resp. $\mathcal{B}'$) against the* OW-CPA *(resp.* IND-CPA*) security of* PKE'*, such that*

$$\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) \leq 4(q_F+q_H+1)^{1.5}\sqrt{\mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{PKE}'}(\mathcal{B})}+16(q_H+1)\sqrt{\delta_{ac}}+\frac{8q_D}{\sqrt{|\mathcal{M}_1|}}+\Delta,$$

$$\begin{aligned}
\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) \leq & 2\sqrt{2(q_H+1)}\sqrt{\mathsf{Adv}^{\text{IND-CPA}}_{\mathsf{PKE}'}(\mathcal{B}') + 16(q_H+1)\sqrt{\delta_{ac}}} \\
& + (8q_D+4(q_F+q_H+1)^{1.5})/\sqrt{|\mathcal{M}_1|} + \Delta,
\end{aligned}$$

where q_F, q_H, q_D are the bounded numbers of $\mathcal{A}$'s queries to the random oracles F, H, and the decapsulation oracle, respectively, and $\Delta \leq 24(q_H+1)/\sqrt{|\mathcal{M}_2|}$.

If PKE' *is deterministic, the bound can be improved to*

$$\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) \leq 2\sqrt{q_H+1}\sqrt{\mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{dPKE}'}(\mathcal{B})} + 20(q_H+1)\sqrt{\delta_{ac}} + \frac{8q_D}{\sqrt{|\mathcal{M}_1|}} + \Delta/2.$$

Theorem 5.5. We define a sequence of games G_j, played by the challenger and the adversary $\mathcal{A}$, as shown in Fig. 12 for $j = 0, \ldots, 4$. In each game, let b be the random bit sampled by the challenger and $\hat{b}$ the bit output by $\mathcal{A}$. Denote by W_j the event that $b = \hat{b}$ in game G_j.

Game G_0. At the beginning of the game, the challenger randomly samples two functions $\mathsf{F} \leftarrow\$ \Omega_F$ and $\mathsf{H} \leftarrow\$ \Omega_H$ to model the random oracles F and H, respectively. We can see that the behavior of the challenger in Game G_0 is identical to that in the IND-CCA game for $\mathsf{KEM} := \mathsf{FOAC}'[\mathsf{PKE}', F, H]$. Hence,

$$|\Pr[W_0] - 1/2| = \mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) .$$

Game G_1. This game first simulates the random oracle H using the compressed oracle $\mathsf{H_c}$ with database register $\mathsf{D_H}$ (as shown in line 13 of Fig. 12), and then modifies the decapsulation oracle. Given a query $c \neq c^*$, after computing $m_1 \| m_2 := \mathsf{Dec}'(\mathsf{sk}, c)$ $(\neq \perp)$ and $r := m_2 \oplus F(m_1)$, the behavior of the decapsulation oracle in Game G_0 is equivalent to first applying a unitary operator

$$|(c, m_1, m_2), r, 0^{\log|\mathcal{K}|}\rangle \mapsto \begin{cases} |(c, m_1, m_2), r, 0^{\log|\mathcal{K}|} \oplus H(r)[2]\rangle , & \\ & \text{if } H(r) \in S_{(c,m_1,m_2),r}, \\ |(c, m_1, m_2), r, 0^{\log|\mathcal{K}|} \oplus \perp\rangle , & \text{otherwise} , \end{cases}$$

Games G_0 to G_4:

1 : $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda),\ r^* \leftarrow\!\!\$\ \mathcal{M}_2$
2 : $\mathsf{F} \leftarrow\!\!\$\ \Omega_F,\ \mathsf{H} \leftarrow\!\!\$\ \Omega_H$
3 : $(m_1^*, \rho^*, k_0) := H(r^*)$ // G_0–G_1
4 : $(m_1^*, \rho^*, k_0) \leftarrow\!\!\$\ \mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}$ // G_2–G_4
5 : $m_2^* := r^* \oplus F(m_1^*)$
6 : $c^* := \mathsf{Enc}'(\mathsf{pk}, m_1^* \| m_2^*; \rho^*)$
7 : $b \leftarrow\!\!\$\ \{0,1\},\ k_1 \leftarrow\!\!\$\ \mathcal{K}$
8 : $\hat{b} \leftarrow \mathcal{A}^{|F\rangle, |H\rangle, O_{\mathsf{Dec}}}(\mathsf{pk}, c^*, k_b)$
9 : **return** $[b = \hat{b}]$

Random Oracle $F(|x, y\rangle)$:

10 : **return** $|x, y \oplus \mathsf{F}(x)\rangle$

Random Oracle $H(|x, y\rangle)$:

11 : **return** $|x, y \oplus \mathsf{H}(x)\rangle$ // G_0
12 : **if** $x = r^*$ **then**
  return $|x, y \oplus (m^*, \rho^*, k_0)\rangle$ // G_2–G_3
13 : **perform** $\mathsf{H_c}$ // G_1–G_4

Decapsulation Oracle $O_{\mathsf{Dec}}(c)$:

14 : **if** $c = c^*$ **then** : **return** $\perp$
15 : $m_1 \| m_2 := \mathsf{Dec}'(\mathsf{sk}, c)$ // G_2
16 : **if** $m_1 \| m_2 = \perp$ **then**
  return $\perp$ // G_0–G_2
17 : $r := m_2 \oplus F(m_1)$ // G_0–G_2
18 : $(m_1', \rho, k) := H(r)$ // G_0
19 : $(m_1', \rho, k) := \mathsf{D_H}(r)$ // G_1–G_2
20 : **if** $m_1 = m_1'$ **and**
  $\mathsf{Enc}'(\mathsf{pk}, m_1' \| m_2; \rho) = c$ **then**
  return k // G_0–G_2
21 : **return** $\perp$ // G_0–G_2
22 : **return** $f_1(c, D_{\mathsf{H}})$ // G_3–G_4

Figure 12: Games G_0 to G_4 for the proof of Theorem 5.5.

where $S_{(c,m_1,m_2),r} := \{(m_1', \rho, k) \mid m_1' = m_1 \wedge \mathsf{Enc}'(\mathsf{pk}, m_1' \| m_2; \rho) = c\}$, and then measuring the last register in the computational basis to obtain the output of O_{Dec}. In this game, we replace $H(r)$ in the unitary with $D_{\mathsf{H}}(r)$, where $|D_{\mathsf{H}}\rangle$ is the computational basis state of $\mathsf{D_H}$. We indicate this change in line 19 of Fig. 12 by writing $(m_1', \rho, k) := \mathsf{D_H}(r)$, meaning that the tuple (m_1', ρ, k) is now from the value stored at position r in $\mathsf{D_H}$, rather than from $H(r)$. Since

$$\begin{aligned}
\Gamma_{S_{(c,m_1,m_2),r}} &= \max_{(c,m_1,m_2),r} \left|\{(m_1', \rho, k) \mid m_1' = m_1 \wedge \mathsf{Enc}'(\mathsf{pk}, m_1' \| m_2; \rho) = c\}\right| \\
&\leq \max_{(c,m_1,m_2),r} \left|\{(m_1', \rho, k) \mid m_1' = m_1\}\right| \\
&= |\{m_1\} \times \mathcal{R} \times \mathcal{K}| = |\mathcal{R} \times \mathcal{K}| \ ,
\end{aligned}$$

by the Switching Lemma (Lemma 5.1), we have

$$|\Pr[W_1] - \Pr[W_0]| \leq 6 q_D \sqrt{\Gamma_{S_{(c,m_1,m_2),r}} / |\mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}|} = 6 q_D \sqrt{1/|\mathcal{M}_1|} \ .$$

Game G_2. Note that in Game G_1, given a query $c \neq c^*$ and the database register $\mathsf{D_H}$, after computing $m_1 \| m_2 := \mathsf{Dec}'(\mathsf{sk}, c)$ $(\neq \perp)$ and $r := m_2 \oplus F(m_1)$, the decapsulation oracle can be viewed as applying a unitary operator

$$|(c, m_1, m_2, r), D_{\mathsf{H}}, 0^{\log|\mathcal{K}|}\rangle \mapsto |(c, m_1, m_2, r), D_{\mathsf{H}}, 0^{\log|\mathcal{K}|} \oplus f\big((c, m_1, m_2, r), D_{\mathsf{H}}\big)\rangle \ ,$$

where

$$f\big((c, m_1, m_2, r), D_{\mathsf{H}}\big) = \begin{cases} D_{\mathsf{H}}(r)[2], & \text{if } m_1' = m_1 \text{ and } \mathsf{Enc}'(\mathsf{pk}, m_1' \| m_2; \rho) = c \\ & \text{with } (m_1', \rho, k) := D_{\mathsf{H}}(r) \ , \\ \perp, & \text{otherwise} \ , \end{cases}$$

and then measuring the last register in the computational basis to obtain the response of O_{Dec}. Observe that by the definition of f we have

$$f\big((c, m_1, m_2, r), D_{\mathsf{H}}\big) = f\big((c, m_1, m_2, r), D_{\mathsf{H}} \cup \{(r^*, (m_1^*, \rho^*, k_0))\}\big)$$

for every D_{H} with $D_{\mathsf{H}}(r^*) = \perp$. Otherwise, there would exist some $c \neq c^*$ such that $r^* = m_2 \oplus F(m_1^*)$ (implying $m_2 = m_2^*$) and $\mathsf{Enc}'(\mathsf{pk}, m_1^* \| m_2; \rho^*) = c$, which contradicts $c \neq c^*$. Hence the function f satisfies the conditions required by the Puncturing Lemma (Lemma 5.3).

Consequently, in this game we sample (m_1^*, ρ^*, k_0) uniformly at random (as shown in line 4 of Fig. 12) and modify the behavior of the random oracle H to

$$|x, y, D_{\mathsf{H}}\rangle \mapsto \begin{cases} |x, y \oplus (m_1^*, \rho^*, k_0), D_{\mathsf{H}}\rangle , & \text{if } x = r^* , \\ H\,|x, y, D_{\mathsf{H}}\rangle , & \text{otherwise} , \end{cases}$$

i.e., we add line 12 in Fig. 12. By the Puncturing Lemma (Lemma 5.3), we have

$$|\Pr[W_2] - \Pr[W_1]| \leq 2q_D / \sqrt{|\mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}|} .$$

Game G_3. Note that in Game G_2, after receiving a query $c \neq c^*$ and given the database register $\mathsf{D_H}$, the decapsulation oracle can be viewed as first applying a unitary operator

$$|(c, D_{\mathsf{H}}, 0^{\log|\mathcal{K}|})\rangle \mapsto |c, D_{\mathsf{H}}, 0^{\log|\mathcal{K}|} \oplus f_0(c, D_{\mathsf{H}})\rangle \ ,$$

with[5]

$$f_0(c, D_{\mathsf{H}}) = \begin{cases} D_{\mathsf{H}}(r)[2], & \text{if } (r, D_{\mathsf{H}}(r)) \in S_c, \\ & \text{where } r = \mathsf{Dec}'(\mathsf{sk}, c)[1] \oplus F(\mathsf{Dec}'(\mathsf{sk}, c)[0]) \ , \\ \perp, & \text{otherwise} , \end{cases}$$

where $S_c := \{(r, (m_1', \rho, k)) \mid m_1' = \mathsf{Dec}'(\mathsf{sk}, c)[0] \wedge \mathsf{Enc}'(\mathsf{pk}, m_1' \| (r \oplus F(m_1')); \rho) = c\}$. Here, $\mathsf{Dec}'(\mathsf{sk}, c)[0]$ (resp. $\mathsf{Dec}'(\mathsf{sk}, c)[1]$) denotes the first part m_1 (resp. the second part m_2) for $\mathsf{Dec}'(\mathsf{sk}, c) = m_1 \| m_2$. The oracle then measures the last register in the computational basis to obtain the classical response of O_{Dec}. In this game, we replace f_0 by

$$f_1(c, D_{\mathsf{H}}) = \begin{cases} D_{\mathsf{H}}(r)[2], & \text{if } \exists (r, D_{\mathsf{H}}(r)) \in S_c', \\ & \text{and } \forall r' < r,\ (r', D_{\mathsf{H}}(r')) \notin S_c' \ , \\ \perp, & \text{otherwise} , \end{cases}$$

where $S_c' := \{(r, (m_1', \rho, k)) \mid \mathsf{Enc}'(\mathsf{pk}, m_1' \| (r \oplus F(m_1')); \rho) = c\}$.

To apply the Lookup Lemma (Lemma 5.2), we first define, for each c, the set

$$T_c := \{(r, (m_1', \rho, k)) \in S_c \mid r = \mathsf{Dec}'(\mathsf{sk}, c)[1] \oplus F(\mathsf{Dec}'(\mathsf{sk}, c)[0])\} .$$

We then introduce the binary relation

$$R := \{(r, (m_1', \rho, k)) \mid \exists\, c \text{ s.t. } (r, (m_1', \rho, k)) \in T_c \triangle S_c'\} .$$

For every $(r, (m_1', \rho, k)) \in T_c$, the relation $\mathsf{Enc}'(\mathsf{pk}, m_1' \| (r \oplus F(m_1')); \rho) = c$ necessarily holds, which implies $(r, (m_1', \rho, k)) \in S_c'$; consequently, $T_c \setminus S_c' = \emptyset$. On the other hand, for any $(r, (m_1', \rho, k)) \in S_c'$ with $(r, (m_1', \rho, k)) \notin T_c$, we must have

$$\mathsf{Dec}'\big(\mathsf{sk}, \mathsf{Enc}'(\mathsf{pk}, m_1' \| (r \oplus F(m_1')); \rho)\big) \neq m_1' \| (r \oplus F(m_1')) .$$

Therefore,

$$\begin{aligned} \Gamma_R &= \max_r \big|\{(m_1', \rho, k) \mid (r, (m_1', \rho, k)) \in R\}\big| \\ &\leq \max_r \big|\{(m_1', \rho, k) \mid \exists\, c \text{ s.t. } (r, (m_1', \rho, k)) \in T_c \setminus S_c'\}\big| \\ &\quad + \max_r \big|\{(m_1', \rho, k) \mid \exists\, c \text{ s.t. } (r, (m_1', \rho, k)) \in S_c' \setminus T_c\}\big| \\ &\leq \max_r |\Theta_r| \cdot |\mathcal{K}| \ , \end{aligned}$$

[5]For technical convenience, we extend the definition of the random oracle F by the convention $F(\perp) := \perp$.

where the set $\Theta_r := \{(m_1', \rho) \mid \mathsf{Dec}'(\mathsf{sk}, \mathsf{Enc}'(\mathsf{pk}, m_1' \| (r \oplus F(m_1')); \rho)) \neq m_1' \| (r \oplus F(m_1'))\}$. For a fixed key pair $(\mathsf{pk}, \mathsf{sk})$ and a fixed function F that is used to simulated the random oracle F, the Lookup Lemma (Lemma 5.2) yields

$$\begin{aligned}&|\Pr[W_3 : (\mathsf{pk}, \mathsf{sk}, \mathsf{F})] - \Pr[W_2 : (\mathsf{pk}, \mathsf{sk}, \mathsf{F})]| \\ \leq\ & 16(q_H + 1)\sqrt{\Gamma_R / |\mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}|} \leq 16(q_H + 1)\sqrt{\max_r |\Theta_r| / |\mathcal{M}_1 \times \mathcal{R}|}\ .\end{aligned}$$

For a given $(\mathsf{pk}, \mathsf{sk})$, define

$$\delta(\mathsf{pk}, \mathsf{sk}) := \Pr[\mathsf{Dec}'(\mathsf{sk}, \mathsf{Enc}'(\mathsf{pk}, m_1 \| m_2; \rho)) \neq m_1 \| m_2]\ ,$$

where the probability is taken over $m_1 \| m_2 \leftarrow\!\!\$\ \mathcal{M}_1 \times \mathcal{M}_2$ and $\rho \leftarrow\!\!\$\ \mathcal{R}$. Note that for any fixed $r \in \mathcal{M}_2$ and any $m_1 \in \mathcal{M}_1$, the variable $r \oplus \mathsf{F}(m_1)$ (with $\mathsf{F} \leftarrow\!\!\$\ \Omega_F$) is also uniformly distributed over $\mathcal{M}_2$. Hence, we have

$$\mathbb{E}_{\mathsf{F} \leftarrow\$\Omega_F}[\,|\Theta_r| / |\mathcal{M}_1 \times \mathcal{R}|\,] \leq \delta(\mathsf{pk}, \mathsf{sk}) \quad \text{for every } r\ .$$

Thus, by averaging over $\mathsf{F} \leftarrow\!\!\$\ \Omega_F$ and using Jensen's inequality, we can obtain

$$\begin{aligned}&|\Pr[W_3 : (\mathsf{pk}, \mathsf{sk})] - \Pr[W_2 : (\mathsf{pk}, \mathsf{sk})]| \\ \leq\ & 16(q_H + 1)\mathbb{E}_{\mathsf{F} \leftarrow\$\Omega_F}[\sqrt{\max_r |\Theta_r| / |\mathcal{M}_1 \times \mathcal{R}|}] \\ \leq\ & 16(q_H + 1)\sqrt{\mathbb{E}_{\mathsf{F} \leftarrow\$\Omega_F}[\max_r |\Theta_r| / |\mathcal{M}_1 \times \mathcal{R}|]} \leq 16(q_H + 1)\sqrt{\delta(\mathsf{pk}, \mathsf{sk})}\ .\end{aligned}$$

Finally, by averaging over the key pair $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$ and applying Jensen's inequality once more, we have

$$\begin{aligned}&|\Pr[W_3] - \Pr[W_2]| \leq 16(q_H + 1)\mathbb{E}_{(\mathsf{pk}, \mathsf{sk})}[\sqrt{\delta(\mathsf{pk}, \mathsf{sk})}] \\ \leq\ & 16(q_H + 1)\sqrt{\mathbb{E}_{(\mathsf{pk}, \mathsf{sk})}[\delta(\mathsf{pk}, \mathsf{sk})]} \overset{(a)}{=} 16(q_H + 1)\sqrt{\delta_{ac}},\end{aligned}$$

where (a) follows from the definition of the δ_{ac}-average-case correctness of PKE'.

Note that from this game onward, O_{Dec} can be simulated without sk.

Game G_4. In this game, we remove line 12 of Fig. 12, which is equivalent to reprogramming the random oracle H at the point r^*. Now, we can construct two oracle algorithms, $A^{|F\rangle, |H\rangle, O_{\mathsf{Dec}}}(z)$ and $A^{|F\rangle, |H'\rangle, O_{\mathsf{Dec}}}(z)$, to simulate Games G_3 and G_2, respectively, and then apply the Semi-Classical One-Way to Hiding Lemma (Lemma 3.2) to bound $|\Pr[W_4] - \Pr[W_3]|$.

Let $z := (\mathsf{pk}, c^*, k_0, k_1, b)$, where $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$, $r^* \leftarrow\!\!\$\ \mathcal{M}_2$, $(m_1^*, \rho^*, k_0) \leftarrow\!\!\$\ \mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}$, $m_2^* := r^* \oplus F(m_1^*)$, $c^* := \mathsf{Enc}'(\mathsf{pk}, m_1^* \| m_2^*; \rho^*)$, $k_1 \leftarrow\!\!\$\ \mathcal{K}$, and $b \leftarrow\!\!\$\ \{0, 1\}$. Let the random oracle H be simulated by the compressed oracle $\mathsf{H_c}$ (as in Game G_4), and let H' be an oracle such that $H'(r) = H(r)$ for all $r \neq r^*$ and $H'(r^*) = (m_1^*, \rho^*, k_0)$ (as in Game G_3). The behaviors of the random oracle F and the decapsulation oracle O_{Dec} are the same as in Game G_3. The algorithm $A^{|F\rangle, |H\rangle, O_{\mathsf{Dec}}}(z)$ (or $A^{|F\rangle, |H'\rangle, O_{\mathsf{Dec}}}(z)$) runs $\hat{b} \leftarrow \mathcal{A}^{|F\rangle, |H\rangle, O_{\mathsf{Dec}}}(z)$ (or $\hat{b} \leftarrow \mathcal{A}^{|F\rangle, |H'\rangle, O_{\mathsf{Dec}}}(z)$) and then outputs $[b = \hat{b}]$. Note that the behaviors of H and O_{Dec} in Games G_4 and G_3 are identical. It is easy to see that

$$\Pr[1 \leftarrow A^{|F\rangle, |H\rangle, O_{\mathsf{Dec}}}(z)] = \Pr[W_4] \text{ and } \Pr[1 \leftarrow A^{|F\rangle, |H'\rangle, O_{\mathsf{Dec}}}(z)] = \Pr[W_3]\ .$$

By the Semi-Classical One-Way to Hiding lemma (Lemma 3.2), we can obtain[6]

$$\begin{aligned}&\left|\Pr[1 \leftarrow A^{|F\rangle, |H\rangle, O_{\mathsf{Dec}}}(z)] - \Pr[1 \leftarrow A^{|F\rangle, |H'\rangle, O_{\mathsf{Dec}}}(z)]\right| \\ &\qquad \leq 2\sqrt{(q_H + 1)\Pr[\mathsf{Find}_{r^*} : A^{|F\rangle, |H \setminus \{r^*\}\rangle, O_{\mathsf{Dec}}}(z)]}\ .\end{aligned}$$

[6]When applying Lemma 3.2, F and O_{Dec} can be treated as $\mathcal{A}$'s internal oracles.

Games G_5 to G_7:	Random Oracle $F(\|x,y\rangle)$:
$1: (\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$	$11:$ **return** $\|x, y \oplus \mathsf{F}(x)\rangle$
$2: \mathsf{F} \leftarrow\!\!\$\ \Omega_F,\ \hat{m}_1^* \leftarrow\!\!\$\ \mathcal{M}_1$	
$3: (m_1^*, \rho^*, k_0) \leftarrow\!\!\$\ \mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}$	Punctured Oracle $H\backslash\{r^*\}(\|x,y\rangle)$:
$4: r^* \leftarrow\!\!\$\ \mathcal{M}_2,\ m_2^* := r^* \oplus F(m_1^*)$ // G_5	$12:$ **perform** $\mathcal{O}^{SC}_{\{r^*\}}$ // G_5
$5: m_2^* \leftarrow\!\!\$\ \mathcal{M}_2$ // G_6–G_8	$13:$ **perform** $\mathcal{O}^{SC}_{\{m_2^* \oplus F(m_1^*)\}}$ // G_6–G_7
$6: c^* := \mathsf{Enc}'(\mathsf{pk}, m_1^* \| m_2^*; \rho^*)$ // G_5–G_6	$14:$ **perform** $\mathsf{H_c}$
$7: c^* := \mathsf{Enc}'(\mathsf{pk}, \hat{m}_1^* \| m_2^*; \rho^*)$ // G_7	
$8: b \leftarrow\!\!\$\ \{0,1\},\ k_1 \leftarrow\!\!\$\ \mathcal{K}$	Decapsulation Oracle $O_{\mathsf{Dec}}(c)$:
$9: \hat{b} \leftarrow \mathcal{A}^{\|F\rangle, \|H\backslash\{r^*\}\rangle, O_{\mathsf{Dec}}}(\mathsf{pk}, c^*, k_b)$	$15:$ **if** $c = c^*$ **then** : **return** $\perp$
$10:$ **if** Find **then** : **return** 1	$16:$ **return** $f_1(c, \mathsf{D_H})$
else : **return** 0	

Figure 13: Games G_5 to G_7 for the proof of Theorem 5.5.

where $H\backslash\{r^*\}$ denotes a punctured oracle that, before each query to H, first applies a semi-classical oracle $\mathcal{O}^{SC}_{\{r^*\}}$ (as defined in Definition 3.8) to the input, and then proceeds with the usual query to H; Find_{r^*} is the classical event that the measurement of $\mathcal{O}^{SC}_{\{r^*\}}$ ever outputs 1 (as defined in Definition 3.8).

Note that in this game, both k_0 and k_1 are sampled uniformly at random and are independent of all other oracle answers and public values; therefore, the bit b is uniformly random and independent of $\mathcal{A}$'s view. Consequently,

$$\Pr[W_4] = 1/2 .$$

We next define a sequence of games G_j (for $j = 5, \ldots, 7$) played between the challenger and the adversary $\mathcal{A}$, as depicted in Fig. 13. In each game, Find denotes the event that the measurement of the semi-classical oracle inside the punctured oracle $H\backslash\{r^*\}$ ever outputs 1. Let Z_j be the event that Find occurs in Game G_j.

Game G_5. This game is defined as in Fig. 13. Compared to the challenger in Game G_4, it additionally samples $\hat{m}_1^* \leftarrow\!\!\$\ \mathcal{M}_1$, which is never used in the game, and swaps the order of sampling (m_1^*, ρ^*, k_0) and r^*. It is easy to see that these changes do not affect the behavior of the challenger from $\mathcal{A}$'s view. By the definitions of the punctured oracle $H \setminus \{r^*\}$ and the event Find in Game G_5 (depicted in Fig. 13), we can see that

$$\Pr[Z_5] = \Pr[\mathsf{Find}_{r^*} : A^{|F\rangle, |H\backslash\{r^*\}\rangle, O_{\mathsf{Dec}}}(z)] .$$

Game G_6. In this game, we make a *conceptual* change to the challenger's description. Recall that in Game G_5 the challenger samples $r^* \leftarrow\!\!\$\ \mathcal{M}_2$ uniformly and then sets $m_2^* := r^* \oplus F(m_1^*)$ (line 4 of Fig. 13). Because r^* is uniform and independent of $F(m_1^*)$, the value m_2^* is itself uniformly distributed over $\mathcal{M}_2$. Hence, we can replace that two-step procedure with the direct sampling $m_2^* \leftarrow\!\!\$\ \mathcal{M}_2$ (now line 5 of Fig. 13), which also implicitly defines $r^* := m_2^* \oplus F(m_1^*)$. Consequently, the semi-classical oracle $\mathcal{O}^{SC}_{\{r^*\}}$ (line 12) is modified to $\mathcal{O}^{SC}_{\{m_2^* \oplus F(m_1^*)\}}$ (line 13). Since this alteration is purely a reformulation of the same process, we have

$$\Pr[Z_6] = \Pr[Z_5] .$$

Now we can bound $\Pr[Z_6]$ if PKE' is OW-CPA. Let $B^{|F\rangle, |H\rangle, O_{\mathsf{Dec}}}(z')$ be an oracle algorithm that takes $z' := (\mathsf{pk}, c^*, k_b)$ as input, randomly picks $i^* \leftarrow\!\!\$\ \{1, \ldots, q_H\}$, runs $\mathcal{A}^{|F\rangle, |H\rangle, O_{\mathsf{Dec}}}(z)$

Games G_6^A to G_7^A:	Random Oracle $F(\lvert x, y\rangle)$:
$1:\ (\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$	$10:$ **return** $\lvert x, y \oplus \mathsf{F}(x)\rangle$ $\quad /\!\!/\ G_6^A$
$2:\ \mathsf{F} \leftarrow\!\$\ \Omega_F,\ \ \hat{m}_1^* \leftarrow\!\$\ \mathcal{M}_1$	$11:$ **perform** $\mathsf{F_c}$ $\quad /\!\!/\ G_7^A$
$3:\ (m_1^*, \rho^*, k_0) \leftarrow\!\$\ \mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}$	
$4:\ m_2^* \leftarrow\!\$\ \mathcal{M}_2$	Random Oracle $H(\lvert x, y\rangle)$:
$5:\ c^* := \mathsf{Enc}'(\mathsf{pk}, m_1^* \\| m_2^*; \rho^*)$	$12:$ **perform** $\mathsf{H_c}$
$6:\ b \leftarrow\!\$\ \{0,1\},\ \ k_1 \leftarrow\!\$\ \mathcal{K}$	
$7:\ \hat{r} \leftarrow B^{\lvert F\rangle, \lvert H\rangle, O_{\mathsf{Dec}}}(\mathsf{pk}, c^*, k_b)$	Decapsulation Oracle $O_{\mathsf{Dec}}(c)$:
$8:$ **return** $[\hat{r} = \big(m_2^* \oplus F(m_1^*)\big)]$ $\quad /\!\!/\ G_6^A$	$13:$ **if** $c = c^*$ **then** $:$ **return** $\perp$
$9:$ **return** $[\hat{r} = \big(m_2^* \oplus \mathsf{D_F}(m_1^*)\big)]$ $\quad /\!\!/\ G_7^A$	$14:$ **return** $f_1(c, \mathsf{D_H})$

Figure 14: Games G_6^A to G_7^A for the proof of Theorem 5.5.

until (just before) the i^*-th query to the random oracle H, measures the input register in the computational basis, and outputs the measurement result. Here H behaves the same as $H \backslash \{r^*\}$, except that it does not query the semi-classical oracle $\mathcal{O}^{SC}_{\{m_2^* \oplus F(m_1^*)\}}$. By Lemma 3.3, we have

$$\Pr[Z_6] \leq 4q_H \cdot \Pr[\hat{r} = \big(m_2^* \oplus F(m_1^*)\big) : \hat{r} \leftarrow B^{\lvert F\rangle, \lvert H\rangle, O_{\mathsf{Dec}}}(z')] \ .$$

Consider a game G_6^A played between the challenger and B, as shown in Fig. 14, and let Z_j^A be the event that the challenger outputs 1 at the end of game G_j^A. It is clear that

$$\Pr[Z_6^A] = \Pr[\hat{r} = \big(m_2^* \oplus F(m_1^*)\big) : \hat{r} \leftarrow B^{\lvert F\rangle, \lvert H\rangle, O_{\mathsf{Dec}}}(z')] \ .$$

Now we modify G_6^A to obtain G_7^A, as shown in Fig. 14. In this game, we further simulate the random oracle F using the compressed oracle $\mathsf{F_c}$ with database register $\mathsf{D_F}$ (line 11 of Fig. 14). Note that in Game G_6^A, after the oracle algorithm B outputs $\hat{r}$, the challenger is equivalent to applying the unitary operator

$$\lvert (m_2^*, \hat{r}), m_1^*, 0\rangle \mapsto \begin{cases} \lvert (m_2^*, \hat{r}), m_1^*, 1\rangle \ , & \text{if } F(m_1^*) \in S_{(m_2^*, \hat{r}), m_1^*} \ , \\ \lvert (m_2^*, \hat{r}), m_1^*, 0\rangle \ , & \text{otherwise} \ , \end{cases}$$

where $S_{(m_2^*, \hat{r}), m_1^*} := \{u \mid m_2^* \oplus u = \hat{r}\}$, and then measures the last register in the computational basis to obtain the final outcome of the game. We replace $F(m_1^*)$ in this unitary by $D_{\mathsf{F}}(m_1^*)$, where $\lvert D_{\mathsf{F}}\rangle$ is the computational-basis state of $\mathsf{D_F}$. We denote the challenger's check on $\hat{r}$ output by B in Game G_7^A as $\hat{r} = \big(m_2^* \oplus \mathsf{D_F}(m_1^*)\big)$ (line 9 of Fig. 14). Clearly, exactly one value $u = m_2^* \oplus \hat{r}$ satisfies $m_2^* \oplus u = \hat{r}$; hence $\Gamma_{S_{(m_2^*, \hat{r}), m_1^*}} = \max_{(m_2^*, \hat{r}), m_1^*} \left| S_{(m_2^*, \hat{r}), m_1^*} \right| = 1$. Therefore, by the Switching Lemma (Lemma 5.1) we obtain

$$\left| \sqrt{\Pr[Z_7^A]} - \sqrt{\Pr[Z_6^A]} \right| \leq 6\sqrt{1/|\mathcal{M}_2|} \ .$$

By the basic principles of quantum computation, the challenger's verification of the value $\hat{r}$ output by B in Game G_7^A is equivalent to first measuring the register $\mathsf{D_F}$ in the computational basis, obtaining a classical database D_{F}, and then checking whether $m_2^* \oplus D_{\mathsf{F}}(m_1^*) = \hat{r}$. If the equality holds, it means there exists an entry $(m_1^*, D_{\mathsf{F}}(m_1^*))$ in D_{F} such that $m_2^* = \hat{r} \oplus D_{\mathsf{F}}(m_1^*)$.

Based on this observation, and since now the values m_1^*, m_2^*, ρ^* are used only to compute $c^* := \mathsf{Enc}'(\mathsf{pk}, m_1^* \| m_2^*; \rho^*)$ while the challenger's simulation does not require sk, we can construct an adversary $\mathcal{B}$ that attacks the OW-CPA security of PKE' whenever the event Z_7^A

Games G_6^B to G_8^B:	Random Oracle $F(\lvert x, y\rangle)$:
$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$	**return** $\lvert x, y \oplus \mathsf{F}(x)\rangle$ // G_6^B
$\mathsf{F} \leftarrow\$ \Omega_F,\ \hat{m}_1^* \leftarrow\$ \mathcal{M}_1$	**perform** $\mathsf{F_c}$ // G_7^B–G_8^B
$(m_1^*, k_0) \leftarrow\$ \mathcal{M}_1 \times \mathcal{K}$	Punctured Oracle $H\backslash\{r^*\}(\lvert x, y\rangle)$:
$m_2^* \leftarrow\$ \mathcal{M}_2$	**perform** $\mathcal{O}^{SC}_{\{m_2^* \oplus F(m_1^*)\}}$ // G_6^B
$c^* := \mathsf{Enc}'(\mathsf{pk}, m_1^* \Vert m_2^*)$	**perform** $\mathcal{O}^{SC}_{\{m_2^* \oplus \mathsf{D_F}(m_1^*)\}}$ // G_7^B
$b \leftarrow\$ \{0,1\},\ k_1 \leftarrow\$ \mathcal{K}$	**perform** $\mathcal{O}^{SC}_{[\mathsf{Enc}'(\mathsf{pk}, m_1 \Vert (\cdot \oplus \mathsf{D_F}(m_1))) = c^*]}$ // G_8^B
$\hat{b} \leftarrow \mathcal{A}^{\lvert F\rangle, \lvert H\backslash\{r^*\}\rangle, O_{\mathsf{Dec}}}(\mathsf{pk}, c^*, k_b)$	**perform** $\mathsf{H_c}$
if Find **then** : **return** 1	Decapsulation Oracle $O_{\mathsf{Dec}}(c)$:
else : **return** 0	**if** $c = c^*$ **then** : **return** $\perp$
	return $f_1(c, \mathsf{D_H})$

Figure 15: Games G_6^B to G_8^B for the proof of Theorem 5.5.

occurs. Concretely, upon receiving the challenge (pk, c^*) from its OW-CPA challenger, $\mathcal{B}$ samples $k_0, k_1 \leftarrow\$ \mathcal{K}$ and $b \leftarrow\$ \{0,1\}$, forms $z' := (\mathsf{pk}, c^*, k_b)$, and interacts with $B^{\lvert F\rangle, \lvert H\rangle, O_{\mathsf{Dec}}}(z')$ by simulating the oracles F, H, O_{Dec} exactly as in Game G_7^A. After B outputs $\hat{r}$, $\mathcal{B}$ measures $\mathsf{D_F}$ in the computational basis to obtain a classical D_{F}, randomly chooses a pair (m_1, u) from D_{F}, computes $m_2 := \hat{r} \oplus u$, and outputs $m_1 \Vert m_2$ as the solution to the OW-CPA challenge. Clearly, if the event $\hat{r} = \big(m_2^* \oplus \mathsf{D_F}(m_1^*)\big)$ holds and $\mathcal{B}$ happens to pick the correct pair $(m_1^*, D_{\mathsf{F}}(m_1^*))$, then $m_1 \Vert m_2$ is indeed a valid solution. This occurs with probability $\Pr[Z_7^A]/|D_{\mathsf{F}}|$.

Recall that during the simulation of the decapsulation oracle, each evaluation of f_1 requires queries to the random oracle F for every entry in D_{H} with $D_{\mathsf{H}}(r) \neq \perp$. Since B makes at most q_H queries to H, we have $|D_{\mathsf{H}}| \leq q_H$. Moreover, by the property of the compressed oracle, all first components m_1 stored in D_{F} are distinct. Consequently, the number of entries added to D_{F} via the F-queries inside O_{Dec} is at most q_H; together with at most q_F direct queries of B to F, we obtain $|D_{\mathsf{F}}| \leq q_F + q_H$. Therefore,

$$\Pr[Z_7^A]/(q_F + q_H) \leq \mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{PKE}'}(\mathcal{B}) \ .$$

Putting all bounds together, we have

$$\begin{aligned}
\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) &\leq 2\sqrt{4q_H(q_H{+}1)}\Big(\sqrt{(q_F + q_H)\mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{PKE}'}(\mathcal{B})} + \frac{6}{\sqrt{|\mathcal{M}_2|}}\Big) \\
&\quad + 16(q_H{+}1)\sqrt{\delta_{ac}} + \frac{6q_D}{\sqrt{|\mathcal{M}_1|}} + \frac{2q_D}{\sqrt{|\mathcal{M}_1 {\times} \mathcal{R} {\times} \mathcal{K}|}} \\
&\leq 4(q_F + q_H + 1)^{1.5}\sqrt{\mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{PKE}'}(\mathcal{B})} + 16(q_H + 1)\sqrt{\delta_{ac}} \\
&\quad + 8q_D/\sqrt{|\mathcal{M}_1|} + 24(q_H + 1)/\sqrt{|\mathcal{M}_2|} \ .
\end{aligned}$$

When PKE' is deterministic, consider the game G_6^B played by the challenger and the adversary $\mathcal{A}$, as depicted in Fig. 15, where the randomness space $\mathcal{R} = \emptyset$ of PKE' is omitted. Let Z_j^B be the event that Find occurs in Game G_j^B. Clearly,

$$\Pr[Z_6^B] = \Pr[Z_6] \ .$$

In Game G_7^B of Fig. 15, the random oracle F is simulated using the compressed oracle $\mathsf{F_c}$ with database register $\mathsf{D_F}$ (line 10 of Fig. 14). Recall that, in Game G_6^B, for a query x, the semi-classical oracle $\mathcal{O}^{SC}_{\{m_2^* \oplus F(m_1^*)\}}$ inside the punctured oracle $H \setminus \{r^*\}$ is equivalent to applying a unitary operator

$$|(m_2^*, x), m_1^*, 0\rangle \mapsto \begin{cases} |(m_2^*, x), m_1^*, 1\rangle\,, & \text{if } F(m_1^*) \in S_{(m_2^*,x),m_1^*}\ , \\ |(m_2^*, x), m_1^*, 0\rangle\,, & \text{otherwise}\ , \end{cases}$$

where $S_{(m_2^*,x),m_1^*} \coloneqq \{u \mid m_2^* \oplus u = x\}$, and then measures the last register in the computational basis. In Game G_7^B, we replace $F(m_1^*)$ in this unitary by $D_{\mathsf{F}}(m_1^*)$, where $|D_{\mathsf{F}}\rangle$ is the computational basis state of D_{F}. We denote the modified semi-classical oracle by $\mathcal{O}^{SC}_{\{m_2^* \oplus \mathsf{D}_{\mathsf{F}}(m_1^*)\}}$ (as shown in line 12 of Fig. 14). Clearly, exactly one value $u = m_2^* \oplus x$ satisfies $m_2^* \oplus u = x$; hence $\Gamma_{S_{(m_2^*,x),m_1^*}} = \max_{(m_2^*,x),m_1^*} \left|S_{(m_2^*,x),m_1^*}\right| = 1$. Therefore, by the Switching Lemma (Lemma 5.1) we obtain

$$\left|\sqrt{\Pr[Z_7^B]} - \sqrt{\Pr[Z_6^B]}\right| \le 6q_H\sqrt{1/|\mathcal{M}_2|}\ .$$

We now modify G_7^B to obtain G_8^B, depicted in Fig. 15. Note that in Game G_7^B, for a query x, the semi-classical oracle $\mathcal{O}^{SC}_{\{m_2^* \oplus \mathsf{D}_{\mathsf{F}}(m_1^*)\}}$ is equivalent to applying the unitary operator

$$|x, D_{\mathsf{F}}, 0\rangle \mapsto \begin{cases} |x, D_{\mathsf{F}}, 1\rangle\,, & \text{if } x \oplus D_{\mathsf{F}}(m_1^*) = m_2^*\ , \\ |x, D_{\mathsf{F}}, 0\rangle\,, & \text{otherwise}\ . \end{cases}$$

In Game G_8^B, we replace the condition in this unitary to obtain

$$|x, D_{\mathsf{F}}, 0\rangle \mapsto \begin{cases} |x, D_{\mathsf{F}}, 1\rangle\,, & \text{if } \exists (m_1, D_{\mathsf{F}}(m_1)) \in D_{\mathsf{F}} \\ & \text{s.t. } \mathsf{Enc}'(\mathsf{pk}, m_1 \| (x \oplus D_{\mathsf{F}}(m_1))) = c^*\ , \\ |x, D_{\mathsf{F}}, 0\rangle\,, & \text{otherwise}\ . \end{cases}$$

We denote by $\mathcal{O}^{SC}_{[\mathsf{Enc}'(\mathsf{pk}, m_1 | (\cdot \oplus \mathsf{DF}(m_1))) = c^*]}$ the resulting modified semi-classical oracle. Clearly, $\mathcal{O}^{SC}_{\{m_2^* \oplus \mathsf{D}_{\mathsf{F}}(m_1^*)\}}$ and $\mathcal{O}^{SC}_{[\mathsf{Enc}'(\mathsf{pk}, m_1 \| (\cdot \oplus \mathsf{D}_{\mathsf{F}}(m_1))) = c^*]}$ are identical unless for the randomly sampled $m_1^* \| m_2^*$ there exists a different pair $m_1 \| m_2$ such that $\mathsf{Enc}'(\mathsf{pk}, m_1^* \| m_2^*) = \mathsf{Enc}'(\mathsf{pk}, m_1 \| m_2)$. For a deterministic PKE', the probability of such a collision is at most $2\delta_{ac}$ according to [16, Lemma G1]. Hence,

$$\left|\Pr[Z_8^B] - \Pr[Z_7^B]\right| \le 2\delta_{ac}\ .$$

Since in Game G_8^B the simulation of all oracles does not require m_1^*, m_2^* nor sk, once Z_8^B occurs we can construct an adversary $\mathcal{B}$ against the OW-CPA security of PKE'. Concretely, upon receiving the challenge (pk, c^*) from its OW-CPA challenger, $\mathcal{B}$ sets $c_1^* \coloneqq c^*$, generates c_2^*, k_0, k_1, b as in Game G_8^B, and interacts with B by simulating the oracles as in that game. When Z_8^B occurs, i.e., the measurement of $\mathcal{O}^{SC}_{[\mathsf{Enc}'(\mathsf{pk}, m_1 \| (\cdot \oplus \mathsf{D}_{\mathsf{F}}(m_1))) = c^*]}$ outputs 1, $\mathcal{B}$ immediately measures $\mathcal{A}$'s input register to obtain $\hat{r}$ and then measures D_{F} to obtain a classical database D_{F}. It then iterates over all entries $(m_1, D_{\mathsf{F}}(m_1))$ in D_{F}, finds one satisfying $\mathsf{Enc}'(\mathsf{pk}, m_1 \| (\hat{r} \oplus D_{\mathsf{F}}(m_1))) = c^*$, and outputs $m_1 \| (\hat{r} \oplus D_{\mathsf{F}}(m_1))$ as the solution to the OW-CPA challenge. Evidently, this candidate equals $m_1^* \| m_2^*$ unless for the randomly sampled $m_1^* \| m_2^* \leftarrow\!\$\, \mathcal{M}_1 \times \mathcal{M}_2$ there exists a different pair $m_1 \| m_2 \neq m_1^* \| m_2^*$ with $\mathsf{Enc}'(\mathsf{pk}, m_1 \| m_2) = \mathsf{Enc}'(\mathsf{pk}, m_1^* \| m_2^*)$. For a deterministic PKE', this probability is at most $2\delta_{ac}$ according to [16, Lemma G1]. Hence,

$$\Pr[Z_8^B] \le \mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{dPKE}'}(\mathcal{B}) + 2\delta_{ac}\ .$$

Putting all bounds together, we have

$$\begin{aligned}
\mathsf{Adv}^{\text{IND-CCA}}_{\mathsf{KEM}}(\mathcal{A}) &\leq 2\sqrt{q_H+1}\Big(\sqrt{\mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{dPKE}'}(\mathcal{B})+4\delta_{ac}}+\frac{6q_H}{\sqrt{|\mathcal{M}_2|}}\Big) \\
&\quad + 16(q_H+1)\sqrt{\delta_{ac}} + \frac{6q_D}{\sqrt{|\mathcal{M}_1|}} + \frac{2q_D}{\sqrt{|\mathcal{M}_1\times\mathcal{R}\times\mathcal{K}|}} \\
&\leq 2\sqrt{q_H+1}\sqrt{\mathsf{Adv}^{\text{OW-CPA}}_{\mathsf{dPKE}'}(\mathcal{B})} + 20(q_H+1)\sqrt{\delta_{ac}} \\
&\quad + 8q_D/\sqrt{|\mathcal{M}_1|} + 12(q_H+1)/\sqrt{|\mathcal{M}_2|}\ .
\end{aligned}$$

Game G_7. We now consider the case where PKE' is IND-CPA secure. In this game, the challenger uses another random message $\hat{m}_1^* \leftarrow\!\!\$\ \mathcal{M}_1$ to compute $c^* := \mathsf{Enc}'(\mathsf{pk}, \hat{m}_1^* \| m_2^*; \rho^*)$ (line 7 of Fig. 13) instead of m_1^* (i.e., line 6 is removed).

Note that in both Games G_6 and G_7, the simulation of the oracles does not require sk. We now argue that if $\mathcal{A}$ can detect this modification, then we can construct a new adversary $\mathcal{B}'$ that attacks the IND-CPA security of PKE'. Upon receiving pk from the IND-CPA challenger, $\mathcal{B}'$ randomly chooses $m_1^*, \hat{m}_1^* \leftarrow\!\!\$\ \mathcal{M}_1$ and $m_2^* \leftarrow\!\!\$\ \mathcal{M}_2$, sets $m_0 := m_1^* \| m_2^*$ and $m_1 := \hat{m}_1^* \| m_2^*$, and sends (m_0, m_1) to the challenger. Once it receives the challenge ciphertext $c^* := \mathsf{Enc}'(\mathsf{pk}, m_{b'})$ (where b' is the challenger's random bit), $\mathcal{B}'$ samples $k_0, k_1 \leftarrow\!\!\$\ \mathcal{K}$ and $b \leftarrow\!\!\$\ \{0,1\}$, sends $z := (\mathsf{pk}, c^*, k_b)$ to $\mathcal{A}$, and simulates the oracles for $\mathcal{A}$ exactly as in Game G_6. Whenever the event Find occurs, $\mathcal{B}'$ outputs $\hat{b}' = 1$; otherwise, it outputs $\hat{b}' = 0$. Clearly, when $b' = 0$, the ciphertext c^* corresponds to m_0, so $\mathcal{B}'$ behaves exactly as the challenger in Game G_6; when $b' = 1$, c^* corresponds to m_1, so $\mathcal{B}'$ behaves as the challenger in Game G_7. Hence, we have $\Pr[\hat{b}' = 1 | b' = 0] = \Pr[Z_6]$ and $\Pr[\hat{b}' = 1 | b' = 1] = \Pr[Z_7]$. Therefore,

$$\begin{aligned}
\mathsf{Adv}^{\text{IND-CPA}}_{\mathsf{PKE}'}(\mathcal{B}') &\geq \Big|\Pr[b' = \hat{b}'] - 1/2\Big| \\
&= \Big|\Pr[b' = \hat{b}' \wedge b' = 0] + \Pr[b' = \hat{b}' \wedge b' = 1] - 1/2\Big| \\
&= \frac{1}{2}\Big|\Pr[\hat{b}' = 0 | b' = 0] + \Pr[\hat{b}' = 1 | b' = 1] - 1\Big| \\
&= \frac{1}{2}\Big|\Pr[\hat{b}' = 1 | b' = 1] - \Pr[\hat{b}' = 1 | b' = 0]\Big| \\
&= \frac{1}{2}|\Pr[Z_7] - \Pr[Z_6]|\ .
\end{aligned}$$

Now we only need to bound $\Pr[Z_7]$ to complete the proof. Let $C^{|F\rangle,|H\rangle,O_{\mathsf{Dec}}}(z')$ be the oracle algorithm that, on input $z' := (\mathsf{pk}, c^*, k_b)$, picks a random $i^* \leftarrow\!\!\$\ \{1, \ldots, q_H\}$, runs $\mathcal{A}^{|F\rangle,|H\rangle,O_{\mathsf{Dec}}}(z')$ until (just before) its i^*-th query to the random oracle H, measures $\mathcal{A}$'s input register in the computational basis, and outputs the measurement outcome. Here, H behaves identically to $H\backslash\{r^*\}$, except that it does not query the semi-classical oracle $\mathcal{O}^{SC}_{\{m_2^*\oplus F(m_1^*)\}}$. By Lemma 3.3,

$$\Pr[Z_7] \leq 4q_H \cdot \Pr[\hat{r} = \big(m_2^* \oplus F(m_1^*)\big) : \hat{r} \leftarrow C^{|F\rangle,|H\rangle,O_{\mathsf{Dec}}}(z')]\ .$$

Consider a game G_7^C played between the challenger and C, as depicted in Fig. 16, and let Z_j^C denote the event that the challenger outputs 1 in Game G_j^C. Clearly,

$$\Pr[Z_7^C] = \Pr[\hat{r} = \big(m_2^* \oplus F(m_1^*)\big) : \hat{r} \leftarrow C^{|F\rangle,|H\rangle,O_{\mathsf{Dec}}}(z')]\ .$$

Now we modify Game G_7^C to obtain G_8^C, as shown in Fig. 16. In this game, the random oracle F is simulated using the compressed oracle $\mathsf{F_c}$ with database register $\mathsf{D_F}$ (line 11 of Fig. 16).

Games G_7^C to G_8^C:	Random Oracle $F(\lvert x,y\rangle)$:
$1:\ (\mathsf{pk},\mathsf{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$	$10:\ \textbf{return}\ \lvert x, y \oplus \mathsf{F}(x)\rangle \quad /\!\!/\ G_7^C$
$2:\ \mathsf{F} \leftarrow\!\!\$\ \Omega_F,\ \hat{m}_1^* \leftarrow\!\!\$\ \mathcal{M}_1$	$11:\ \textbf{perform}\ \mathsf{F_c} \quad /\!\!/\ G_8^C$
$3:\ (m_1^*, \rho^*, k_0) \leftarrow\!\!\$\ \mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}$	
$4:\ m_2^* \leftarrow\!\!\$\ \mathcal{M}_2$	Random Oracle $H(\lvert x,y\rangle)$:
$5:\ c^* := \mathsf{Enc}'(\mathsf{pk}, \hat{m}_1^* \| m_2^*; \rho^*)$	$12:\ \textbf{perform}\ \mathsf{H_c}$
$6:\ b \leftarrow\!\!\$\ \{0,1\},\ k_1 \leftarrow\!\!\$\ \mathcal{K}$	
$7:\ \hat{r} \leftarrow C^{\lvert F\rangle, \lvert H\rangle, O_{\mathsf{Dec}}}(\mathsf{pk}, c^*, k_b)$	Decapsulation Oracle $O_{\mathsf{Dec}}(c)$:
$8:\ \textbf{return}\ [\hat{r} = \big(m_2^* \oplus F(m_1^*)\big)] \quad /\!\!/\ G_7^C$	$13:\ \textbf{if}\ c = c^*\ \textbf{then}\ :\textbf{return}\ \perp$
$9:\ \textbf{return}\ [\hat{r} = \big(m_2^* \oplus \mathsf{D_F}(m_1^*)\big)] \quad /\!\!/\ G_8^C$	$14:\ \textbf{return}\ f_1(c, \mathsf{D_H})$

Figure 16: Games G_7^C to G_8^C for the proof of Theorem 5.5.

Note that in Game G_7^C, after C outputs $\hat{r}$, the challenger effectively applies the unitary operator

$$\lvert (m_2^*, \hat{r}), m_1^*, 0\rangle \mapsto \begin{cases} \lvert (m_2^*, \hat{r}), m_1^*, 1\rangle\,, & \text{if } F(m_1^*) \in S_{(m_2^*,\hat{r}),m_1^*}\,, \\ \lvert (m_2^*, \hat{r}), m_1^*, 0\rangle\,, & \text{otherwise}\,, \end{cases}$$

where $S_{(m_2^*,\hat{r}),m_1^*} := \{u \mid m_2^* \oplus u = \hat{r}\}$, and then measures the last register in the computational basis to obtain the final outcome of the game. We replace $F(m_1^*)$ in this unitary by $D_{\mathsf{F}}(m_1^*)$, where $\lvert D_{\mathsf{F}}\rangle$ is the computational-basis state of $\mathsf{D_F}$. The modified check is denoted by $\hat{r} = \big(m_2^* \oplus \mathsf{D_F}(m_1^*)\big)$ (line 9 of Fig. 16). Since only one value $u = m_2^* \oplus \hat{r}$ satisfies $m_2^* \oplus u = \hat{r}$, we have $\Gamma_{S_{(m_2^*,\hat{r}),m_1^*}} = \max_{(m_2^*,\hat{r}),m_1^*} \left| S_{(m_2^*,\hat{r}),m_1^*} \right| = 1$. Therefore, by the Switching Lemma (Lemma 5.1),

$$\left| \sqrt{\Pr[Z_8^C]} - \sqrt{\Pr[Z_7^C]} \right| \le 6\sqrt{1/|\mathcal{M}_2|}\ .$$

Note that in Game G_8^C, when Z_8^C occurs, it means that if we further measure the register $\mathsf{D_F}$ in the computational basis, any resulting classical database D_{F} must necessarily contain m_1^*. However, in Game G_8^C, m_1^* is *not* used during the interaction between the challenger and C, so the probability that m_1^* appears in D_{F} is at most $|D_{\mathsf{F}}|/|\mathcal{M}_1|$ by a union bound. According to the earlier analysis in Game $G_7^{\mathcal{A}}$, we have $|D_{\mathsf{F}}| \le q_F + q_H$. Consequently,

$$\Pr[Z_8^C] \le (q_F + q_H)/|\mathcal{M}_1|\ .$$

Combining all bounds, we obtain

$$\begin{aligned} \mathsf{Adv}_{\mathsf{KEM}}^{\text{IND-CCA}}(\mathcal{A}) &\le 2\sqrt{q_H+1}\Big(\sqrt{2\mathsf{Adv}_{\mathsf{PKE}'}^{\text{IND-CPA}}(\mathcal{B}')} + \sqrt{4q_H}\big(\frac{\sqrt{q_F+q_H}}{\sqrt{|\mathcal{M}_1|}} + \frac{6}{\sqrt{|\mathcal{M}_2|}}\big)\Big) \\ &\quad + 16(q_H+1)\sqrt{\delta_{ac}} + \frac{6q_D}{\sqrt{|\mathcal{M}_1|}} + \frac{2q_D}{\sqrt{|\mathcal{M}_1 \times \mathcal{R} \times \mathcal{K}|}} \\ &\le 2\sqrt{2(q_H+1)}\sqrt{\mathsf{Adv}_{\mathsf{PKE}'}^{\text{IND-CPA}}(\mathcal{B}')} + 16(q_H+1)\sqrt{\delta_{ac}} \\ &\quad + (8q_D + 4(q_F+q_H+1)^{1.5})/\sqrt{|\mathcal{M}_1|} + 24(q_H+1)/\sqrt{|\mathcal{M}_2|}\ . \end{aligned}$$

□

References

[1] Ambainis, A., Hamburg, M., Unruh, D.: Quantum security proofs using semi-classical oracles. In: Boldyreva, A., Micciancio, D. (eds.) CRYPTO 2019, Part II. LNCS, vol. 11693, pp. 269–295. Springer, Cham (Aug 2019). https://doi.org/10.1007/978-3-030-26951-7_10

[2] Bellare, M., Rogaway, P.: Random oracles are practical: A paradigm for designing efficient protocols. In: Denning, D.E., Pyle, R., Ganesan, R., Sandhu, R.S., Ashby, V. (eds.) ACM CCS 93. pp. 62–73. ACM Press (Nov 1993). https://doi.org/10.1145/168588.168596

[3] BIKE team : Bike: Bit flipping key encapsulation (2024), available at https://bikesuite.org/files/v5.2/BIKE_Spec.2024.10.10.1.pdf

[4] Bindel, N., Hamburg, M., Hövelmanns, K., Hülsing, A., Persichetti, E.: Tighter proofs of CCA security in the quantum random oracle model. In: Hofheinz, D., Rosen, A. (eds.) TCC 2019, Part II. LNCS, vol. 11892, pp. 61–90. Springer, Cham (Dec 2019). https://doi.org/10.1007/978-3-030-36033-7_3

[5] Boneh, D., Dagdelen, Ö., Fischlin, M., Lehmann, A., Schaffner, C., Zhandry, M.: Random oracles in a quantum world. In: Lee, D.H., Wang, X. (eds.) ASIACRYPT 2011. LNCS, vol. 7073, pp. 41–69. Springer, Berlin, Heidelberg (Dec 2011). https://doi.org/10.1007/978-3-642-25385-0_3

[6] Boneh, D., Shoup, V.: A Graduate Course in Applied Cryptography (2023), available at http://toc.cryptobook.us/

[7] Chen, J., Wang, Y., Chen, R., Huang, X., Peng, W.: Tighter proofs for PKE-to-KEM transformation in the quantum random oracle model. In: Chung, K.M., Sasaki, Y. (eds.) ASIACRYPT 2024, Part IV. LNCS, vol. 15487, pp. 101–133. Springer, Singapore (Dec 2024). https://doi.org/10.1007/978-981-96-0894-2_4

[8] Classic McEliece team : Classic mceliece: conservative code-based cryptography: cryptosystem specification (2022), available at https://classic.mceliece.org/mceliece-spec-20221023.pdf

[9] Dent, A.W.: A designer's guide to KEMs. In: Paterson, K.G. (ed.) 9th IMA International Conference on Cryptography and Coding. LNCS, vol. 2898, pp. 133–151. Springer, Berlin, Heidelberg (Dec 2003). https://doi.org/10.1007/978-3-540-40974-8_12

[10] Don, J., Fehr, S., Majenz, C., Schaffner, C.: Online-extractability in the quantum random-oracle model. In: Dunkelman, O., Dziembowski, S. (eds.) EUROCRYPT 2022, Part III. LNCS, vol. 13277, pp. 677–706. Springer, Cham (May / Jun 2022). https://doi.org/10.1007/978-3-031-07082-2_24

[11] Duman, J., Hövelmanns, K., Kiltz, E., Lyubashevsky, V., Seiler, G., Unruh, D.: A thorough treatment of highly-efficient NTRU instantiations. In: Boldyreva, A., Kolesnikov, V. (eds.) PKC 2023, Part I. LNCS, vol. 13940, pp. 65–94. Springer, Cham (May 2023). https://doi.org/10.1007/978-3-031-31368-4_3

[12] Fujisaki, E., Okamoto, T.: Secure integration of asymmetric and symmetric encryption schemes. In: Wiener, M.J. (ed.) CRYPTO'99. LNCS, vol. 1666, pp. 537–554. Springer, Berlin, Heidelberg (Aug 1999). https://doi.org/10.1007/3-540-48405-1_34

[13] Fujisaki, E., Okamoto, T.: Secure integration of asymmetric and symmetric encryption schemes. Journal of Cryptology **26**(1), 80–101 (Jan 2013). https://doi.org/10.1007/s00145-011-9114-1

[14] Ge, J., Liao, H., Xue, R.: Measure-rewind-extract: Tighter proofs of one-way to hiding and CCA security in the quantum random oracle model. In: Chung, K.M., Sasaki, Y. (eds.) ASIACRYPT 2024, Part IV. LNCS, vol. 15487, pp. 3–34. Springer, Singapore (Dec 2024). https://doi.org/10.1007/978-981-96-0894-2_1

[15] Ge, J., Shan, T., Xue, R.: Tighter QCCA-secure key encapsulation mechanism with explicit rejection in the quantum random oracle model. In: Handschuh, H., Lysyanskaya, A. (eds.) CRYPTO 2023, Part V. LNCS, vol. 14085, pp. 292–324. Springer, Cham (Aug 2023). https://doi.org/10.1007/978-3-031-38554-4_10

[16] Ge, J., Yang, K., Yu, Y., Yu, Y.: Fujisaki-okamoto transformation under average-case decryption error: Tighter and more general proofs with applications to PQC. Cryptology ePrint Archive, Report 2025/1641 (2025), https://eprint.iacr.org/2025/1641

[17] Guo, F., Chen, R., Susilo, W., Lai, J., Yang, G., Mu, Y.: Optimal security reductions for unique signatures: Bypassing impossibilities with a counterexample. In: Katz, J., Shacham, H. (eds.) CRYPTO 2017, Part II. LNCS, vol. 10402, pp. 517–547. Springer, Cham (Aug 2017). https://doi.org/10.1007/978-3-319-63715-0_18

[18] Hoffstein, J., Pipher, J., Silverman, J.H.: NTRU: A ring-based public key cryptosystem. In: Third Algorithmic Number Theory Symposium (ANTS). LNCS, vol. 1423, pp. 267–288. Springer (Jun 1998)

[19] Hofheinz, D., Hövelmanns, K., Kiltz, E.: A modular analysis of the Fujisaki-Okamoto transformation. In: Kalai, Y., Reyzin, L. (eds.) TCC 2017, Part I. LNCS, vol. 10677, pp. 341–371. Springer, Cham (Nov 2017). https://doi.org/10.1007/978-3-319-70500-2_12

[20] Hövelmanns, K.: Generic constructions of quantum-resistant cryptosystems. Ph.D. thesis, Ruhr University Bochum, Germany (2021), available at https://hss-opus.ub.ruhr-uni-bochum.de/opus4/frontdoor/deliver/index/docId/7758/file/diss.pdf

[21] Hövelmanns, K., Hülsing, A., Majenz, C.: Failing gracefully: Decryption failures and the Fujisaki-Okamoto transform. In: Agrawal, S., Lin, D. (eds.) ASIACRYPT 2022, Part IV. LNCS, vol. 13794, pp. 414–443. Springer, Cham (Dec 2022). https://doi.org/10.1007/978-3-031-22972-5_15

[22] Hövelmanns, K., Kiltz, E., Schäge, S., Unruh, D.: Generic authenticated key exchange in the quantum random oracle model. In: Kiayias, A., Kohlweiss, M., Wallden, P., Zikas, V. (eds.) PKC 2020, Part II. LNCS, vol. 12111, pp. 389–422. Springer, Cham (May 2020). https://doi.org/10.1007/978-3-030-45388-6_14

[23] HQC team: Hamming quasi-cyclic (HQC) (2025), available at http://pqc-hqc.org/doc/hqc_specifications_2025_08_22.pdf

[24] Huguenin-Dumittan, L., Vaudenay, S.: On IND-qCCA security in the ROM and its applications - CPA security is sufficient for TLS 1.3. In: Dunkelman, O., Dziembowski, S. (eds.) EUROCRYPT 2022, Part III. LNCS, vol. 13277, pp. 613–642. Springer, Cham (May / Jun 2022). https://doi.org/10.1007/978-3-031-07082-2_22

[25] Jiang, H., Ma, Z., Zhang, Z.: Post-quantum security of key encapsulation mechanism against CCA attacks with a single decapsulation query. In: Guo, J., Steinfeld, R. (eds.) ASIACRYPT 2023, Part IV. LNCS, vol. 14441, pp. 434–468. Springer, Singapore (Dec 2023). https://doi.org/10.1007/978-981-99-8730-6_14

[26] Jiang, H., Zhang, Z., Chen, L., Wang, H., Ma, Z.: IND-CCA-secure key encapsulation mechanism in the quantum random oracle model, revisited. In: Shacham, H., Boldyreva, A. (eds.) CRYPTO 2018, Part III. LNCS, vol. 10993, pp. 96–125. Springer, Cham (Aug 2018). https://doi.org/10.1007/978-3-319-96878-0_4

[27] Jiang, H., Zhang, Z., Ma, Z.: Key encapsulation mechanism with explicit rejection in the quantum random oracle model. In: Lin, D., Sako, K. (eds.) PKC 2019, Part II. LNCS, vol. 11443, pp. 618–645. Springer, Cham (Apr 2019). https://doi.org/10.1007/978-3-030-17259-6_21

[28] Jiang, H., Zhang, Z., Ma, Z.: Tighter security proofs for generic key encapsulation mechanism in the quantum random oracle model. In: Ding, J., Steinwandt, R. (eds.) Post-Quantum Cryptography - 10th International Conference, PQCrypto 2019. pp. 227–248. Springer, Cham (2019). https://doi.org/10.1007/978-3-030-25510-7_13

[29] Jiang, H., Zhang, Z., Ma, Z.: On the non-tightness of measurement-based reductions for key encapsulation mechanism in the quantum random oracle model. In: Tibouchi, M., Wang, H. (eds.) ASIACRYPT 2021, Part I. LNCS, vol. 13090, pp. 487–517. Springer, Cham (Dec 2021). https://doi.org/10.1007/978-3-030-92062-3_17

[30] Kim, J., Park, J.H.: NTRU++: compact construction of NTRU using simple encoding method. IEEE Trans. Inf. Forensics Secur. **18**, 4760–4774 (2023)

[31] Kuchta, V., Sakzad, A., Stehlé, D., Steinfeld, R., Sun, S.: Measure-rewind-measure: Tighter quantum random oracle model proofs for one-way to hiding and CCA security. In: Canteaut, A., Ishai, Y. (eds.) EUROCRYPT 2020, Part III. LNCS, vol. 12107, pp. 703–728. Springer, Cham (May 2020). https://doi.org/10.1007/978-3-030-45727-3_24

[32] Liao, H., Ge, J., Cao, S., Xue, R.: New proof for plain OAEP: Post-quantum security without parameter restrictions or collision-resistance. Cryptology ePrint Archive, Report 2025/1288 (2025), https://eprint.iacr.org/2025/1288

[33] Lyubashevsky, V., Seiler, G.: NTTRU: Truly fast NTRU using NTT. IACR TCHES **2019**(3), 180–201 (2019). https://doi.org/10.13154/tches.v2019.i3.180-201, https://tches.iacr.org/index.php/TCHES/article/view/8293

[34] National Institute of Standards and Technology: Post-quantum cryptography project (2017), available at https://www.nist.gov/pqcrypto

[35] National Institute of Standards and Technology: Module-lattice-based key-encapsulation mechanism standard. Tech. Rep. Federal Information Processing Standards Publications (FIPS) 203, U.S. Department of Commerce, Washington, D.C. (2024). https://doi.org/10.6028/NIST.FIPS.203

[36] Nielsen, M.A., Chuang, I.L.: Quantum Computation and Quantum Information (10th Anniversary edition). Cambridge University Press (2016)

[37] NSR, NIS: National security research institute and national intelligence service: Korean post-quantum cryptography (KpqC) competition (2021), available at https://kpqc.or.kr/competition.html

[38] NSR, NIS: National security research institute and national intelligence service: Selected algorithms from the KpqC competition round 2 (2025), available at https://kpqc.or.kr/competition_02.html

[39] Pan, J., Riepel, D., Zeng, R.: Key exchange with tight (full) forward secrecy via key confirmation. In: Joye, M., Leander, G. (eds.) EUROCRYPT 2024, Part VII. LNCS, vol. 14657, pp. 59–89. Springer, Cham (May 2024). https://doi.org/10.1007/978-3-031-58754-2_3

[40] Rackoff, C., Simon, D.R.: Non-interactive zero-knowledge proof of knowledge and chosen ciphertext attack. In: Feigenbaum, J. (ed.) CRYPTO'91. LNCS, vol. 576, pp. 433–444. Springer, Berlin, Heidelberg (Aug 1992). https://doi.org/10.1007/3-540-46766-1_35

[41] Saito, T., Xagawa, K., Yamakawa, T.: Tightly-secure key-encapsulation mechanism in the quantum random oracle model. In: Nielsen, J.B., Rijmen, V. (eds.) EUROCRYPT 2018, Part III. LNCS, vol. 10822, pp. 520–551. Springer, Cham (Apr / May 2018). https://doi.org/10.1007/978-3-319-78372-7_17

[42] Yamakawa, T., Zhandry, M.: Classical vs quantum random oracles. In: Canteaut, A., Standaert, F.X. (eds.) EUROCRYPT 2021, Part II. LNCS, vol. 12697, pp. 568–597. Springer, Cham (Oct 2021). https://doi.org/10.1007/978-3-030-77886-6_20

[43] Zhandry, M.: How to record quantum queries, and applications to quantum indifferentiability. In: Boldyreva, A., Micciancio, D. (eds.) CRYPTO 2019, Part II. LNCS, vol. 11693, pp. 239–268. Springer, Cham (Aug 2019). https://doi.org/10.1007/978-3-030-26951-7_9

[44] Zhang, J., Yu, Y., Feng, D., Fan, S., Zhang, Z.: On the (quantum) random oracle methodology: New separations and more. Cryptology ePrint Archive, Report 2019/1101 (2019), https://eprint.iacr.org/2019/1101

A Supporting Material: Proof for the Lemmas in Sec. 5.1

A.1 Auxiliary Lemma

For density operators ρ and σ, the fidelity is defined as

$$F(\rho, \sigma) := \operatorname{tr}\sqrt{\rho^{1/2}\sigma\rho^{1/2}}\ , \qquad [36, (9.53)]$$

and the Bures distance is defined as

$$B(\rho, \sigma) := \sqrt{2 - 2F(\rho, \sigma)}\ . \qquad [1, \text{Section 5.1}]$$

Lemma A.1 (Bures Distance Bound for Convex Mixtures)**.** *Let $\{p_i\}_i$ be a probability distribution, and let $\{|\psi_i\rangle\}_i$, $\{|\phi_i\rangle\}_i$ be two families of pure states. Define the corresponding density operators*

$$\rho_1 := \sum_i p_i |\psi_i\rangle\langle\psi_i|\,, \quad \rho_2 := \sum_i p_i |\phi_i\rangle\langle\phi_i|\ .$$

Then the Bures distance between ρ_1 and ρ_2 satisfies

$$B(\rho_1, \rho_2) \le \max_i \| |\psi_i\rangle - |\phi_i\rangle \|\ .$$

Lemma A.1. Using the relation $B(\rho,\sigma) = \sqrt{2 - 2F(\rho,\sigma)}$, we have

$$\begin{aligned} B(\rho_1,\rho_2)^2 &= 2 - 2F\Big(\sum_i p_i \left|\psi_i\right\rangle\!\left\langle\psi_i\right|, \sum_i p_i \left|\phi_i\right\rangle\!\left\langle\phi_i\right|\Big) \\ &\overset{(a)}{\leq} 2 - 2\sum_i p_i\, F\big(\left|\psi_i\right\rangle\!\left\langle\psi_i\right|, \left|\phi_i\right\rangle\!\left\langle\phi_i\right|\big) \ , \end{aligned}$$

where (a) follows from the joint concavity of fidelity [36, (9.95)]. For pure states,

$$F\big(\left|\psi_i\right\rangle\!\left\langle\psi_i\right|, \left|\phi_i\right\rangle\!\left\langle\phi_i\right|\big) \geq 1 - \frac{1}{2}\left\|\left|\psi_i\right\rangle - \left|\phi_i\right\rangle\right\|^2 \ ,$$

as shown in [1, Lemma 3]. Combining this with the previous step, we obtain

$$B(\rho_1,\rho_2)^2 \leq \sum_i p_i \left\|\left|\psi_i\right\rangle - \left|\phi_i\right\rangle\right\|^2 \ .$$

Because the weights $\{p_i\}$ form a probability distribution, the right-hand side is at most $\max_i \left\|\left|\psi_i\right\rangle - \left|\phi_i\right\rangle\right\|^2$. Taking the square root completes the proof. □

A.2 Proof for the Switching Lemma

Lemma 5.1 (Switching Lemma)**.** *Let $\mathsf{H} : \mathcal{X} \to \mathcal{Y}$ be a quantum-accessible random oracle, and let O_0 be a quantum oracle acting on the input registers WX (defined over $\mathcal{W} \times \mathcal{X}$) and an output register $\bar{\mathsf{Y}}$ (defined over $\bar{\mathcal{Y}}$) as*

$$\mathsf{O}_0 : \left|w,x,\bar{y}\right\rangle_{\mathsf{WX\bar{Y}}} \mapsto \left|w,x,\bar{y} \oplus f(\mathsf{H}(x))\right\rangle_{\mathsf{WX\bar{Y}}} \ \textit{with}\ f(y) = \begin{cases} g_{w,x}(y), & \textit{if } y \in S_{w,x} \\ g_{w,x}(\perp), & \textit{otherwise} \ , \end{cases}$$

where both the function $g_{w,x} : \mathcal{Y} \cup \{\perp\} \to \bar{\mathcal{Y}}$ and the set $S_{w,x} \subseteq \mathcal{Y}$ are determined by (w,x). Let $\mathsf{H_c} : \mathcal{X} \to \mathcal{Y}$ be a compressed standard oracle with a database register D initialized to $\bigotimes_{i=1}^{q_{\mathsf{H}}} \left|(\perp, 0^{\log|\mathcal{Y}|})\right\rangle$, and let O_1 be the quantum oracle

$$\mathsf{O}_1 : \left|w,x,D,\bar{y}\right\rangle_{\mathsf{WXD\bar{Y}}} \mapsto \left|w,x,D,\bar{y} \oplus f(D(x))\right\rangle_{\mathsf{WXD\bar{Y}}}$$

which uses the same input/output registers $\mathsf{WX}/\bar{\mathsf{Y}}$ as O_0 and additionally employs D as a work register.

Let A be a quantum oracle algorithm (not necessarily unitary) making at most q_{H} queries to $\mathsf{H}/\mathsf{H_c}$ and at most q_{O} queries to $\mathsf{O}_0/\mathsf{O}_1$. For an arbitrary classical event Ev and a random z distributed according to $\mathfrak{D}_z$, define

$$P_{\text{left}} \coloneqq \Pr[\mathsf{Ev} : A^{|\mathsf{H}\rangle,|\mathsf{O}_0\rangle}(z)] \ \textit{and}\ P_{\text{right}} \coloneqq \Pr[\mathsf{Ev} : A^{|\mathsf{H_c}\rangle,|\mathsf{O}_1\rangle}(z)] \ ,$$

where the probability is taken over $z \leftarrow \mathfrak{D}_z$ and the internal randomness of A (including all quantum measurements). Setting $\Gamma_S \coloneqq \max_{w,x}|S_{w,x}|$, we have

$$|P_{\text{left}} - P_{\text{right}}| \leq 6q_{\mathsf{O}}\sqrt{\Gamma_S/|\mathcal{Y}|} \quad \textit{and} \quad \left|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}\right| \leq 6q_{\mathsf{O}}\sqrt{\Gamma_S/|\mathcal{Y}|} \ .$$

Lemma 5.1. For the quantum oracle algorithm $A^{|\mathsf{H}\rangle,|\mathsf{O}_0\rangle}(z)$ with quantum access to the random oracle $\mathsf{H} : \mathcal{X} \to \mathcal{Y}$ and the quantum oracle O_0, we perform the following replacement. Let q_{H} and q_{O} be the bounded number of queries that A makes to H and to O_0, respectively. First, H is replaced with the compressed standard oracle $\mathsf{H_c}$, which is implemented with a database register D initialized to $\bigotimes_{i=1}^{q_{\mathsf{H}}+2q_{\mathsf{O}}} |(\perp, 0^{\log|\mathcal{Y}|})\rangle$. Then, every call to H inside the oracle O_0 is replaced by a call to H_{c}; the resulting oracle is denoted by $\mathsf{O}_0^{\mathsf{c}}$. Note that, in addition to the at most q_{H} direct queries that A makes to H_{c}, each execution of $\mathsf{O}_0^{\mathsf{c}}$ itself requires two queries to H_{c} (as will be shown below). Consequently, during the whole execution of A, the oracle H_{c} is invoked at most $q_{\mathsf{H}}+2q_{\mathsf{O}}$ times, which explains the chosen dimension of the database register D. Since this modification only replaces calls to H with calls to H_{c} (and leaves all other operations unchanged), Lemma 3.1 directly gives

$$\Pr[\mathsf{Ev} : A^{|\mathsf{H}\rangle,|\mathsf{O}_0\rangle}(z)] = \Pr[\mathsf{Ev} : A^{|\mathsf{H}_{\mathsf{c}}\rangle,|\mathsf{O}_0^{\mathsf{c}}\rangle}(z)] \ . \tag{3}$$

Without loss of generality, the execution of $A^{|\mathsf{H}_{\mathsf{c}}\rangle,|\mathsf{O}_0^{\mathsf{c}}\rangle}(z)$ can be formulated as $U_{q_{\mathsf{O}}+1} \circ \prod_{i=1}^{q_{\mathsf{O}}} \left(\mathsf{O}_0^{\mathsf{c}} \circ U_i\right) |\psi_z\rangle$, where $|\psi_z\rangle$ is the initial state of $A^{|\mathsf{H}_{\mathsf{c}}\rangle,|\mathsf{O}_0^{\mathsf{c}}\rangle}(z)$ that depends on the input z, and for $i = 1, \dots, q_{\mathsf{O}} + 1$, each U_i is a unitary operator[7] that encompasses all operations of A (including queries to the random oracle H_{c}) except the queries to $\mathsf{O}_0^{\mathsf{c}}$. Furthermore, by the *principle of deferred measurement* [36], any intermediate measurement can be postponed to the end of the circuit. Thus, the classical event Ev can be regarded as the outcome of a projective measurement on the final state of $A^{|\mathsf{H}_{\mathsf{c}}\rangle,|\mathsf{O}_0^{\mathsf{c}}\rangle}(z)$.

Let $f_0 : \mathcal{W} \times \mathcal{X} \times \mathcal{Y} \to \bar{\mathcal{Y}}$ be the function defined as

$$f_0(w, x, y) = \begin{cases} g_{w,x}(y), & \text{if } y \in S_{w,x} \ , \\ g_{w,x}(\perp), & \text{otherwise} \ , \end{cases}$$

where the functions $g_{w,x} : \mathcal{Y} \cup \{\perp\} \to \bar{\mathcal{Y}}$ and sets $S_{w,x}$ are as defined in Lemma 5.1. Since f_0 can be computed efficiently, the unitary operation $U_{f_0} : |w, x, y, \bar{y}\rangle \mapsto |w, x, y, \bar{y} \oplus f_0(w, x, y)\rangle$ can also be implemented efficiently based on the quantum computation theory. Let CStO be the unitary operator that implements H_{c}. Then the quantum circuit realizing the oracle $\mathsf{O}_0^{\mathsf{c}}$ can be constructed as shown in Fig. 17, i.e.,

1. Initialize an internal register Y defined over $\mathcal{Y}$ to $|0^{\log|\mathcal{Y}|}\rangle$.
2. Apply CStO to registers XYD, with Y as the target output register.
3. Apply U_{f_0} to registers $\mathsf{WXY\bar{Y}}$, with $\mathsf{\bar{Y}}$ as the target output register.
4. Apply CStO again to registers XYD, with Y as the target output register. This step uncomputes Y, returning it to $|0^{\log|\mathcal{Y}|}\rangle$, so it can be discarded.

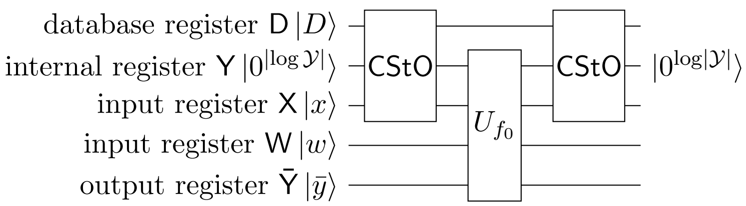

Figure 17: The quantum circuit diagram for $\mathsf{O}_0^{\mathsf{c}}$.

Let $|\Psi\rangle$ denote the state of $A^{|\mathsf{H}_{\mathsf{c}}\rangle,|\mathsf{O}_0^{\mathsf{c}}\rangle}(z)$ immediately before an (arbitrary) query to the quantum oracle $\mathsf{O}_0^{\mathsf{c}}$, i.e.,

$$|\Psi\rangle := U_j \circ \prod_{i=1}^{j-1} \left(\mathsf{O}_0^{\mathsf{c}} \circ U_i\right) |\psi_z\rangle$$

[7]This follows from the fact that any quantum oracle algorithm can be transformed into a *unitary* quantum oracle algorithm with only constant factor computational overhead and the same number of oracle queries [1,15].

for some $j \leq q_{\mathsf{O}}$. This state resides in registers $\mathsf{WXD\bar{Y}P}$, where $\mathsf{WX}/\mathsf{\bar{Y}}$ are the input/output registers of $\mathsf{O}_0^{\mathsf{c}}$, D is the database register used by CStO, and P contains all remaining registers. Let $n := \log|\mathcal{Y}|$. We can decompose $|\Psi\rangle$ into three mutually orthogonal components $|\Psi\rangle = |\Psi_1\rangle + |\Psi_2\rangle + |\Psi_3\rangle$, with

$$|\Psi_1\rangle = \sum\nolimits_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp}} \beta_{\bar{y}p}^{wxD} |w, x, D, \bar{y}, p\rangle \ ,$$

$$|\Psi_2\rangle = \sum\nolimits_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp, r\in\mathcal{Y}\setminus\{0^n\}}} \frac{\beta_{\bar{y}pr}^{wxD}}{\sqrt{2^n}} \sum\nolimits_{y_1\in\mathcal{Y}} (-1)^{r\cdot y_1} |w, x, D \cup (x, y_1), \bar{y}, p\rangle \ ,$$

$$|\Psi_3\rangle = \sum\nolimits_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp, r=0^n}} \frac{\beta_{\bar{y}p0}^{wxD}}{\sqrt{2^n}} \sum\nolimits_{y_1\in\mathcal{Y}} |w, x, D \cup (x, y_1), \bar{y}, p\rangle \ .$$

Since D is an internal register of CStO, no operation in U_i other than CStO acts on D. As shown in Fig. 17, the register D accessed by $\mathsf{O}_0^{\mathsf{c}}$ is likewise only used by CStO. Thus, it follows from [10, Section 2.2] that the component $|\Psi_3\rangle$ does not appear in $|\Psi\rangle$. Hence,

$$|\Psi\rangle = |\Psi_1\rangle + |\Psi_2\rangle \ .$$

Based on the analysis in section A.3, we have

$$\|(\mathsf{O}_0^{\mathsf{c}} - \mathsf{O}_1)\,|\Psi\rangle\| \leq 6\sqrt{\Gamma_S/|\mathcal{Y}|} \ . \tag{4}$$

For a fixed input z, we denote the final state of $A^{|\mathsf{H}_{\mathsf{c}}\rangle,|\mathsf{O}_0^{\mathsf{c}}\rangle}(z)$ by $|\Psi_z^0\rangle := U_{q_{\mathsf{O}}+1} \circ \prod_{i=1}^{q_{\mathsf{O}}}\big(\mathsf{O}_0^{\mathsf{c}} \circ U_i\big)\,|\psi_z\rangle$. Replacing every call to $\mathsf{O}_0^{\mathsf{c}}$ with O_1 yields the corresponding final state of $A^{|\mathsf{H}_{\mathsf{c}}\rangle,|\mathsf{O}_1\rangle}(z)$, defined as $|\Psi_z^1\rangle := U_{q_{\mathsf{O}}+1} \circ \prod_{i=1}^{q_{\mathsf{O}}}\big(\mathsf{O}_1 \circ U_i\big)\,|\psi_z\rangle$. We now aim to bound $\big\||\Psi_z^0\rangle - |\Psi_z^1\rangle\big\|$. To simplify the notation, we define $\Delta_j^k := \prod_{i=j}^{k}\big(\mathsf{O}_0^{\mathsf{c}} \circ U_i\big)$ and $\Upsilon_j^k := \prod_{i=j}^{k}\big(\mathsf{O}_1 \circ U_i\big)$, with the convention that $\Delta_j^k = \Upsilon_j^k = I$ (the identity operator) whenever $k < j$. Then, we have

$$\begin{aligned}
&\big\||\Psi_z^0\rangle - |\Psi_z^1\rangle\big\| \\
&= \|U_{q_{\mathsf{O}}+1} \circ \Delta_1^{q_{\mathsf{O}}}\,|\psi_z\rangle - U_{q_{\mathsf{O}}+1} \circ \Upsilon_1^{q_{\mathsf{O}}}\,|\psi_z\rangle\| \\
&\overset{(a)}{=} \|\Delta_1^{q_{\mathsf{O}}}\,|\psi_z\rangle - \Upsilon_1^{q_{\mathsf{O}}}\,|\psi_z\rangle\| \\
&\overset{(b)}{=} \left\|\sum_{i=1}^{q_{\mathsf{O}}}\Big(\Upsilon_{i+1}^{q_{\mathsf{O}}} \circ (\mathsf{O}_0^{\mathsf{c}} \circ U_i) \circ \Delta_1^{i-1}\,|\psi_z\rangle - \Upsilon_{i+1}^{q_{\mathsf{O}}} \circ (\mathsf{O}_1 \circ U_i) \circ \Delta_1^{i-1}\,|\psi_z\rangle)\Big)\right\| \\
&\overset{(c)}{\leq} \sum_{i=1}^{q_{\mathsf{O}}}\big\|\Upsilon_{i+1}^{q_{\mathsf{O}}} \circ (\mathsf{O}_0^{\mathsf{c}} \circ U_i) \circ \Delta_1^{i-1}\,|\psi_z\rangle - \Upsilon_{i+1}^{q_{\mathsf{O}}} \circ (\mathsf{O}_1 \circ U_i) \circ \Delta_1^{i-1}\,|\psi_z\rangle\big\| \\
&\overset{(d)}{=} \sum_{i=1}^{q_{\mathsf{O}}}\big\|(\mathsf{O}_0^{\mathsf{c}} - \mathsf{O}_1) \circ U_i \circ \Delta_1^{i-1}\,|\psi_z\rangle\big\| \overset{(e)}{\leq} \sum_{i=1}^{q_{\mathsf{O}}} 6\sqrt{\Gamma_S/|\mathcal{Y}|} = 6q_{\mathsf{O}}\sqrt{\Gamma_S/|\mathcal{Y}|} \ ,
\end{aligned}$$

where (a) and (d) follow from the norm-preserving property of the unitary operators $U_{q_{\mathsf{O}}+1}$ and $\Upsilon_{i+1}^{q_{\mathsf{O}}}$; (b) follows from $\Upsilon_{i+1}^{q_{\mathsf{O}}} \circ (\mathsf{O}_0^{\mathsf{c}} \circ U_i) \circ \Delta_1^{i-1}\,|\psi_z\rangle = \Upsilon_{i+2}^{q_{\mathsf{O}}} \circ (\mathsf{O}_1 \circ U_{i+1}) \circ \Delta_1^{i}\,|\psi_z\rangle$ for $i = 1, \ldots, q_{\mathsf{O}}-1$; (c) uses the triangle inequality; and (e) applies (4), with the observation that $U_i \circ \Delta_1^{i-1}\,|\psi_z\rangle$ is exactly the state of $A^{|\mathsf{H}_{\mathsf{c}}\rangle,|\mathsf{O}_0^{\mathsf{c}}\rangle}(z)$ just before the i-th call to $\mathsf{O}_0^{\mathsf{c}}$.

For a random input $z \leftarrow \mathfrak{D}_z$, the final states of $A^{|\mathsf{H}_{\mathsf{c}}\rangle,|\mathsf{O}_0^{\mathsf{c}}\rangle}(z)$ and $A^{|\mathsf{H}_{\mathsf{c}}\rangle,|\mathsf{O}_1\rangle}(z)$ are the density operators $\rho_0 := \sum_z p_z\,|\Psi_z^0\rangle\langle\Psi_z^0|$ and $\rho_1 := \sum_z p_z\,|\Psi_z^1\rangle\langle\Psi_z^1|$ respectively, where $p_z = \Pr_{z\leftarrow\mathfrak{D}_z}[z]$. By Lemma A.1, we have

$$B(\rho_0, \rho_1) \leq \max_z \big\||\Psi_z^0\rangle - |\Psi_z^1\rangle\big\| \leq 6q_{\mathsf{O}}\sqrt{\Gamma_S/|\mathcal{Y}|} \ .$$

Consequently, we can obtain

$$\begin{aligned} \left|\Pr[\mathsf{Ev} : A^{|\mathsf{H_c}\rangle,|\mathsf{O}_0^{\mathsf{c}}\rangle}(z)] - \Pr[\mathsf{Ev} : A^{|\mathsf{H_c}\rangle,|\mathsf{O}_1\rangle}(z)]\right| &\overset{(f)}{\leq} B(\rho_0,\rho_1) \leq 6q_{\mathsf{O}}\sqrt{\Gamma_S/|\mathcal{Y}|} \ , \\ \left|\sqrt{\Pr[\mathsf{Ev} : A^{|\mathsf{H_c}\rangle,|\mathsf{O}_0^{\mathsf{c}}\rangle}(z)]} - \sqrt{\Pr[\mathsf{Ev} : A^{|\mathsf{H_c}\rangle,|\mathsf{O}_1\rangle}(z)]}\right| &\overset{(g)}{\leq} B(\rho_0,\rho_1) \leq 6q_{\mathsf{O}}\sqrt{\Gamma_S/|\mathcal{Y}|} \ , \end{aligned} \tag{5}$$

where (f) and (g) follow from [1, Lemma 4]. Note that the execution of O_1 does *not* require calls to the random oracle $\mathsf{H_c}$. Consequently, the database register D used by $\mathsf{H_c}$ within $A^{|\mathsf{H_c}\rangle,|\mathsf{O}_1\rangle}(z)$ only needs q_{H} entries. Combining (5) with (3), we complete the proof of Lemma 5.1. $\square$

A.3 Bound on $\|(\mathsf{O}_0^{\mathsf{c}} - \mathsf{O}_1)\,|\psi\rangle\|$

From the decryption of $\mathsf{O}_0^{\mathsf{c}}$, O_1, $|\Psi_1\rangle$, and $|\Psi_2\rangle$ in the proof of Lemma 5.1 and the definition of CStO in Definition 3.7, we obtain[8]

$$\begin{aligned} \mathsf{O}_0^{\mathsf{c}}\,|\Psi_1\rangle &= \mathsf{CStO} \circ U_{f_0} \circ \mathsf{CStO} \sum\nolimits_{\substack{w,x,D,\bar{y},p,\\ D(x)=\bot}} \beta_{\bar{y}p}^{wxD}\,|w,x,D,\bar{y},p\rangle \\ &= \sum\nolimits_{\substack{w,x,D,\bar{y},p,\\ D(x)=\bot}} \frac{\beta_{\bar{y}p}^{wxD}}{\sqrt{2^n}} \sum\nolimits_{y_1\in S_{w,x}} F_x\,|w,x,D\cup(x,y_1),\bar{y}\oplus g(y_1),p\rangle \\ &\quad + \sum\nolimits_{\substack{w,x,D,\bar{y},p,\\ D(x)=\bot}} \frac{\beta_{\bar{y}p}^{wxD}}{\sqrt{2^n}} \sum\nolimits_{y_1\notin S_{w,x}} F_x\,|w,x,D\cup(x,y_1),\bar{y}\oplus g(\bot),p\rangle \\ &= \sum\nolimits_{\substack{w,x,D,\bar{y},p,\\ D(x)=\bot}} \frac{\beta_{\bar{y}p}^{wxD}}{\sqrt{2^n}} \sum\nolimits_{y_1\in S_{w,x}} F_x\,|w,x,D\cup(x,y_1),\bar{y}\oplus g(y_1),p\rangle \\ &\quad + \sum\nolimits_{\substack{w,x,D,\bar{y},p,\\ D(x)=\bot}} \beta_{\bar{y}p}^{wxD}\,|w,x,D,\bar{y}\oplus g(\bot),p\rangle \\ &\quad - \sum\nolimits_{\substack{w,x,D,\bar{y},p,\\ D(x)=\bot}} \frac{\beta_{\bar{y}p}^{wxD}}{\sqrt{2^n}} \sum\nolimits_{y_1\in S_{w,x}} F_x\,|w,x,D\cup(x,y_1),\bar{y}\oplus g(\bot),p\rangle \ , \end{aligned}$$

$$\mathsf{O}_1\,|\Psi_1\rangle = \mathsf{O}_1 \sum\nolimits_{\substack{w,x,D,\bar{y},p,\\ D(x)=\bot}} \beta_{\bar{y}p}^{wxD}|w,x,D,\bar{y},p\rangle = \sum\nolimits_{\substack{w,x,D,\bar{y},p,\\ D(x)=\bot}} \beta_{\bar{y}p}^{wxD}|w,x,D,\bar{y}\oplus g(\bot),p\rangle \ ,$$

$$\mathsf{O}_0^{\mathsf{c}}\,|\Psi_2\rangle = \mathsf{CStO} \circ U_{f_0} \circ \mathsf{CStO} \sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\bot,r\in\mathcal{Y}\setminus\{0^n\}}} \frac{\beta_{\bar{y}pr}^{wxD}}{\sqrt{2^n}} \sum\nolimits_{y_1\in\mathcal{Y}} (-1)^{r\cdot y_1}\,|w,x,D\cup(x,y_1),\bar{y},p\rangle$$

$$\begin{aligned} = \sum\nolimits_{\substack{w,x,D,\bar{y},p,\\ D(x)=\bot,r\in\mathcal{Y}\setminus\{0^n\}}} \frac{\beta_{\bar{y}pr}^{wxD}}{\sqrt{2^n}} \Big(&\sum\nolimits_{y_1\in S_{w,x}} (-1)^{r\cdot y_1} F_x\,|w,x,D\cup(x,y_1),\bar{y}\oplus g(y_1),p\rangle \\ &+ \sum\nolimits_{y_1\notin S_{w,x}} (-1)^{r\cdot y_1} F_x\,|w,x,D\cup(x,y_1),\bar{y}\oplus g(\bot),p\rangle \Big) \end{aligned}$$

$$\begin{aligned} = &\sum\nolimits_{\substack{w,x,D,\bar{y},p,\\ D(x)=\bot,r\in\mathcal{Y}\setminus\{0^n\}}} \frac{\beta_{\bar{y}pr}^{wxD}}{\sqrt{2^n}} \Big(\sum\nolimits_{y_1\in S_{w,x}} (-1)^{r\cdot y_1}\,|w,x,D\cup(x,y_1),\bar{y}\oplus g(y_1),p\rangle \\ &\qquad + \sum\nolimits_{y_1\notin S_{w,x}} (-1)^{r\cdot y_1}\,|w,x,D\cup(x,y_1),\bar{y}\oplus g(\bot),p\rangle \Big) \\ &+ \sum\nolimits_{\substack{w,x,D,\bar{y},p,\\ D(x)=\bot,r\in\mathcal{Y}\setminus\{0^n\}}} \frac{\beta_{\bar{y}pr}^{wxD}}{2^n} \Big(\sum\nolimits_{y_1\in S_{w,x}} (-1)^{r\cdot y_1}\,|w,x,D,\bar{y}\oplus g(y_1),p\rangle \\ &\qquad - \sum\nolimits_{y_1\in S_{w,x}} (-1)^{r\cdot y_1} \sum\nolimits_{y_2\in\mathcal{Y}} \frac{1}{\sqrt{2^n}}\,|w,x,D\cup(x,y_2),\bar{y}\oplus g(y_1),p\rangle \\ &\qquad + \sum\nolimits_{y_1\notin S_{w,x}} (-1)^{r\cdot y_1}\,|w,x,D,\bar{y}\oplus g(\bot),p\rangle \\ &\qquad - \sum\nolimits_{y_1\notin S_{w,x}} (-1)^{r\cdot y_1} \sum\nolimits_{y_2\in\mathcal{Y}} \frac{1}{\sqrt{2^n}}\,|w,x,D\cup(x,y_2),\bar{y}\oplus g(\bot),p\rangle \Big) \end{aligned}$$

[8] In the following derivation, we omit the subscript of $g_{w,x}$ for notational simplicity.

$$
\begin{aligned}
&\overset{(a)}{=} \sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp, r\in\mathcal{Y}\setminus\{0^n\}}} \frac{\beta^{wxD}_{\bar{y}pr}}{\sqrt{2^n}} \Big(\sum_{y_1\in S_{w,x}} (-1)^{r\cdot y_1} \left|w,x,D\cup(x,y_1),\bar{y}\oplus g(y_1),p\right\rangle \\
&\qquad + \sum_{y_1\notin S_{w,x}} (-1)^{r\cdot y_1} \left|w,x,D\cup(x,y_1),\bar{y}\oplus g(\perp),p\right\rangle \Big) \\
&+ \sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp, r\in\mathcal{Y}\setminus\{0^n\}}} \frac{\beta^{wxD}_{\bar{y}pr}}{2^n} \Big(\sum_{y_1\in S_{w,x}} (-1)^{r\cdot y_1} \left|w,x,D,\bar{y}\oplus g(y_1),p\right\rangle \\
&\qquad - \sum_{y_1\in S_{w,x}} (-1)^{r\cdot y_1} \sum_{y_2\in\mathcal{Y}} \frac{1}{\sqrt{2^n}} \left|w,x,D\cup(x,y_2),\bar{y}\oplus g(y_1),p\right\rangle \\
&\qquad - \sum_{y_1\in S_{w,x}} (-1)^{r\cdot y_1} \left|w,x,D,\bar{y}\oplus g(\perp),p\right\rangle \\
&\qquad + \sum_{y_1\in S_{w,x}} (-1)^{r\cdot y_1} \sum_{y_2\in\mathcal{Y}} \frac{1}{\sqrt{2^n}} \left|w,x,D\cup(x,y_2),\bar{y}\oplus g(\perp),p\right\rangle \Big) ,
\end{aligned}
$$

$$
\begin{aligned}
\mathsf{O}_1 \left|\Psi_2\right\rangle = &\sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp, r\in\mathcal{Y}\setminus\{0^n\}}} \frac{\beta^{wxD}_{\bar{y}pr}}{\sqrt{2^n}} \Big(\sum_{y_1\in S_{w,x}} (-1)^{r\cdot y_1} \left|w,x,D\cup(x,y_1),\bar{y}\oplus g(y_1),p\right\rangle \\
&+ \sum_{y_1\notin S_{w,x}} (-1)^{r\cdot y_1} \left|w,x,D\cup(x,y_1),\bar{y}\oplus g(\perp),p\right\rangle \Big) ,
\end{aligned}
$$

where F_x is s the decompression procedure applied to D as defined in Definition 3.7, and (a) uses the fact that for any $r \neq 0^n$, $\sum_{y_1\in\mathcal{Y}}(-1)^{r\cdot y_1} = 0$.

Then we have

$$
\begin{aligned}
\left\|(\mathsf{O}_0^{\mathsf{c}} - \mathsf{O}_1)\left|\Psi_1\right\rangle\right\|^2 = &\Big\| \sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp}} \frac{\beta^{wxD}_{\bar{y}p}}{\sqrt{2^n}} \sum_{y_1\in S_{w,x}} F_x \left|w,x,D\cup(x,y_1),\bar{y}\oplus g(y_1),p\right\rangle \\
&- \sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp}} \frac{\beta^{wxD}_{\bar{y}p}}{\sqrt{2^n}} \sum_{y_1\in S_{w,x}} F_x \left|w,x,D\cup(x,y_1),\bar{y}\oplus g(\perp),p\right\rangle \Big\|^2 \\
\overset{(b)}{\leq} &\, 2\Big\| \sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp}} \frac{\beta^{wxD}_{\bar{y}p}}{\sqrt{2^n}} \sum_{y_1\in S_{w,x}} F_x \left|w,x,D\cup(x,y_1),\bar{y}\oplus g(y_1),p\right\rangle \Big\|^2 \\
&+ 2\Big\| \sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp}} \frac{\beta^{wxD}_{\bar{y}p}}{\sqrt{2^n}} \sum_{y_1\in S_{w,x}} F_x \left|w,x,D\cup(x,y_1),\bar{y}\oplus g(\perp),p\right\rangle \Big\|^2 \\
= &\, 2\sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp}} \frac{\left|\beta^{wxD}_{\bar{y}p}\right|^2}{2^n} \sum_{y_1\in S_{w,x}} \Big\| F_x \left|w,x,D\cup(x,y_1),\bar{y}\oplus g(y_1),p\right\rangle \Big\|^2 \\
&+ 2\sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp}} \frac{\left|\beta^{wxD}_{\bar{y}p}\right|^2}{2^n} \sum_{y_1\in S_{w,x}} \Big\| F_x \left|w,x,D\cup(x,y_1),\bar{y}\oplus g(\perp),p\right\rangle \Big\|^2 \\
= &\, 4 \sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp}} \frac{\left|\beta^{wxD}_{\bar{y}p}\right|^2}{2^n} |S_{w,x}| \leq \frac{4}{2^n}\Gamma_S \sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp}} \left|\beta^{wxD}_{\bar{y}p}\right|^2 \overset{(c)}{=} \frac{4}{2^n}\Gamma_S \left\|\left|\Psi_1\right\rangle\right\|^2 ,
\end{aligned}
$$

$$
\begin{aligned}
&\left\|(\mathsf{O}_0^{\mathsf{c}} - \mathsf{O}_1)\left|\Psi_2\right\rangle\right\|^2 \\
&= \Big\| \sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp, r\in\mathcal{Y}\setminus\{0^n\}}} \frac{\beta^{wxD}_{\bar{y}pr}}{2^n} \Big(\sum_{y_1\in S_{w,x}} (-1)^{r\cdot y_1} \left|w,x,D,\bar{y}\oplus g(y_1),p\right\rangle \\
&\qquad - \sum_{y_1\in S_{w,x}} (-1)^{r\cdot y_1} \sum_{y_2\in\mathcal{Y}} \frac{1}{\sqrt{2^n}} \left|w,x,D\cup(x,y_2),\bar{y}\oplus g(y_1),p\right\rangle \\
&\qquad - \sum_{y_1\in S_{w,x}} (-1)^{r\cdot y_1} \left|w,x,D,\bar{y}\oplus g(\perp),p\right\rangle \\
&\qquad + \sum_{y_1\in S_{w,x}} (-1)^{r\cdot y_1} \sum_{y_2\in\mathcal{Y}} \frac{1}{\sqrt{2^n}} \left|w,x,D\cup(x,y_2),\bar{y}\oplus g(\perp),p\right\rangle \Big) \Big\|^2
\end{aligned}
$$

$$
\begin{aligned}
&\overset{(d)}{\leq} 4\Big\|\sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp, r\in\mathcal{Y}\setminus\{0^n\}}} \frac{\beta^{wxD}_{\bar{y}pr}}{2^n}\sum_{y_1\in S_{w,x}}(-1)^{r\cdot y_1}\left|w,x,D,\bar{y}\oplus g(y_1),p\right\rangle\Big\|^2\\
&\quad + 4\Big\|\sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp, r\in\mathcal{Y}\setminus\{0^n\}}} \frac{\beta^{wxD}_{\bar{y}pr}}{2^n}\sum_{y_1\in S_{w,x}}(-1)^{r\cdot y_1}\sum_{y_2\in\mathcal{Y}}\frac{1}{\sqrt{2^n}}\left|w,x,D\cup(x,y_2),\bar{y}\oplus g(y_1),p\right\rangle\Big\|^2\\
&\quad + 4\Big\|\sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp, r\in\mathcal{Y}\setminus\{0^n\}}} \frac{\beta^{wxD}_{\bar{y}pr}}{2^n}\sum_{y_1\in S_{w,x}}(-1)^{r\cdot y_1}\left|w,x,D,\bar{y}\oplus g(\perp),p\right\rangle\Big\|^2\\
&\quad + 4\Big\|\sum_{\substack{w,x,D,\bar{y},p,\\ D(x)=\perp, r\in\mathcal{Y}\setminus\{0^n\}}} \frac{\beta^{wxD}_{\bar{y}pr}}{2^n}\sum_{y_1\in S_{w,x}}(-1)^{r\cdot y_1}\sum_{y_2\in\mathcal{Y}}\frac{1}{\sqrt{2^n}}\left|w,x,D\cup(x,y_2),\bar{y}\oplus g(\perp),p\right\rangle\Big\|^2
\end{aligned}
$$

$$
\begin{aligned}
&= 8\sum_{\substack{w,x,D,\bar{y},\\ p,D(x)=\perp}}\sum_{\bar{y}_1\in g(\mathcal{Y})}\Big|\sum_{\substack{y_1\in S_{w,x}\\ g(y_1)=\bar{y}_1}}\sum_{r\in\mathcal{Y}\setminus\{0^n\}}\frac{\beta^{wxD}_{\bar{y}pr}}{2^n}(-1)^{r\cdot y_1}\Big|^2\\
&\qquad\qquad + 8\sum_{\substack{w,x,D,\bar{y},\\ p,D(x)=\perp}}\Big|\sum_{y_1\in S_{w,x}}\sum_{r\in\mathcal{Y}\setminus\{0^n\}}\frac{\beta^{wxD}_{\bar{y}pr}}{2^n}(-1)^{r\cdot y_1}\Big|^2
\end{aligned}
$$

$$
\overset{(e)}{\leq} 16\sum_{\substack{w,x,D,\bar{y},\\ p,D(x)=\perp}}\sum_{y_1\in S_{w,x}}|S_{w,x}|\Big|\sum_{r\in\mathcal{Y}\setminus\{0^n\}}\frac{\beta^{wxD}_{\bar{y}pr}}{2^n}(-1)^{r\cdot y_1}\Big|^2
$$

$$
\leq \frac{16}{2^n}\sum_{\substack{w,x,D,\bar{y},\\ p,D(x)=\perp}}\sum_{y_1\in\mathcal{Y}}\Big|\sum_{r\in\mathcal{Y}\setminus\{0^n\}}\frac{\beta^{wxD}_{\bar{y}pr}}{\sqrt{2^n}}(-1)^{r\cdot y_1}\Big|^2 \overset{(f)}{=} \frac{16}{2^n}\Gamma_S\||\Psi_2\rangle\|^2 .
$$

Here, (b) and (d) follow from $\|\sum_{i=1}^q|\psi_i\rangle\|^2 \leq q\sum_{i=1}^q\||\psi_i\rangle\|^2$ for arbitrary states $|\psi_i\rangle$; (c) and (f) arise from the definitions of $\||\Psi_1\rangle\|^2$ and $\||\Psi_2\rangle\|^2$, respectively; and (e) is obtained by first applying Jensen's inequality and then bounding the coefficient of the former term by $|S_{w,x}|$. Therefore, we obtain

$$
\begin{aligned}
\|(\mathsf{O}_0^{\mathsf{c}}-\mathsf{O}_1)\,|\Psi\rangle\| &\overset{(g)}{\leq} \|(\mathsf{O}_0^{\mathsf{c}}-\mathsf{O}_1)\,|\Psi_1\rangle\| + \|(\mathsf{O}_0^{\mathsf{c}}-\mathsf{O}_1)\,|\Psi_2\rangle\|\\
&= 2\sqrt{\Gamma_S/2^n}\|\Psi_1\| + 4\sqrt{\Gamma_S/2^n}\|\Psi_2\| \overset{(h)}{\leq} 6\sqrt{\Gamma_S/|\mathcal{Y}|} .
\end{aligned}
$$

Here, (g) comes from the triangle inequality, and (h) uses the fact that $\||\Psi_i\rangle\| \leq 1$. This completes the proof. □

A.4 Proof for the Lookup Lemma

Lemma 5.2 (Lookup Lemma)**.** *Let $\mathsf{H_c}:\mathcal{X}\to\mathcal{Y}$ be a compressed standard oracle with a database register D initialized to $\bigotimes_{i=1}^{q_{\mathsf{H}}}|(\perp,0^{\log|\mathcal{Y}|})\rangle$, and let O_0 be a quantum oracle acting on the input register W (defined over $\mathcal{W}$), the database register D, and an output register $\bar{\mathsf{Y}}$ (defined over $\bar{\mathcal{Y}}$) as*

$$
\mathsf{O}_0 : |w,D,\bar{y}\rangle_{\mathsf{WD\bar{Y}}} \mapsto |w,D,\bar{y}\oplus f_0(w,D)\rangle_{\mathsf{WD\bar{Y}}}
$$

with

$$
f_0(w,D) = \begin{cases} g_{w,D}(x), & \text{if } (x,D(x))\in S_w, \text{ where } x := d(w) ,\\ g_{w,D}(\perp), & \text{otherwise} ,\end{cases}
$$

where each function $g_{w,D}:\mathcal{X}\cup\{\perp\}\to\bar{\mathcal{Y}}$ and each set $S_w\subseteq\mathcal{X}\times\mathcal{Y}$ are determined by (w,D) and by w, respectively, and $d:\mathcal{W}\to\mathcal{X}$ is a fixed function. Let O_1 be the quantum oracle

$$
\mathsf{O}_1 : |w,D,\bar{y}\rangle_{\mathsf{WD\bar{Y}}} \mapsto |w,D,\bar{y}\oplus f_1(w,D)\rangle_{\mathsf{WD\bar{Y}}}
$$

with

$$f_1(w,D) = \begin{cases} g_{w,D}(x), & \text{if } \exists (x,y) \in D \text{ s.t. } (x,y) \in S'_w \\ & \text{and } \forall x' < x,\ (x',y') \in D \implies (x',y') \notin S'_w \ , \\ g_{w,D}(\bot), & \text{otherwise} \ , \end{cases}$$

which uses the same registers $\mathsf{WD\bar{Y}}$ *and the same function* $g_{w,D}$ *as* O_0*. It should be note that the set* $S'_w \subseteq \mathcal{X} \times \mathcal{Y}$ *determined by* w *differs from* S_w *used in* O_0*.*

Let A *be a quantum oracle algorithm (not necessarily unitary) making at most* q_{H} *queries to* $\mathsf{H_c}$ *and at most* q_{O} *queries to* $\mathsf{O}_0/\mathsf{O}_1$*. For an arbitrary classical event* Ev *and a random* z *distributed according to* $\mathfrak{D}_z$*, define*

$$P_{\text{left}} \coloneqq \Pr[\mathsf{Ev} : A^{|\mathsf{H_c}\rangle,|\mathsf{O}_0\rangle}(z)] \text{ and } P_{\text{right}} \coloneqq \Pr[\mathsf{Ev} : A^{|\mathsf{H_c}\rangle,|\mathsf{O}_1\rangle}(z)] \ ,$$

where the probability is taken over $z \leftarrow \mathfrak{D}_z$ *and the internal randomness of* A *(including all quantum measurements). For each* $w \in \mathcal{W}$*, define the set*

$$T_w \coloneqq \{(x,y) \mid (x,y) \in S_w \text{ and } x = d(w)\} \ .$$

Then, let $R \subseteq \mathcal{X} \times \mathcal{Y}$ *be the binary relation given by*

$$R \coloneqq \{(x,y) \mid \exists w \in \mathcal{W} \text{ s.t. } (x,y) \in T_w \triangle S'_w\} \ ,$$

where $\triangle$ *denotes the symmetric difference, i.e.,* $T_w \triangle S'_w \coloneqq (T_w \setminus S'_w) \cup (S'_w \setminus T_w)$*. Let* $\Gamma_R \coloneqq \max_x |\{y \mid (x,y) \in R\}|$*. Then, we have*

$$|P_{\text{left}} - P_{\text{right}}| \leq 16(q_{\mathsf{H}}+1)\sqrt{\Gamma_R/|\mathcal{Y}|}, \ \left|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}\right| \leq 16(q_{\mathsf{H}}+1)\sqrt{\Gamma_R/|\mathcal{Y}|} \ .$$

Lemma 5.2. We will prove this lemma by means of the compressed semi-classical one-way to hiding lemma (Lemma 3.4). Let $\mathcal{D}$ be the set of classical database D in register D (as given in Definition 3.7). Using the relation R, we define the subset

$$S_R \coloneqq \{D \in \mathcal{D} \mid \exists (x, D(x)) \in R\} \ .$$

Let $\hat{A}$ be a quantum oracle algorithm that behaves exactly like A, except that after each query to $\mathsf{H_c}$ it immediately makes a query to the compressed semi-classical oracle $\mathcal{O}^{CSC}_{S_R}$ on register D. At the end, if Ev occurs and the event Find (defined in Definition 3.9) never occurs, $\hat{A}$ outputs 1; otherwise, it outputs 0. It is clear that Find does not occur if $\mathcal{O}^{CSC}_{S_R}$ is never queried. Thus, we have

$$\begin{aligned} \Pr[1 \leftarrow \hat{A}^{|\mathsf{H_c}\setminus S_R\rangle,|\mathsf{O}_0\rangle}(z)] &= \Pr[\mathsf{Ev} \wedge \overline{\mathsf{Find}} : A^{|\mathsf{H_c}\setminus S_R\rangle,|\mathsf{O}_0\rangle}(z)] \ , \\ \Pr[1 \leftarrow \hat{A}^{|\mathsf{H_c}\setminus S_R\rangle,|\mathsf{O}_1\rangle}(z)] &= \Pr[\mathsf{Ev} \wedge \overline{\mathsf{Find}} : A^{|\mathsf{H_c}\setminus S_R\rangle,|\mathsf{O}_1\rangle}(z)] \ , \\ \Pr[1 \leftarrow \hat{A}^{|\mathsf{H_c}\rangle,|\mathsf{O}_0\rangle}(z)] &= \Pr[\mathsf{Ev} : A^{|\mathsf{H_c}\rangle,|\mathsf{O}_0\rangle}(z)] \ , \\ \Pr[1 \leftarrow \hat{A}^{|\mathsf{H_c}\rangle,|\mathsf{O}_1\rangle}(z)] &= \Pr[\mathsf{Ev} : A^{|\mathsf{H_c}\rangle,|\mathsf{O}_1\rangle}(z)] \ . \end{aligned} \tag{6}$$

Note that if $D \notin S_R$, then $\mathsf{O}_0 |w,D,\bar{y}\rangle = \mathsf{O}_1 |w,D,\bar{y}\rangle$. Hence, if Find never occurs, the behaviours of $A^{|\mathsf{H_c}\setminus S_R\rangle,|\mathsf{O}_0\rangle}(z)$ and $A^{|\mathsf{H_c}\setminus S_R\rangle,|\mathsf{O}_1\rangle}(z)$ are identical. Thus,

$$\Pr[\mathsf{Ev} \wedge \overline{\mathsf{Find}} : A^{|\mathsf{H_c}\setminus S_R\rangle,|\mathsf{O}_0\rangle}(z)] = \Pr[\mathsf{Ev} \wedge \overline{\mathsf{Find}} : A^{|\mathsf{H_c}\setminus S_R\rangle,|\mathsf{O}_1\rangle}(z)] \ . \tag{7}$$

By the compressed semi-classical one-way to hiding lemma (Lemma 3.4), we have

$$\begin{aligned}
\left|\Pr[1 \leftarrow \hat{A}^{|\mathsf{H_c}\rangle,|\mathsf{O}_0\rangle}(z)] - \Pr[1 \leftarrow \hat{A}^{|\mathsf{H_c}\setminus S_R\rangle,|\mathsf{O}_0\rangle}(z)]\right| &\le \sqrt{q_{\mathsf{H}}(q_{\mathsf{H}}+1)\Delta}\ , \\
\left|\Pr[1 \leftarrow \hat{A}^{|\mathsf{H_c}\rangle,|\mathsf{O}_1\rangle}(z)] - \Pr[1 \leftarrow \hat{A}^{|\mathsf{H_c}\setminus S_R\rangle,|\mathsf{O}_1\rangle}(z)]\right| &\le \sqrt{q_{\mathsf{H}}(q_{\mathsf{H}}+1)\Delta}\ , \\
\left|\sqrt{\Pr[1 \leftarrow \hat{A}^{|\mathsf{H_c}\rangle,|\mathsf{O}_0\rangle}(z)]} - \sqrt{\Pr[1 \leftarrow \hat{A}^{|\mathsf{H_c}\setminus S_R\rangle,|\mathsf{O}_0\rangle}(z)]}\right| &\le \sqrt{q_{\mathsf{H}}(q_{\mathsf{H}}+1)\Delta}\ , \\
\left|\sqrt{\Pr[1 \leftarrow \hat{A}^{|\mathsf{H_c}\rangle,|\mathsf{O}_1\rangle}(z)]} - \sqrt{\Pr[1 \leftarrow \hat{A}^{|\mathsf{H_c}\setminus S_R\rangle,|\mathsf{O}_1\rangle}(z)]}\right| &\le \sqrt{q_{\mathsf{H}}(q_{\mathsf{H}}+1)\Delta}\ ,
\end{aligned} \tag{8}$$

where Δ denotes $\mathbb{E}[\|[J_{S_R}, \mathsf{CStO}]\|^2]$, $J_{S_R} \coloneqq \sum_{D\in S_R} |D\rangle\langle D|$, CStO is the unitary operator implementing $\mathsf{H_c}$, and the expectation is taken over the randomness of (S_R, z). Define the projection $\Sigma^\perp \coloneqq \sum_{D\in\mathcal{D}\,:\,\nexists(x,D(x))\in R} |D\rangle\langle D|$. Clearly $I - J_{S_R} = \Sigma^\perp$ with $I \coloneqq \sum_{D\in\mathcal{D}} |D\rangle\langle D|$. Hence, we have

$$\|[J_{S_R}, \mathsf{CStO}]\| \overset{(a)}{=} \|[I - J_{S_R}, \mathsf{CStO}]\| = \left\|[\Sigma^\perp, \mathsf{CStO}]\right\| \overset{(b)}{\le} 8\sqrt{\Gamma_R/|\mathcal{Y}|}\ , \tag{9}$$

where (a) follows from a basic property of commutators and (b) is given by [15, Lemma 4], with $\Gamma_R \coloneqq \max_x |\{y \mid (x,y) \in R\}|$. Finally, set

$$P_{\text{left}} \coloneqq \Pr[\mathsf{Ev} : A^{|\mathsf{H_c}\rangle,|\mathsf{O}_0\rangle}(z)] \text{ and } P_{\text{right}} \coloneqq \Pr[\mathsf{Ev} : A^{|\mathsf{H_c}\rangle,|\mathsf{O}_1\rangle}(z)]\ .$$

Combining (6)–(9) along with the triangle inequality, we obtain

$$|P_{\text{left}} - P_{\text{right}}| \le 16(q_{\mathsf{H}}+1)\sqrt{\Gamma_R/|\mathcal{Y}|},\ \left|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}\right| \le 16(q_{\mathsf{H}}+1)\sqrt{\Gamma_R/|\mathcal{Y}|}\ ,$$

which completes the proof. □

A.5 Proof for the Puncturing Lemma

Lemma 5.3 (Puncturing Lemma). *Let $\mathsf{H_c} : \mathcal{X} \to \mathcal{Y}$ be a compressed standard oracle with a database register D initialized to $\bigotimes_{i=1}^{q_{\mathsf{H}}} |(\perp, 0^{\log|\mathcal{Y}|})\rangle$. For a fixed pair $(x^*, y^*) \in \mathcal{X} \times \mathcal{Y}$, define a punctured oracle*

$$\mathsf{H}'_{\mathsf{c}} : |x, y, D\rangle \mapsto \begin{cases} |x, y \oplus y^*, D\rangle\ , & \text{if } x = x^*\ , \\ \mathsf{H_c}\,|x, y, D\rangle\ , & \text{otherwise}\ , \end{cases}$$

which answers queries to x^* without updating the database D. Let O_{x^*,y^*} be a quantum oracle acting on the input register W (defined over $\mathcal{W}$), the database D, and an output register $\bar{\mathsf{Y}}$ (defined over $\bar{\mathcal{Y}}$) as

$$\mathsf{O}_{x^*,y^*} : |w, D, \bar{y}\rangle_{\mathsf{WD\bar{Y}}} \mapsto |w, D, \bar{y} \oplus f_{x^*,y^*}(w, D)\rangle_{\mathsf{WD\bar{Y}}}\ ,$$

where $f_{x^*,y^*} : \mathcal{W} \times \mathcal{D} \to \bar{\mathcal{Y}}$ satisfies

$$f_{x^*,y^*}(w, D) = f_{x^*,y^*}\big(w, D \cup (x^*, y^*)\big) \ \text{ for every } D \text{ with } D(x^*) = \perp\ .$$

Let A be a quantum oracle algorithm (not necessarily unitary) making at most q_{H} queries to $\mathsf{H_c}/\mathsf{H}'_{\mathsf{c}}$ and at most q_{O} queries to O_{x^*,y^*}. For an arbitrary classical event Ev, define

$$\begin{aligned}
P_{\text{left}} &\coloneqq \Pr[\mathsf{Ev} : x^* \leftarrow\!\!\$\, \mathcal{X},\ y^* \coloneqq \mathsf{H_c}(x^*),\ z \leftarrow \mathfrak{D}_z,\ A^{|\mathsf{H_c}\rangle,|\mathsf{O}_{x^*,y^*}\rangle}(y^*, z)]\ , \\
P_{\text{right}} &\coloneqq \Pr[\mathsf{Ev} : x^* \leftarrow\!\!\$\, \mathcal{X},\ y^* \leftarrow\!\!\$\, \mathcal{Y},\ z \leftarrow \mathfrak{D}_z,\ A^{|\mathsf{H}'_{\mathsf{c}}\rangle,|\mathsf{O}_{x^*,y^*}\rangle}(y^*, z)]\ ,
\end{aligned}$$

where the probability is taken over all random coins. Then, we have

$$|P_{\text{left}} - P_{\text{right}}| \le 2q_{\mathsf{O}}/\sqrt{|\mathcal{Y}|} \quad \textit{and} \quad \left|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}\right| \le 2q_{\mathsf{O}}/\sqrt{|\mathcal{Y}|}\ .$$

Lemma 5.3. Let CStO denote the unitary operator implementing $\mathsf{H_c}$. First, we can modify the generation of y^* in P_{left} as follows: sample $y^* \leftarrow\!\!\$\ \mathcal{Y}$, then immediately apply $F_{x^*}U_{x^*}^{y^*}$ to the database register D. Here, F_{x^*} is the decompression procedure at point x^* (as defined in Definition 3.7) applied to the database register D, and $U_{x^*}^{y^*}$ is a unitary operator acting on D defined by

$$U_{x^*}^{y^*}\,|D\rangle_{\mathsf{D}} \mapsto \begin{cases} |D \cup (x^*, y^*)\rangle_{\mathsf{D}}\,, & \text{if } D(x^*) = \perp\,, \\ |D \setminus (x^*, y^*)\rangle_{\mathsf{D}}\,, & \text{if } D(x^*) = y^*, \\ |D\rangle_{\mathsf{D}}\,, & \text{otherwise}\,. \end{cases}$$

Recall that initially D is set to $|D_\perp\rangle$, where $D_\perp$ denotes a database containing q_{H} pairs $(\perp, 0^{\log|\mathcal{Y}|})$. In P_{left}, for a classical $\mathsf{H_c}(x^*)$ query, we first prepare the state $|x^*, 0^{\log|\mathcal{Y}|}\rangle_{\mathsf{XY}}$ on the input/output registers X/Y of $\mathsf{H_c}$, then apply CStO to $|x^*, 0^{\log|\mathcal{Y}|}\rangle_{\mathsf{XY}}\,|D_\perp\rangle_{\mathsf{D}}$ to obtain

$$\begin{aligned} \mathsf{CStO}\,|x^*, 0^{\log|\mathcal{Y}|}\rangle\,|D_\perp\rangle &= F_{x^*}O_{x^*}F_{x^*}\,|x^*, 0^{\log|\mathcal{Y}|}\rangle\,|D_\perp\rangle \\ &= \frac{1}{\sqrt{|\mathcal{Y}|}}F_{x^*}O_{x^*}\sum_{y_1 \in \mathcal{Y}}|x^*, 0^{\log|\mathcal{Y}|}\rangle\,|D_\perp \cup (x^*, y_1)\rangle \\ &= \frac{1}{\sqrt{|\mathcal{Y}|}}F_{x^*}\sum_{y_1 \in \mathcal{Y}}|x^*, y_1\rangle\,|D_\perp \cup (x^*, y_1)\rangle\ . \end{aligned}$$

Then we measure Y in the computational basis, obtaining some $y^* \in \mathcal{Y}$ with probability $\frac{1}{\sqrt{|\mathcal{Y}|}}$, and the database register collapses to $F_{x^*}\,|D_\perp \cup (x^*, y^*)\rangle$. Obviously, sampling y^* directly uniformly from $\mathcal{Y}$ also has probability $\frac{1}{\sqrt{|\mathcal{Y}|}}$, and directly applying $F_{x^*}U_{x^*}^{y^*}$ to D yields

$$F_{x^*}U_{x^*}^{y^*}\,|D_\perp\rangle = F_{x^*}\,|D_\perp \cup (x^*, y^*)\rangle\ .$$

Hence, this modification does not affect the joint distribution of y^* and the state of D before A starts. Let

$$P_1 \coloneqq \Pr[\mathsf{Ev} : x^* \leftarrow\!\!\$\ \mathcal{X},\ y^* \leftarrow\!\!\$\ \mathcal{Y},\ F_{x^*}U_{x^*}^{y^*}\mathsf{D},\ z \leftarrow\!\!\$\ \mathfrak{D}_z,\ A^{|\mathsf{H_c}\rangle, |\mathsf{O}_{x^*,y^*}\rangle}(y^*, z)]\ ,$$

where the notation $F_{x^*}U_{x^*}^{y^*}\mathsf{D}$ indicates the application of $F_{x^*}U_{x^*}^{y^*}$ to register D. Therefore we have

$$P_{\text{left}} = P_1\ . \tag{10}$$

Recall that D is an internal register of $\mathsf{H_c}$ and O_{x^*,y^*} does not modify D. Thus, apart from $\mathsf{H_c}$ queries, $A^{|\mathsf{H_c}\rangle, |\mathsf{O}_{x^*,y^*}\rangle}(y^*, z)$ never performs any operation on D. Since for any $D_{q_{\mathsf{H}}-1} \in \mathcal{D}_{q_{\mathsf{H}}-1}$ with $D(x^*) = \perp$ we have

$$\begin{aligned} &\mathsf{CStO}\big(|x, y, D_{q_{\mathsf{H}}-1}\rangle \otimes F_{x^*}\,|(x^*, y^*)\rangle\big) \\ &\qquad = \begin{cases} F_xO_xF_x(|x, y, D_{q_{\mathsf{H}}-1}\rangle) \otimes F_{x^*}\,|(x^*, y^*)\rangle\,, & \text{if } x \neq x^*, \\ |x, y \oplus y^*, D_{q_{\mathsf{H}}-1}\rangle \otimes F_{x^*}\,|(x^*, y^*)\rangle\,, & \text{if } x = x^*, \end{cases} \end{aligned}$$

the database register D always contains the component $F_{x^*}\,|(x^*, y^*)\rangle$ during the execution of A. Note that the unitary operator implementing H'_{c} can be written as $\mathsf{CStO}_{x^*}^{y^*} \coloneqq |x^*\rangle\langle x^*| \otimes O^{y^*} + \sum_{x \in \mathcal{X}\setminus\{x^*\}}|x\rangle\langle x| \otimes F_xO_xF_x$, where $O^{y^*} : |y\rangle_{\mathsf{Y}} \mapsto |y \oplus y^*\rangle_{\mathsf{Y}}$. Thus, for any $D_{q_{\mathsf{H}}-1} \in \mathcal{D}_{q_{\mathsf{H}}-1}$, if $x \neq x^*$, we have

$$\mathsf{CStO}_{x^*}^{y^*}\big(|x, y, D_{q_{\mathsf{H}}-1}\rangle \otimes F_{x^*}\,|(x^*, y^*)\rangle\big) = F_xO_xF_x(|x, y, D_{q_{\mathsf{H}}-1}\rangle) \otimes F_{x^*}\,|(x^*, y^*)\rangle\ ,$$

and if $x = x^*$, we have

$$\mathsf{CStO}_{x^*}^{y^*}\left(|x, y, D_{q_\mathsf{H}-1}\rangle \otimes F_{x^*}\,|(x^*, y^*)\rangle\right) = |x, y \oplus y^*, D_{q_\mathsf{H}-1}\rangle \otimes F_{x^*}\,|(x^*, y^*)\rangle \ .$$

Hence, from A's perspective, $\mathsf{H_c}$ and $\mathsf{H_c'}$ are indistinguishable. Let

$$P_2 \coloneqq \Pr[\mathsf{Ev} : x^* \leftarrow\!\$\ \mathcal{X},\ y^* \leftarrow\!\$\ \mathcal{Y},\ F_{x^*}U_{x^*}^{y^*}\mathsf{D},\ z \leftarrow\!\$\ \mathfrak{D}_z,\ A^{|\mathsf{H_c'}\rangle,|\mathsf{O}_{x^*,y^*}\rangle}(y^*, z)] \ .$$

We have

$$P_1 = P_2 \ . \tag{11}$$

Next, we replace the oracle O_{x^*,y^*} accessed by A with $\hat{\mathsf{O}}_{x^*,y^*} \coloneqq (F_{x^*}U_{x^*}^{y^*}) \circ \mathsf{O}_{x^*,y^*} \circ (U_{x^*}^{y^*}F_{x^*})$. Without loss of generality, the execution of $A^{|\mathsf{H_c'}\rangle,|\mathsf{O}_{x^*,y^*}\rangle}(y^*, z)$ can be represented as $U_{q_\mathsf{O}+1} \circ \prod_{i=1}^{q_\mathsf{O}}(\mathsf{O}_{x^*,y^*} \circ U_i)\,|\psi_{y^*,z}\rangle$, where $|\psi_{y^*,z}\rangle$ is the initial state of $A^{|\mathsf{H_c'}\rangle,|\mathsf{O}_{x^*,y^*}\rangle}(y^*, z)$ depending on input (y^*, z), and for $i = 1, \ldots, q_\mathsf{O}+1$, each U_i is a unitary operator[9] encompassing all operations of A (including queries to the random oracle $\mathsf{H_c'}$) except the O_{x^*,y^*} queries. Moreover, by the principle of deferred measurement [36], any intermediate measurement can be postponed to the end of the circuit. Thus, the classical event Ev can be viewed as the outcome of a projective measurement on the final state of $A^{|\mathsf{H_c'}\rangle,|\mathsf{O}_{x^*,y^*}\rangle}(y^*, z)$. Let $|\Psi\rangle$ denote the state of $A^{|\mathsf{H_c'}\rangle,|\mathsf{O}_{x^*,y^*}\rangle}(y^*, z)$ immediately before an arbitrary query to O_{x^*,y^*}, i.e.,

$$|\Psi\rangle \coloneqq U_j \circ \prod_{i=1}^{j-1}(\mathsf{O}_{x^*,y^*} \circ U_i)\,|\psi_{y^*,z}\rangle$$

for some $j \leq q_\mathsf{O}$. This state resides in registers $\mathsf{WD\bar{Y}P}$, where $\mathsf{W}/\mathsf{\bar{Y}}$ are the input/output registers of O_{x^*,y^*}, D is the database register used by $\mathsf{H_c'}$, and P contains all remaining registers. Since the database register D always contains the component $F_{x^*}\,|(x^*, y^*)\rangle$ during the execution of A, we can write $|\Psi\rangle$ as

$$|\Psi\rangle = \sum_{\substack{w,\bar{y},p,D,\\ D(x^*)=\perp}} \beta_{p,D}^{w,\bar{y}}\,|w, \bar{y}, p\rangle_{\mathsf{W\bar{Y}P}} \otimes F_{x^*}|D \cup (x^*, y^*)\rangle_\mathsf{D} \ .$$

From the analysis in section A.6, we have

$$\left\|(\mathsf{O}_{x^*,y^*} - \hat{\mathsf{O}}_{x^*,y^*})\,|\Psi\rangle\right\| \leq 2/\sqrt{|\mathcal{Y}|} \ . \tag{12}$$

For a fixed input (y^*, z), denote the final state of $A^{|\mathsf{H_c'}\rangle,|\mathsf{O}_{x^*,y^*}\rangle}(y^*, z)$ by $|\Psi_{y^*,z}^0\rangle \coloneqq U_{q_\mathsf{O}+1} \circ \prod_{i=1}^{q_\mathsf{O}}(\mathsf{O}_{x^*,y^*} \circ U_i)\,|\psi_{y^*,z}\rangle$. Replacing every call to O_{x^*,y^*} with $\hat{\mathsf{O}}_{x^*,y^*}$ yields the final state of $A^{|\mathsf{H_c'}\rangle,|\hat{\mathsf{O}}_{x^*,y^*}\rangle}(y^*, z)$, denoted by $|\Psi_{y^*,z}^1\rangle \coloneqq U_{q_\mathsf{O}+1} \circ \prod_{i=1}^{q_\mathsf{O}}(\hat{\mathsf{O}}_{x^*,y^*} \circ U_i)\,|\psi_{y^*,z}\rangle$. We now aim to bound $\left\||\Psi_{y^*,z}^0\rangle - |\Psi_{y^*,z}^1\rangle\right\|$. To simplify notation, define $\Delta_j^k \coloneqq \prod_{i=j}^{k}(\mathsf{O}_{x^*,y^*} \circ U_i)$ and $\Upsilon_j^k \coloneqq \prod_{i=j}^{k}(\hat{\mathsf{O}}_{x^*,y^*} \circ U_i)$, with the convention that $\Delta_j^k = \Upsilon_j^k = I$ (the identity operator) whenever

[9] This follows because any quantum oracle algorithm can be transformed into a unitary quantum oracle algorithm with only a constant factor computational overhead and the same number of oracle queries [1,15].

$k < j$. Then,

$$
\begin{aligned}
&\left\| |\Psi^0_{y^*,z}\rangle - |\Psi^1_{y^*,z}\rangle \right\| \\
&= \| U_{q_O+1} \circ \Delta_1^{q_O} |\psi_{y^*,z}\rangle - U_{q_O+1} \circ \Upsilon_1^{q_O} |\psi_{y^*,z}\rangle \| \\
&\overset{(a)}{=} \| \Delta_1^{q_O} |\psi_{y^*,z}\rangle - \Upsilon_1^{q_O} |\psi_{y^*,z}\rangle \| \\
&\overset{(b)}{=} \left\| \sum_{i=1}^{q_O} \Big(\Upsilon_{i+1}^{q_O} \circ (\mathsf{O}_{x^*,y^*} \circ U_i) \circ \Delta_1^{i-1} |\psi_{y^*,z}\rangle - \Upsilon_{i+1}^{q_O} \circ (\hat{\mathsf{O}}_{x^*,y^*} \circ U_i) \circ \Delta_1^{i-1} |\psi_{y^*,z}\rangle \Big) \right\| \\
&\overset{(c)}{\leq} \sum_{i=1}^{q_O} \left\| \Upsilon_{i+1}^{q_O} \circ (\mathsf{O}_{x^*,y^*} \circ U_i) \circ \Delta_1^{i-1} |\psi_{y^*,z}\rangle - \Upsilon_{i+1}^{q_O} \circ (\hat{\mathsf{O}}_{x^*,y^*} \circ U_i) \circ \Delta_1^{i-1} |\psi_{y^*,z}\rangle \right\| \\
&\overset{(d)}{=} \sum_{i=1}^{q_O} \left\| (\mathsf{O}_{x^*,y^*} - \hat{\mathsf{O}}_{x^*,y^*}) \circ U_i \circ \Delta_1^{i-1} |\psi_{y^*,z}\rangle \right\| \\
&\overset{(e)}{\leq} \sum_{i=1}^{q_O} 2/\sqrt{|\mathcal{Y}|} = 2q_O/\sqrt{|\mathcal{Y}|} \ ,
\end{aligned}
$$

where (a) and (d) use the norm-preserving property of the unitaries U_{q_O+1} and $\Upsilon_{i+1}^{q_O}$; (b) follows from a telescoping sum decomposition; (c) applies the triangle inequality; and (e) uses (12), noting that $U_i \circ \Delta_1^{i-1} |\psi_{y^*,z}\rangle$ is exactly the state of $A^{|\mathsf{H}'_c\rangle, |\mathsf{O}_{x^*,y^*}\rangle}(y^*, z)$ just before the i-th call to O_{x^*,y^*}.

For random $y^* \leftarrow\$ \mathcal{Y}$ and $z \leftarrow \mathfrak{D}_z$, the final states of $A^{|\mathsf{H}'_c\rangle, |\mathsf{O}_{x^*,y^*}\rangle}(y^*, z)$ and $A^{|\mathsf{H}'_c\rangle, |\hat{\mathsf{O}}_{x^*,y^*}\rangle}(y^*, z)$ are the density operators $\rho_0 \coloneqq \sum_{y^*,z} p_{y^*,z} |\Psi^0_{y^*,z}\rangle\langle\Psi^0_{y^*,z}|$ and $\rho_1 \coloneqq \sum_{y^*,z} p_{y^*,z} |\Psi^1_{y^*,z}\rangle\langle\Psi^1_{y^*,z}|$, respectively, with $p_{y^*,z} = \Pr_{y^* \leftarrow\$ \mathcal{Y}, z \leftarrow \mathfrak{D}_z}[(y^*, z)]$. By Lemma A.1, we have

$$B(\rho_0, \rho_1) \leq \max_{y^*,z} \left\| |\Psi^0_{y^*,z}\rangle - |\Psi^1_{y^*,z}\rangle \right\| \leq 2q_O/\sqrt{|\mathcal{Y}|} \ .$$

Let

$$P_3 \coloneqq \Pr[\mathsf{Ev} : x^* \leftarrow\$ \mathcal{X},\ y^* \leftarrow\$ \mathcal{Y},\ F_{x^*} U_{x^*}^{y^*} \mathsf{D},\ z \leftarrow\$ \mathfrak{D}_z,\ A^{|\mathsf{H}'_c\rangle, |\hat{\mathsf{O}}_{x^*,y^*}\rangle}(y^*, z)] \ .$$

Then, we have

$$|P_2 - P_3| \overset{(f)}{\leq} B(\rho_0, \rho_1) \leq 2q_O\sqrt{|\mathcal{Y}|} \text{ and } |\sqrt{P_2} - \sqrt{P_3}| \overset{(g)}{\leq} B(\rho_0, \rho_1) \leq 2q_O\sqrt{|\mathcal{Y}|} \ , \tag{13}$$

where (f) and (g) follow from [1, Lemma 4].

Note that the final state of $A^{|\mathsf{H}'_c\rangle, |\hat{\mathsf{O}}_{x^*,y^*}\rangle}(y^*, z)$ in P_3 is

$$
\begin{aligned}
|\Psi^1_{y^*,z}\rangle &\coloneqq U_{q_O+1} \circ \prod_{i=1}^{q_O} (\hat{\mathsf{O}}_{x^*,y^*} \circ U_i) |\psi_{y^*,z}\rangle \\
&\overset{(h)}{=} U_{q_O+1} \big((F_{x^*} U_{x^*}^{y^*}) \mathsf{O}_{x^*,y^*} (U_{x^*}^{y^*} F_{x^*}) U_{q_O} \big) \big((F_{x^*} U_{x^*}^{y^*}) \mathsf{O}_{x^*,y^*} (U_{x^*}^{y^*} F_{x^*}) U_{q_O-1} \big) \\
&\qquad \cdots \big((F_{x^*} U_{x^*}^{y^*}) \mathsf{O}_{x^*,y^*} (U_{x^*}^{y^*} F_{x^*}) U_2 \big) \big((F_{x^*} U_{x^*}^{y^*}) \mathsf{O}_{x^*,y^*} (U_{x^*}^{y^*} F_{x^*}) U_1 \big) |\psi_{y^*,z}\rangle \\
&\overset{(i)}{=} (F_{x^*} U_{x^*}^{y^*}) U_{q_O+1} \big(\mathsf{O}_{x^*,y^*} (U_{x^*}^{y^*} F_{x^*}) (F_{x^*} U_{x^*}^{y^*}) U_{q_O} \big) \big(\mathsf{O}_{x^*,y^*} (U_{x^*}^{y^*} F_{x^*}) (F_{x^*} U_{x^*}^{y^*}) U_{q_O-1} \big) \\
&\qquad \cdots \big(\mathsf{O}_{x^*,y^*} (U_{x^*}^{y^*} F_{x^*}) (F_{x^*} U_{x^*}^{y^*}) U_2 \big) \big(\mathsf{O}_{x^*,y^*} U_1 (U_{x^*}^{y^*} F_{x^*}) \big) |\psi_{y^*,z}\rangle \\
&= (F_{x^*} U_{x^*}^{y^*}) U_{q_O+1} \circ \prod_{i=1}^{q_O} (\mathsf{O}_{x^*,y^*} \circ U_i) (U_{x^*}^{y^*} F_{x^*}) |\psi_{y^*,z}\rangle \ ,
\end{aligned}
$$

where (h) follows from the definition of $\hat{\mathsf{O}}_{x^*,y^*}$, and (i) holds because F_{x^*} and $U_{x^*}^{y^*}$ act only on the database register D and both commute with $\mathsf{CStO}_{x^*}^{y^*}$, hence they commute with all U_i.

Note that this is equivalent to saying that the algorithm A in P_3 replaces each query to $\hat{\mathsf{O}}_{x^*,y^*}$ with a query to O_{x^*,y^*} while inserting an extra $(U_{x^*}^{y^*} F_{x^*})$ before the algorithm starts and a final $(F_{x^*} U_{x^*}^{y^*})$ after it ends. Note that applying $(U_{x^*}^{y^*} F_{x^*})$ to the database register D before the execution is equivalent to uncomputing $(F_{x^*}U_{x^*}^{y^*})\mathsf{D}$; moreover, the final $(F_{x^*}U_{x^*}^{y^*})$ does not act on A's registers and therefore does not affect the probability of the event Ev. Consequently,

$$\Pr\left[\mathsf{Ev} : x^* \leftarrow\$ \mathcal{X},\ y^* \leftarrow\$ \mathcal{Y},\ z \leftarrow \mathfrak{D}_z,\ A^{|\mathsf{H}'_{\mathsf{c}}\rangle, |\mathsf{O}_{x^*,y^*}\rangle}(y^*, z)\right] = P_3 \ ,$$

that is,

$$P_{\text{right}} = P_3. \tag{14}$$

Combining equations (10), (11), (13) and (14), we conclude

$$|P_{\text{left}} - P_{\text{right}}| \leq 2q_{\mathsf{O}}/\sqrt{|\mathcal{Y}|} \quad \text{and} \quad \left|\sqrt{P_{\text{left}}} - \sqrt{P_{\text{right}}}\right| \leq 2q_{\mathsf{O}}/\sqrt{|\mathcal{Y}|} \ .$$

$\square$

A.6 Bound on $\left\|(\mathsf{O}_{x^*,y^*} - \hat{\mathsf{O}}_{x^*,y^*})\,|\Psi\rangle\right\|$

Let $n := \log|\mathcal{Y}|$. Define $|\hat{D}_x\rangle := \frac{1}{\sqrt{2^n}} \sum_{y_1 \in \mathcal{Y}} |D \cup (x, y_1)\rangle$ for any D such that $D(x) = \perp$. Following O_{x^*,y^*}, $\hat{\mathsf{O}}_{x^*,y^*}$, and $|\Psi\rangle$ in the proof of Lemma 5.3, we have

$$\begin{aligned}
\mathsf{O}_{x^*,y^*}\,|\Psi\rangle &= \mathsf{O}_{x^*,y^*} \sum_{\substack{w,\bar{y},p,D,\\ D(x^*)=\perp}} \beta_{p,D}^{w,\bar{y}}\,|w,\bar{y},p\rangle \otimes F_{x^*}|D \cup (x^*,y^*)\rangle \\
&= \mathsf{O}_{x^*,y^*} \sum_{\substack{w,\bar{y},p,D,\\ D(x^*)=\perp}} \beta_{p,D}^{w,\bar{y}}\,|w,\bar{y},p\rangle \otimes \Big(|D \cup (x^*,y^*)\rangle + \frac{1}{\sqrt{2^n}}\,|D\rangle - \frac{1}{\sqrt{2^n}}\,|\hat{D}_{x^*}\rangle\Big) \\
&\overset{(a)}{=} \sum\nolimits_{\substack{w,\bar{y},p,D,\\ D(x^*)=\perp}} \Big(\beta_{p,D}^{w,\bar{y}}\,|w, \bar{y} \oplus f_{x^*,y^*}(w,D), p, D \cup (x^*,y^*)\rangle \\
&\qquad + \frac{\beta_{p,D}^{w,\bar{y}}}{\sqrt{2^n}}\,|w, \bar{y} \oplus f_{x^*,y^*}(w,D), p, D\rangle - \frac{\beta_{p,D}^{w,\bar{y}}}{\sqrt{2^n}} O_{x^*,y^*}\,|w,\bar{y},p,\hat{D}_{x^*}\rangle\Big) \ ,
\end{aligned}$$

and

$$\begin{aligned}
\hat{\mathsf{O}}_{x^*,y^*}\,|\Psi\rangle &= (F_{x^*}U_{x^*}^{y^*})\mathsf{O}_{x^*,y^*}(U_{x^*}^{y^*}F_{x^*}) \sum_{\substack{w,\bar{y},p,D,\\ D(x^*)=\perp}} \beta_{p,D}^{w,\bar{y}}\,|w,\bar{y},p\rangle \otimes F_{x^*}\,|D \cup (x^*,y^*)\rangle \\
&\overset{(b)}{=} (F_{x^*}U_{x^*}^{y^*})\mathsf{O}_{x^*,y^*} \sum\nolimits_{\substack{w,\bar{y},p,D,\\ D(x^*)=\perp}} \beta_{p,D}^{w,\bar{y}}\,|w,\bar{y},p\rangle \otimes |D\rangle \\
&\overset{(c)}{=} (F_{x^*}U_{x^*}^{y^*}) \sum\nolimits_{\substack{w,\bar{y},p,D,\\ D(x^*)=\perp}} \beta_{p,D}^{w,\bar{y}}\,|w, \bar{y} \oplus f_{x^*,y^*}(w,D), p\rangle \otimes |D\rangle \\
&\overset{(d)}{=} \sum\nolimits_{\substack{w,\bar{y},p,D,\\ D(x^*)=\perp}} \Big(\beta_{p,D}^{w,\bar{y}}\,|w, \bar{y} \oplus f_{x^*,y^*}(w,D), p, D \cup (x^*,y^*)\rangle \\
&+ \frac{\beta_{p,D}^{w,\bar{y}}}{\sqrt{2^n}}\,|w, \bar{y} \oplus f_{x^*,y^*}(w,D), p, D\rangle - \frac{\beta_{p,D}^{w,\bar{y}}}{\sqrt{2^n}}\,|w, \bar{y} \oplus f_{x^*,y^*}(w,D), p, \hat{D}_{x^*}\rangle\Big),
\end{aligned}$$

where (a) and (c) follow from the definition of O_{x^*,y^*}; (b) and (d) follow from the definitions of $U_{x^*}^{y^*}$ and F_{x^*}.

Consequently,

$$
\begin{aligned}
&\left\|(\mathsf{O}_{x^*,y^*}-\hat{\mathsf{O}}_{x^*,y^*})\left|\Psi\right\rangle\right\|^2\\
&=\Big\|O_{x^*,y^*}\sum_{\substack{w,\bar{y},p,D,\\D(x^*)=\perp}}\frac{\beta^{w,\bar{y}}_{p,D}}{\sqrt{2^n}}\left|w,\bar{y},p,\hat{D}_{x^*}\right\rangle-\sum_{\substack{w,\bar{y},p,D,\\D(x^*)=\perp}}\frac{\beta^{w,\bar{y}}_{p,D}}{\sqrt{2^n}}\left|w,\bar{y}\oplus f_{x^*,y^*}(w,D),p,\hat{D}_{x^*}\right\rangle\Big\|^2\\
&\overset{(e)}{\leq} 2\Big\|O_{x^*,y^*}\sum_{\substack{w,\bar{y},p,D,\\D(x^*)=\perp}}\frac{\beta^{w,\bar{y}}_{p,D}}{\sqrt{2^n}}\left|w,\bar{y},p,\hat{D}_{x^*}\right\rangle\Big\|^2+2\Big\|\sum_{\substack{w,\bar{y},p,D,\\D(x^*)=\perp}}\frac{\beta^{w,\bar{y}}_{p,D}}{\sqrt{2^n}}\left|w,\bar{y}\oplus f_{x^*,y^*}(w,D),p,\hat{D}_{x^*}\right\rangle\Big\|^2\\
&\overset{(f)}{=}\frac{2}{2^n}\Big\|\sum_{\substack{w,\bar{y},p,D,\\D(x^*)=\perp}}\beta^{w,\bar{y}}_{p,D}\left|w,\bar{y},p,\hat{D}_{x^*}\right\rangle\Big\|^2+\frac{2}{2^n}\Big\|\sum_{\substack{w,\bar{y},p,D,\\D(x^*)=\perp}}\beta^{w,\bar{y}}_{p,D}\left|w,\bar{y}\oplus f_{x^*,y^*}(w,D),p,\hat{D}_{x^*}\right\rangle\Big\|^2\\
&\overset{(g)}{=}\frac{2}{2^n}\sum_{\substack{w,\bar{y},p,D,\\D(x^*)=\perp}}\Big\|\beta^{w,\bar{y}}_{p,D}\left|w,\bar{y},p,\hat{D}_{x^*}\right\rangle\Big\|^2+\frac{2}{2^n}\sum_{\substack{w,\bar{y},p,D,\\D(x^*)=\perp}}\Big\|\beta^{w,\bar{y}}_{p,D}\left|w,\bar{y}\oplus f_{x^*,y^*}(w,D),p,\hat{D}_{x^*}\right\rangle\Big\|^2\\
&=\frac{2}{2^n}\sum\nolimits_{\substack{w,\bar{y},p,D,\\D(x^*)=\perp}}\left|\beta^{w,\bar{y}}_{p,D}\right|^2+\frac{2}{2^n}\sum\nolimits_{\substack{w,\bar{y},p,D,\\D(x^*)=\perp}}\left|\beta^{w,\bar{y}}_{p,D}\right|^2\overset{(h)}{\leq}\frac{4}{2^n}\|\left|\Psi\right\rangle\|^2=4/|\mathcal{Y}|\ ,
\end{aligned}
$$

where (e) uses $\|\left|\psi\right\rangle-\left|\phi\right\rangle\|^2\leq 2\|\left|\psi\right\rangle\|^2+2\|\left|\phi\right\rangle\|^2$ for any states $\left|\psi\right\rangle,\left|\phi\right\rangle$; (f) holds because the unitary O_{x^*,y^*} preserves the norm; (g) follows since $\|\left|\psi\right\rangle+\left|\phi\right\rangle\|^2=\|\left|\psi\right\rangle\|^2+\|\left|\phi\right\rangle\|^2$ for any orthogonal states $\left|\psi\right\rangle,\left|\phi\right\rangle$; and (h) uses the definition $\|\left|\Psi\right\rangle\|^2=\sum_{w,\bar{y},p,D,D(x^*)=\perp}\|\beta^{w,\bar{y}}_{p,D}\|^2$. Taking the square root completes the proof. □